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ADVANCED

# IT FRONTIERS

## Advanced Series

**Teacher's Edition**

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**First Edition**

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Teacher's Edition

## Author's Note

This series reflects my conviction that computer science education must return to its scientific foundations. In an era of rapidly evolving technologies, the focus should remain on enduring concepts, presented here through clear structure, practical examples, and sound pedagogy that encourage curiosity and critical thinking. Digital literacy emerges naturally through the activities, yet it should not be confined to computer science alone — it is a shared responsibility across all disciplines. While programming is important, the greater value lies in cultivating algorithmic thinking — the ability to analyze, design, and reason about processes. To support this, I developed Jargon, a pseudocode framework that minimizes the distractions of syntax and semantics, enabling learners to focus on conceptual clarity and problem-solving. My aim is to make computer science accessible, rigorous, and relevant for young learners, while preparing them to adapt to technologies that will continue to change.

## Digital Literacy in Education

Digital literacy is not the sole responsibility of computer science education but a shared responsibility across all disciplines. Each subject carries its own digital tools, applications, and nuances of use: from data analysis in science, to dynamic modeling in mathematics, to research and composition in language learning. While computer science provides the foundation for understanding these technologies, their most meaningful application emerges within the disciplines where they are best used.

## Grounded in Research

This program reflects a growing consensus in computer science education: that we must balance practical coding with the enduring foundations of the discipline. Researchers and educators emphasize that computer science is not only about learning tools, which change rapidly, but about cultivating computational thinking and algorithmic reasoning — skills that remain constant.

The K-12 Computer Science Framework highlights computational thinking as essential for all learners, while international assessments such as PISA now include these skills alongside mathematics and science. Scholars from MIT and the ACM community have cautioned against reducing computer science to vocational training, noting that its true power lies in analysis, problem-solving, and conceptual clarity.

This publication embraces these principles: programming is introduced as a valuable skill, but the greater emphasis is on nurturing algorithmic thinking and understanding the scientific concepts that endure beyond changing technologies.

*This series seeks to make computer science accessible and enduring, grounded in science, informed by research, and integrated across education.*

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# 1

# Computers

# Lesson 1: Introducing Computers

Computers are all around us. We use them in almost every aspect of our lives. But what are they? How do we best define them? What do they do for us?

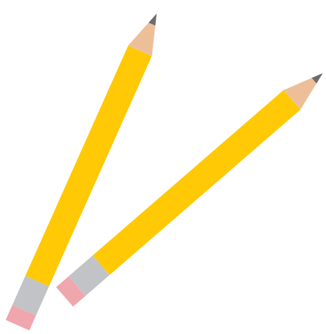
A COMPUTER IS A TECHNOLOGY WHOSE PURPOSE IS TO PROCESS INFORMATION ACCORDING TO A STORED PROGRAM.

## Technology

The word technology refers to the tools and machines we use to help us perform various tasks. Every technology has a specific purpose; it is designed for a particular task.

### Example 1.1

A pencil is a form of technology. It is a tool designed to help us perform the task of writing. Even though there are many different types of pencils, they are all designed to perform the same task, writing.



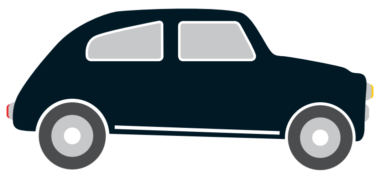
**Notes**

The term 'technology' encompasses more than simply tools and machines; it includes any practical application of knowledge.

The Merriam-Webster Dictionary defines technology as "the practical application of knowledge, especially in a particular area."

### Example 1.2

A car is also a form of technology. It is a machine designed to help us travel from one place to another on land. Regardless of the shape, size and color, all cars are designed for the specific task of transporting people from one place to another on land.



**Every technology is designed for a purpose.**

## Processing

To process something means to change it from one form to another. Processing is not any kind of change; it is a very structured form of change that can be replicated. A process is structured according to a specific set of steps; if given the same starting conditions and materials, these steps will provide the same end result every time they are performed.

### Example 1.3

Processed food is food that undergoes change starting from certain raw materials. Bread is processed food. It starts as wheat, a raw material that is unprocessed. The processing of wheat to make bread must follow a specific set of steps.

**A process is a series of steps that complete a particular task and can be replicated.**

Anyone who replicates these steps will make bread from wheat. Regardless of how many times the steps are repeated and regardless of who performs these steps, the processing of wheat will always result in bread.

**Processing of Wheat to Bread**

STEPS

1. *Clean the wheat grains*
2. *Grind the wheat grains*
3. *Sift the flour to remove the husk*
4. *Send the flour to the bakery*
5. *Use flour to make bread dough*
6. *Bake the dough*



## Information

In general, the term information refers to anything meaningful that tells us something about ourselves or the world around us. This book contains information. The information is conveyed in the words and pictures of this book.

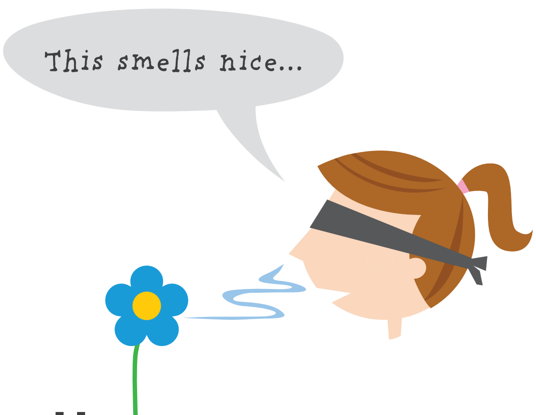
As humans, we process information. We see, feel and hear things that we process to gain information about the world around us. We retain this information in our memories. Later, we may use this information to interact with the world.

### Example 1.4

Sarah and her friends were playing the “Guess the Object” game. Sarah was blindfolded, and her friends laid out objects in front of her. Using her senses, she had to gather information about these objects to guess what they were.

Even without seeing the objects, Sarah could identify them. In addition to each object's shape, which she could feel, the flowers smelled nice, the keys jingled and the ice cube felt cold.

Sarah's senses helped her gather information about each object. But that was not enough to identify each object. Sarah must have had in her memory some information about the characteristics of each object against which she could compare. The new information gathered by her senses was compared with the information she had in her memory.




**[1]**

Sarah gets information by smelling the object.

**[2]**

Sarah processes the information by comparing it with previous information in her memory.

To identify the object, Sarah's brain had to process information. Sarah's brain had to go through a series of steps in which she compared the information acquired through her senses with the information she had in her memory. By making these comparisons, Sarah was able to identify the object and state its name. The information acquired through her senses was processed; it was changed to the name of the object.


**[3]**
Sarah informs her friends that the object is a flower.

Sarah followed the same steps to identify the rest of the objects. Her senses acquired information about each object. She compared that information with that in her memory. An object was identified when a match was found. She would then inform her friends of the name of each object.



**[4]**
Sarah follows the same steps to identify the other two objects.

# Computers

Computers are a form of technology designed for the purpose of processing information. This includes creating information, sharing information and storing information, all of which require some form of processing.

## Creating Information

People use computers to create new information. When we type a text message, email or essay, in each case, we are creating new information. When we record a voice message or video, we are creating new information. In this context, information is considered to be any meaningful content. Making new content, whether in the form of text, audio or images, is all made possible by computers.

Creating new information is not limited to typing an essay, recording a voice message or capturing an image. Creating new information also includes the processing of data either through calculation or analysis.

### Example 1.5

Every school keeps track of its students' grades. At the end of each term, the school issues a report card to every student. The computer helps create every report card. Teachers enter the grades for each subject into the computer system. The computer then processes this information to provide an average for each student. The average is the new information.

| Subject | Grade |
|---|---|
| Math | 75 |
| English | 78 |
| Science | 82 |
| Social Studies | 74 |
| IT Studies | 80 |
| Art | 83 |
| Music | 70 |
| PE | 82 |
| Average | 78 |

Data is processed to calculate the average.

The school ranks its students. To do so, the list of students needs to be sorted from the highest average to the lowest. The computer processed this information and sorted the list of students. This provides the school with the student ranks, which is new information.

| Student | Average |
|---|---|
| Sarah | 88 |
| Omar | 81 |
| Leena | 75 |
| Pradeep | 82 |
| Jenna | 72 |
| Raj | 90 |
| Mario | 70 |
| Ahmad | 77 |

| Student | Average | Rank |
|---|---|---|
| Raj | 90 | 1 |
| Sarah | 88 | 2 |
| Pradeep | 82 | 3 |
| Omar | 81 | 4 |
| Ahmad | 77 | 5 |
| Leena | 75 | 6 |
| Jenna | 72 | 7 |
| Mario | 70 | 8 |

Data is processed to sort the list of students from highest to lowest average. The rank is new information.

## Sharing Information

Creating information and new content becomes more valuable when it can be shared with others. Computers have tremendously increased our ability to share information.

Sharing information is made possible by the Internet, a complex network of telecommunication technologies that spans the globe and allows computers to connect to each other and to other digital devices.

> The internet is a complex network of telecommunications technologies that connects computers across the world.

All information accessible through the Internet is referred to as the World Wide Web or simply as the Web.

Information on the Web is organized into files. The content of each file can be text, audio or images. This content is displayed as webpages, music, pictures or video. Each file has a specific address, a web address, that is used to locate it.

**An ADDRESS is used to locate information.**

Information on the Web is shared as either private or public, which determines who gains access to this information. Access to private information is restricted and requires a password. In contrast, public information does not require a password and is accessible to everyone.

### Example 1.6

Your school may have a website. The school website is made up of several web pages that contain information made public by your school. In addition, the school may offer its students a portal through which they can sign in with their passwords to access their private information, such as their grades.

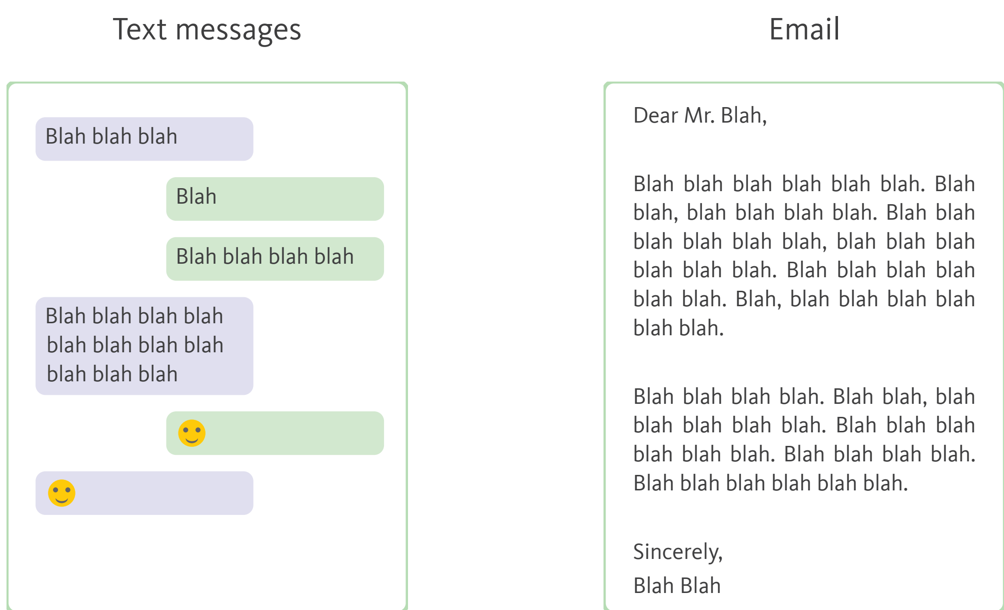
Username:

Password:

**COMMUNICATION IS A FORM OF SHARING INFORMATION THAT RESULTS IN AN EXCHANGE BETWEEN TWO OR MORE PEOPLE.**

Another form of sharing information is communication. Communication is sharing that results in an exchange of information between two or more people.

Two methods of communication facilitated by computers are messaging and emailing. Both methods allow for the exchange of information in the form of text and audiovisual content. Messaging is better suited for shorter and more frequent exchanges of messages, whereas emailing is better suited for more prolonged and less frequent exchanges of messages.




Messaging and emailing are two forms of communication facilitated by computers.

## Storing Information

Once information has been created, it must be stored. Storage of information is important. If information were volatile, meaning it "evaporates" whenever a computer was powered off, no information would be left. All our information would be lost; every message, every essay and every photo would be lost.

Every computer must be capable of storing information. Storage provides a permanent repository for information. Computers not only allow us to store our information, but they also provide us with the facilities to search and find specific items of information kept in storage.

**Storage is the permanent repository for information.**

All information is stored in the form of files. Every file is given a filename by which it can be identified. Every file is also given a specific address at which it can be located. The size of a file is highly variable; it can contain 1 byte or several gigabytes of information. The content of a file may be mostly instructions, which is the case for apps, or it may only be data, as is the case for photos.

**A file is a unit of storable information.**

## The Stored Program

The stored program concept is an important concept that made the modern computer possible. Imagine your computer without any apps. What could it possibly do? Would it be of any use to you?

The fact is that every app on your computer performs a particular task. If you want to compose an essay, you use a word processing app. If you want to record a video, you use a camera app. If you want to search for information on the Web, you use a web browser app. Each of these apps is a program stored in your computer, and each program contains the instructions the computer must follow to perform that particular task.

A PROGRAM IS A SET OF INSTRUCTIONS INTENDED TO PERFORM A PARTICULAR TASK.

In addition to the apps you use, there are other stored programs on your computer that you do not see and do not directly use, but they are critical for making your computer work the way it does. Most of these programs are part of the operating system, which is a group of programs that work together to allow your computer to use a variety of apps and to function properly.

## Summary

The word technology refers to the tools and machines we use to help us perform various tasks. Every technology is designed for a purpose.

Processing is a structured form of change that can be replicated. Computers are designed for the purpose of processing information.

Computers allow us to create, share and store information.

The Internet is a complex network of telecommunications technologies that connects computers across the world. It is used to share and exchange information. Access to shared information is either restricted, private or unrestricted, public.

An address is used to locate information. A web address is an address used to locate information on the World Wide Web.

Communication is a form of sharing information that results in an exchange between two or more people. Emailing and messaging are two examples of methods of communication.

Storage is the permanent repository for information. All information is stored as files. A file is a unit of storable information. Every file has a filename by which it is identified and an address by which it is located.

A program is a set of instructions intended to perform a particular task. Stored programs allow the computer to perform different tasks. An app is an example of a stored program.

A computer is a technology whose purpose is to process information according to a stored program.

## Activity 1.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F | STATEMENT |
|---|---|
| T | Information is any meaningful content. |
| F | Computers are not a form of technology. |
| T | Every technology has a purpose. |
| T | Processing generally means changing something in accordance with a series of steps. |
| F | Information cannot be processed. |
| F | Computers allow us to only share information. |
| T | Storage is the permanent repository for information. |
| T | A program is a set of instructions intended to perform a particular task. |
| F | An application is not an example of a stored program. |
| T | Stored programs made modern day computers possible. |

## Activity 1.2

Identify machines, appliances and devices for each of the forms of technology based on its stated purpose.

1. Which two of these machines are forms of technology designed for the purpose of transportation?
   a) ship
   b) microwave oven
   c) airplane
   d) computer

2. Which two of these appliances are forms of technology designed for the purpose of cooling food?
   a) washing machine
   b) freezer
   c) air conditioner
   d) refrigerator

3. Which two of these devices are forms of technology designed for the purpose of generating electrical energy?
   a) wind turbine
   b) phone battery
   c) solar panel
   d) electric motor

4. Which two of these devices are forms of technology designed for the purpose of processing information?
   a) laptop
   b) television
   c) printer
   d) desktop

## Activity 1.3

Answer the questions below.

1. Transportation technologies can be further categorized based on the medium within which they are intended to travel. List two machines designed for the purpose of traveling in each medium.

   Land : car, train, bus

   Air : airplane, helicopter, hot-air balloon

   Water : ship, submarine, yacht

2. Investigate the two forms of technology. Identify and state the purpose of each.

   Nanotechnology : its purpose is to manipulate matter at the tiniest scale, the nanoscale (1 -100 nm), to make new products.

   Biotechnology : its purpose is to use biological processes, living cells and organisms to make new products.

3. "Recognizing the purpose of any particular form of technology is important." Analyze this statement, investigate and provide at least two supporting reasons.

   Recognizing the purpose behind a particular technology allows us to:

   1) understand how to use it.

   2) determine if it will be useful to us.

   3) avoid potential risks.

## Activity 1.4

Answer the questions below.

1. What information do drivers get when they see a red traffic light?
   a) go
   b) stop
   c) turn left
   d) turn right

2. What information do drivers get when they hear the sirens of an ambulance behind them?
   a) move out of the way
   b) block the road
   c) get help
   d) cover your ears

3. If you were blindfolded and asked to guess if an object in front of you was an ice cube, what information would you be searching for?
   a) cold, wet, smooth
   b) warm, dry, rough
   c) cold, wet, rough
   d) warm, wet, smooth

4. For each of the questions above, state the sense through which information was acquired.
   Question 1: sight
   Question 2: hearing
   Question 3: touch

## Activity 1.5

Answer the questions below.

1. Which of the following best describes a process?
   a) A random sequence of events
   b) A type of machine used in manufacturing
   c) A clear sequence of steps for completing a task
   d) A finished product

2. Which of the following is a characteristic of a well-defined process?
   a) It is constantly changing
   b) It is repeatable and consistent
   c) It does not have a clear beginning
   d) It does not have clear steps to follow

3. Which of the following is not an example of a process?
   a) Baking a cake
   b) Getting dressed
   c) Watching a movie
   d) Brushing teeth

4. Why do you think it is important to understand a process?
   a) It helps us complete the task quickly and properly
   b) It allows us to skip steps in the process
   c) It makes the process more complicated than it needs to be
   d) It ensures that you don't need to follow any specific steps

## Activity 1.6

Answer the questions below.

1. What information do you need to bake a cake?
   a) Which ingredients go into making a cake
   b) What steps need to be performed
   c) Which utensils are required to perform the steps
   d) All the above

2. What item of information would be useful for you to navigate your way out of your classroom?
   a) The color of the walls
   b) The location of the exit
   c) The names of the students
   d) The number of open windows

3. "Information is critical to the process of decision-making." Is this statement true? Why? Give an example.

   Yes, this statement is true. Having the correct information allows us to make good decisions and to avoid making mistakes. The lack of information or having the wrong information can lead to unwanted outcomes. For example, when going on a road trip, having the wrong location for the destination or not having the location may result in us getting lost and not arriving at the destination.

## Activity 1.7

Match each word with the statement that best describes it.

**File** **Filename** **Program** **Address**

**Web Address** **Internet** **Storage**

1. A telecommunications technology that spans the globe Internet
2. A unit of storable information File
3. It is used to locate a file Address
4. It is used to identify a file Filename
5. It is used to locate a file across the Internet Web Address
6. A computer's permanent repository of information Storage
7. A set of instructions intended to perform a task Program

## Activity 1.8

For each of the following scenarios, identify how information was created, stored and shared.

1. Karl watches a video on YouTube.

Created : The video is the information created by someone.

Stored : The video is stored on a YouTube computer (server).

Shared : The video is viewed by people who find it , like Karl.

2. Jenna uses her smartphone to send a voice message to her friend.

Created : The voice message is the information created.

Stored : The voice message is stored on Jenna's smartphone.

Shared : The voice message is sent through the Internet to her friend's phone.

3. A teacher enters her students' grades. Later, parents view their children's report cards which contain their general average and rank.

Created : 1) Grades are entered. 2) General averages are calculated. 3) Ranks are identified by sorting based on general averages.

Stored : Student grades, general averages and ranks are stored in the school's main computer (server).

Shared : Parents access the school's main computer using the Internet and view their child's grades, general average and rank.

4. Describe your own example of using a computer to create, store and share information.

## Activity 1.9

Match each word with the statement that best describes it.

**Program** **Private** **Messaging** **Public**

**Email** **Applications** **Images**

1. Frequent and short exchanges of information Messaging
2. Information that has restricted access Private
3. Information that has unrestricted access Public
4. Communication suitable for long, infrequent exchanges Email
5. Examples of files that contain only data Images
6. Examples of files that contain both instructions and data Applications
7. A set of instructions that perform a particular task Program

## Activity 1.10

Answer the questions below.

1. Two common methods of communication are messaging and email. Describe how the two methods differ.

   Messaging may be more effective for quick and informal communication with friends and classmates, while email may be more effective for more formal or professional communication with teachers or business partners.

2. In the previous question, the differences between messaging and email were described. Describe how the two methods are similar.

   1) Both are forms of digital communication.

   2) Both require the Internet to transfer information.

   3) Both can be used to send information in the form of text, audio and video.

3. Every application performs a particular task. List two applications and what you use them for.

   Answers may vary. For example, I use PowerPoint to prepare a presentation for my classmates about my science project.

4. List at least three ways you will use computers to help you learn.

   Possible answers may include reference to: 1) access to educational resources, 2) collaboration and communication, 3) interactive learning, 4) personalized learning and 5) research and data analysis.

# Lesson 2: Types of Computers

There are many types of computers. In general, computers can be classified based on their intended purpose into two broad categories: general-purpose and specific-purpose.

> Computers can be classified based on their purpose as general-purpose or specific-purpose.

## General Purpose

General-purpose computers are the most common and widely used computers. They are our personal computers: the desktop, laptop, tablet and smartphone. These computers are designed to run a wide variety of applications which allows them to perform a wide variety of tasks that depend on the needs of the user.

So rather than being designed to perform one specific task or set of tasks, these general-purpose computers are designed to be highly flexible and to perform a wide variety of tasks. The choice of the task depends on the user and the availability of an application to perform the required task.

> Personal computers are general-purpose computers.

### Example 2.1

Karl has a laptop that he uses for many different purposes. Here is a list of some of the tasks he can perform using the various applications he has on his laptop.

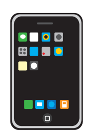
- **Writing assignments**: Using a word processing application, such as MS Word, Karl can type essays, reports or other school projects.
- **Multimedia presentations**: Using applications, such as MS PowerPoint, Karl can prepare slideshows with images and videos for his school projects.
- **Research**: Using a web browser application, such as Google Chrome, Karl can find information in the forms of articles, e-books, images and videos.
- **Collaboration**: Using applications, such as Google Drive, Karl can share his projects with his classmates so that they can complete them as a team.
- **Video Conferencing**: Using applications, such as Google Meet, Karl can communicate with his classmates to enhance collaboration on projects.
- **Gaming**: Karl has a variety of games that he can play alone or with his friends online, such as Minecraft and Fortnite.
- **Entertainment**: Using applications, such as Netflix, Karl can watch movies for entertainment.
- **Time Management**: Using applications, such as Calendar, Karl can schedule his assignments and ensure they are completed on time.

These are some of the many tasks for which Karl uses his computer. The general-purpose nature of his laptop allows him to adapt it to his various needs and preferences, which makes it a versatile and useful machine.


Desktop


Laptop


Tablet


Smartphone

## Personal Computers

Personal computers are classified into four types: desktop, laptop, tablet and smartphone computers. Flexibility is a feature common to all personal computers. Two features that distinguish these computers are portability and capability.

> PERSONAL COMPUTERS CAN BE DISTINGUISHED BASED ON PORTABILITY AND CAPABILITY.

Portability refers to the ease with which a computer can be carried around. This depends on three factors: size, weight and power supply. Certainly, the size and weight of a computer affect the ease with which it is carried around, making smaller and lighter computers more portable than larger and heavier ones. In addition, having a portable electrical power source, a battery, makes portability possible. Computers that need to remain connected to a power outlet remain stationary and are not portable.

> PORTABILITY DEPENDS ON THE SIZE, WEIGHT AND POWER SOURCE OF THE COMPUTER.

Capability refers to the complexity of tasks a computer is capable of performing. Complex tasks typically require a larger computer with a larger screen and, more importantly, greater computational power. The larger screen allows the user to clearly see the various visual elements displayed made possible by the greater computational power which allows the computer to process a lot of information and to perform many complex calculations very fast. However, computational power requires electrical power. The more information a computer processes per second, the more electrical energy it requires per second.

> CAPABILITY DEPENDS MAINLY ON COMPUTATIONAL POWER.

1. **Desktops** are large, stationary computers that are designed to perform more complex tasks. Therefore, they have larger screens and greater computational power, both of which demand more electrical energy. This makes the battery an inconvenient source of electrical power as it will require frequent recharging. In addition, since complex tasks are typically performed with the user seated at a desk, a battery is not required.

2. **Laptops** are portable computers with capabilities close to those of a desktop. To achieve portability, laptops are made smaller by combining the screen and keyboard into one unit rather than being separate parts, as in desktops. In addition, laptops are powered by a rechargeable battery which can handle the energy demands of the smaller screen and the less complex tasks. Do note that certain laptops can match the capabilities of some desktops.

3. **Tablets** are not only portable; they are handheld devices that are larger than smartphones but smaller than laptops. Their smaller size is achieved through touchscreen technology which abates the need for a physical keyboard and pointing devices, such as the mouse and touchpad. They are more portable but less powerful and, therefore, less energy-demanding than laptops.

4. **Smartphones** are portable handheld devices designed primarily for communication but can also be used for a wide range of other tasks, such as browsing the internet, taking pictures and playing games. They are the most portable of the personal computers and the least capable.

In general, personal computers are designed to be usable by a wide range of people and in a wide range of settings. People of all ages, backgrounds and professions find the personal computer very useful in their lives. So not only are personal computers designed to be highly flexible, highly usable and with decent computational power, but they are also designed to deliver all that at an affordable price.

> Personal computers are generally designed to balance capability, flexibility and usability with affordability.

## Specific Purpose

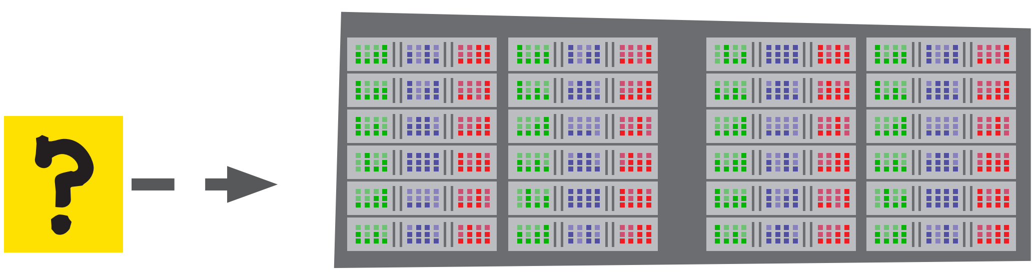
In contrast to general-purpose computers, specific-purpose computers are designed to perform a specific set of functions or tasks. These computers are often specialized and designed to run specific applications rather than a broad range of applications.

### Supercomputers

The supercomputer is a type of computer that is designed to solve complex problems. Typically, a supercomputer will handle one complex task at a time. Such a task requires massive amounts of computational power in order to handle the massive amounts of information that must be processed. Therefore, supercomputers are currently massive in size and demand immense amounts of electrical energy, making them very expensive to have and maintain.

Usually, these computers are used by major institutions, corporations and governments for the purposes of scientific research, weather forecasting, oil and gas exploration, molecular modeling and other large-scale applications.

**SUPERCOMPUTERS ARE DESIGNED TO HANDLE ONE COMPLEX TASK AT A TIME.**



A supercomputer focuses all its computational power on addressing one big task.

## EXAMPLE 2.2

The IBM Blue Gene used by the National Oceanic and Atmospheric Administration (NOAA) in the United States is a supercomputer used for weather forecasting. The Blue Gene can perform over 1 quadrillion calculations per second, making it one of the most powerful computers in the world.

**SUPERCOMPUTERS ARE OPTIMIZED FOR PERFORMANCE.**

The Blue Gene can analyze vast amounts of meteorological data, including satellite images, weather station readings and oceanic data, in order to create detailed models of the atmosphere. These models can then be used to forecast weather patterns, including the trajectory of storms, the likelihood of severe weather events and long-term climate trends.

This allows for more accurate and timely predictions of severe weather events, which has important implications for disaster preparedness and response, as well as for industries such as agriculture and transportation that are heavily impacted by weather patterns.

The IBM Blue Gene/P supercomputer installation at the Argonne Leadership Angela Yang Computing Facility located in the Argonne National Laboratory in Lemont, Illinois, USA

## Servers

The server is a computer that is designed to handle many simple tasks at the same time, rather than one complex task at a time, as would the supercomputer. The majority of these simple tasks are requests for sharing information, such as videos on the YouTube server or for processing information, such as the calculation of student averages in the database of the school server.

**Servers are designed to handle multiple tasks concurrently.**


A server focuses all its computational power on handling as many requests as it can simultaneously.

Servers vary in size depending on the workload they have to handle. Servers that are expected to receive heavy workloads are combined together, forming a data center. The combined servers are large enough to have the required computational power but require a lot of energy and are expensive. On the other hand, servers that are expected to receive light workloads tend to be smaller in size, have less computational power, require less energy and are far less expensive. Servers, therefore, vary from data centers the size of a supercomputer to a single server computer the size of a desktop.

## Example 2.3

The number of requests received by a Google data center, which houses its servers, varies depending on a range of factors, such as the time of day, the location of the data center and the specific services being used. However, it is estimated that Google processes billions of search queries each day, translating to tens of thousands of requests per second across its global network of data centers. In general, each data center may handle millions or even billions of requests per day, depending on the demand for Google's services in that particular region.

One of Google's many data centers is located in St. Ghislain, Belgium. The servers in the data center run Google products such as Search, Gmail and YouTube, as well as provide support for Google users across Europe and around the world.

## Example 2.4

Jenna's school uses a server located on-site to store and share educational resources. Jenna's teachers upload course materials, assignments and student grades to the server, which students like Jenna can access from computers on the school's network. The server also provides Jenna's parents access to her grades, attendance records and other important information about her academic progress.

Jenna's school is a medium-sized school attended by a few hundred students. For a school that size, its server would likely handle a few hundred to, at most, a few thousand requests per day. This number depends on the demand placed on the server by students, teachers and parents.

The server at Jenna's school can handle a few hundred to a few thousand requests per day, depending on the demand placed on it by the students, teachers and parents.

Servers are classified as specific-purpose devices because they are designed to perform a specific set of functions or services. Servers are labeled according to the type of function they are dedicated to performing.

| | |
|---|---|
| **Web servers** | Host websites and deliver web pages |
| **Email servers** | Manage email accounts and deliver email |
| **File servers** | Store and share files |
| **Database servers** | Manage and store data in database |
| **Game servers** | Host online games and manage player accounts |

Servers are dedicated to serving other computers and their users. In general, most users expect servers to respond quickly, be available any time they send a request and, most importantly, to keep their private information secure. This means that servers are designed for high performance, reliability and security.

> SERVERS ARE OPTIMIZED FOR PERFORMANCE, RELIABILITY AND SECURITY.

### Embedded Computers

Embedded computers are computers within other devices or systems, such as cars, medical devices and home appliances and are typically hidden from view. They are classified as specific-purpose because they are designed to perform specific tasks, particularly controlling and optimizing the function of the machine in which they are embedded. In general, embedded computers are optimized for energy efficiency, small size and high reliability.

> EMBEDDED COMPUTERS ARE OPTIMIZED FOR LOW POWER CONSUMPTION, SMALL SIZE AND HIGH RELIABILITY.

1. **Low power consumption**: Embedded computers are often designed to operate in environments where power may be limited, such as in remote areas or in devices that rely on battery power. To conserve power and extend battery life, embedded computers are typically designed to be energy efficient and have low power consumption.

2. **Small size**: Embedded computers are often used in small devices or appliances with limited space. This means that they must be designed to use less space and have small components that can be densely packed.

3. **High reliability**: Embedded computers are often used in devices where reliability is critical, such as in automotive or medical devices. They must be designed to operate without interruption or failure, even in harsh environments or extreme conditions. This requires components that can withstand high temperatures, shock, vibration and other stresses.

### Example 2.5

Jenna's father has a heart condition. His cardiologist informed him that his condition requires a pacemaker. The pacemaker is a device that is placed inside the body and is connected to the heart; its task is to regulate the beating of the heart.

The pacemaker has an embedded computer. The computer is tasked with reading the electrical activity of the heart and sending regular electrical pulses to maintain a steady rhythm.

| | |
|---|---|
| **Low power consumption** | Since the pacemaker is placed inside the body, it must be powered by a battery. Recharging the battery requires a trip to the doctor's office, so having a long battery life is convenient. |
| **Small size** | Since the pacemaker has to fit inside the body, it should be small in size. This means its embedded computer should also be small in size. |
| **High reliability** | Since the pacemaker is responsible for keeping Jenna's father alive, it must be reliable. It must not fail. If it fails, his life will be at risk. |

Overall, embedded computers are designed to meet the specific needs and enhance the functionality of the device or machine they are being used in.

## Summary

Computers can be classified into two broad categories: general-purpose and specific-purpose.

General-purpose computers are flexible and can run a wide range of applications and therefore perform various tasks and serve many purposes. Personal computers fall into the category of general-purpose; they are: desktops, laptops, tablets and smartphones.

Personal computers are distinguished by portability and capability. Portability depends on the size, weight and power source. Capability is largely determined by computational power. The desktop ranks the lowest in portability but the highest in capability. The smartphone ranks the highest in portability but the lowest in capability, with the laptop and tablet ranking in between. Personal computers are designed to have reasonable capability, high flexibility, easy portability and affordability.

Specific-purpose computers are designed to perform a specific set of functions or tasks. Supercomputers, servers and embedded computers are types of specific-purpose computers.

Supercomputers are massive computers that are optimized for high performance regardless of the energy consumption and the high expenses required to own, run and maintain them. They handle one complex task at a time.

Servers vary in size from data centers as large as a supercomputer to a single server computer the size of a desktop, depending on the workload they are required to handle. In all cases, these computers are optimized for high performance, high reliability and high security.

Embedded computers are computers placed inside machines and devices for specific purposes that optimize their overall function. Embedded computers are optimized for energy efficiency, small size and high reliability. They are ideal for use in devices where space and power are limited and reliability is critical.

## Activity 2.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F | STATEMENT |
|---|---|
| F | General-purpose computers are designed to perform specific tasks. |
| T | Desktop computers are generally more powerful than laptops. |
| T | Size, weight and power supply are the three main factors that determine the portability of a computer. |
| F | Supercomputers can handle multiple complex tasks at the same time. |
| F | Servers can handle only one task at a time. |
| F | Portability is an important feature of servers. |
| T | Embedded computers are designed to be part of other devices. |
| T | Servers are typically more powerful than personal computers. |
| T | Embedded computers are optimized to be energy efficient, small in size and highly reliable. |
| T | There are different types of servers, each dedicated to a particular task. |

## Activity 2.2

Answer the questions below.

1. Which type of computer is designed to perform a wide range of tasks?
   a) Specific-purpose computer
   b) General-purpose computer
   c) Embedded computer

2. Which of the following is an example of a specific-purpose computer?
   a) Laptop
   b) Desktop
   c) Supercomputer

3. What is the primary function of a server?
   a) To perform a wide range of tasks
   b) To provide resources or services to other computers on a network
   c) To control the engine of a car

4. What is the primary function of an embedded computer?
   a) To perform a wide range of tasks
   b) To provide resources or services to other computers on a network
   c) To perform a specific set of functions within a larger device or system

5. Which type of computer is designed for portability and convenience?
   a) Desktop
   b) Laptop
   c) Server

## Activity 2.3

Answer the questions below.

1. Which type of computer is optimized for advanced scientific and engineering simulations?
   a) Embedded computer
   b) Supercomputer
   c) Server

2. Which type of computer is designed to control the engine, monitor the fuel efficiency or manage the entertainment system in a car?
   a) Smartphone
   b) Server
   c) Embedded computer

3. Which type of computer is best suited for managing email accounts?
   a) Database server
   b) Email server
   c) Game server

4. Which type of computer is designed to be small and energy efficient?
   a) Desktop
   b) Supercomputer
   c) Embedded computer

5. Which type of computer is optimized for high performance, reliability and security?
   a) Supercomputer
   b) Server
   c) Embedded computer

## Activity 2.4

Match each computer with the statement that best describes it.

**Smartphone** **Desktop** **Laptop**

**Supercomputer** **Server** **Embedded computer**

1. It is a personal computer that is not portable. Desktop
2. It is massive and handles one complex task at a time. Supercomputer
3. It is small in size, highly reliable and energy efficient. Embedded computer
4. It is the smallest and most portable personal computer. Smartphone
5. It is used to provide services for other computers. Server
6. A personal computer that is portable but not handheld. Laptop

## Activity 2.5

Match each computer with the task for which it is best optimized.

**Desktop** **Server** **Smartphone**

**Supercomputer** **Tablet** **Embedded computer**

1. Constructing a map of the visible universe Supercomputer
2. Getting real-time directions for a destination Smartphone
3. Playing an online video game Desktop
4. Controlling the doors of an elevator Embedded computer
5. Hosting websites for various clients Server
6. Drawing an illustration using a stylus Tablet

## Activity 2.6

Answer the questions below about personal computers.

1. Why are personal computers considered general-purpose?

   Personal computers are considered to be general-purpose because they are de signed to be flexible and to run a wide range of applications. This allows them to perform a wide variety of tasks, such as word processing, browsing the internet, playing games, watching videos and more.

2. "Personal computers are designed for reasonable capability, high flexibility and easy portability at an affordable price." Is this statement true? Explain.

   This statement is true. Personal computers provide the capabilities needed by the average person. Personal computers are designed to be easy to use, even for those who are not computer experts. They are flexible, with a wide range of applications available that can be used for different purposes. They are priced to be affordable for the average consumer, although the cost can vary depending on the specific features and capabilities of the device.

## Activity 2.7

Answer the questions below about supercomputers and servers.

1. How would you distinguish between the supercomputer and the server?

1) Purpose: Supercomputers are designed to handle one complex task at a time, such as weather forecasting. Servers are designed to handle many simpler requests from other computers simultaneously.

2) Size: Supercomputers are typically very large and complex. Servers vary in size from large data centers to single computers the size of a desktop.

2. A security breach can happen with any computer system. Compare the impact of a security breach on a server and a supercomputer.

Security breaches are serious threats and can cause significant harm depending on the information being stolen, damaged or lost. A security breach for a server involves the theft, loss or damage to personal, medical or corporate private information stored on the server. A security breach for a supercomputer involves the theft, loss or damage to information related to national security or scientific research. This may compromise the security of the nation or the accuracy and reliability of the scientific calculations and simulations.

## Activity 2.8

Answer the questions below about embedded computers. Note that they can be referred to as 'control modules', 'controllers' or 'micro-controllers', among other names.

1. List five common machines or devices in which embedded computers are found.

1) Home appliances (e.g. refrigerators, washing machines)
2) Medical devices (e.g. pacemakers, MRI machines)
3) Automotive systems (e.g. engine control units, anti-lock brakes)
4) Aircraft systems (e.g. flight control computers, navigation systems)
5) Industrial equipment (e.g. elevators, assembly line machines)
6) Smart home devices (e.g. thermostats, security systems)
7) Wearable devices (e.g. smartwatches, fitness trackers)
8) Consumer electronics (e.g. gaming consoles, digital TVs)
9) ATMs and point-of-sale systems (e.g. credit card readers, cash registers)
10) Traffic control systems (e.g. traffic lights, toll booths)

2. Choose one of the machines or devices you listed in Question 1, and describe the function of its embedded computer(s).

1) Engine control module (ECM) - controls the operation of the engine
2) Anti-lock brake system (ABS) module - controls the brakes
3) Transmission control module (TCM) - controls transmission and gear shifting
4) Airbag control module (ACM) - deploys the airbag in the event of a collision
5) Infotainment system - controls audio and other entertainment functions
6) Climate control module - regulates the air conditioning system

## Activity 2.9

Answer the questions below about reliability. For computers, reliability refers to the ability of a computer to perform its tasks consistently without failure.

1. Compare and discuss the importance of reliability for the operation of a supercomputer used for scientific research, a server handling financial transactions and an embedded computer in an airplane. What would be the adverse consequences of failure?

Reliability is critical for the operation of any computer system, but its importance can vary depending on the application and context.

1) For a supercomputer used for scientific research, failure could result in delays in the research, loss of important information, loss of money spent on the research and loss of trust in the institution conducting the research.

2) For a server handling financial transactions, failure could result in a loss of money for customers, an opportunity for hackers to breach the system and a loss of trust in the financial institution.

3) For an embedded computer in an airplane, failure could result in a plane crash with loss of life of passengers and crew, loss of money as well as loss of trust in the airlines.

## Activity 2.10

Describe a typical day for yourself and the various interactions with computers you may have. Include in your example supercomputers, servers, personal computers and embedded computers.

In the morning, I wake up to the alarm I set on my smartphone (personal computer). I check if I have any messages from my friends (server). Before I get dressed, I check the weather forecast for the day (supercomputer, server). My dad drives me to school in his car (embedded computer). In class, the teacher uses the classroom desktop (personal computer) to project the day's lesson on the board. In the afternoon, I use the microwave oven (embedded computer) to make popcorn. Mom puts laundry into the washing machine (embedded computer). I use my laptop (personal computer) to sign into my school account and check my homework assignments (server). Before bed, I watch television (embedded computer).

# LESSON 3: SYSTEMS & INTERFACES

The concept of a system is fundamental to our understanding of ourselves and the world around us. In this lesson, we will explore computers as technological systems and relate them to humans as biological systems, with the commonality being that they are both capable of information processing. The interaction that occurs between computers and humans underlies the technological developments that have led to the tremendous evolution in our way of life.

## SYSTEMS & SIGNALS

A system is a set of interconnected components that work together to achieve a particular task or purpose. These parts or components interact with each other in a coordinated manner, allowing the system to function as a cohesive whole.

**A SYSTEM IS A SET OF INTERCONNECTED COMPONENTS THAT WORK TOGETHER AS A WHOLE, SERVING A PARTICULAR PURPOSE.**

In order for a system to work as a whole, all its parts must interact with each other and coordinate their functions and activities. In the case of systems that depend on information processing, such as humans and computers, this means that the individual parts of a system must be able to communicate with each other; they must be able to exchange information.

**Notes**

Humans are one example of a biological system capable of information processing. In fact, any living organism with a central nervous system qualifies to be described as an information processing biological system.

A signal is a message that is carried in physical form; it is essentially the transfer of energy in a fashion that carries some meaningful content interpretable by some information processing system.

A SIGNAL IS A MESSAGE CARRIED IN PHYSICAL FORM.

In computers, the form of energy that typically acts as a signal is electrical energy; more precisely, it is electrical potential energy. Humans also use electrical potential energy for internal signaling. In both cases, changes in voltage are propagated along pathways that connect the various parts of the system. For computers, these signals are called digital electronic signals, and they are propagated along conducting wires. Whereas for humans, these signals are called action potentials, and they are propagated along nerves.

Action potential signals

Digital electronic signals

Signals play a crucial role in the coordination of the internal components of a system, as they enable communication between different parts and ensure proper functioning.

SIGNALS ENSURE COMMUNICATION AND COORDINATION BETWEEN THE PARTS OF A SYSTEM.

## Interfaces & Translation

The boundary of a system where it interacts with the outside world is called its interface. For information processing systems, interaction entails an exchange of signals. Signals going into the system are referred to as input, and in response, the signals that come out are output.

> The interface is the boundary at which a system interacts with the external world.

Every information processing system has its own form of internal signals; humans use action potentials, and computers use digital electronic signals. These are the only signals they can use to communicate and process internally.

External signals of the outside world are quite different from internal signals; they must be converted into internal signals. Light, sound, pressure and any other signal of the outside world must be converted into either action potentials or digital electronic signals. This conversion from one form of signal to another must preserve the integrity of the message itself. In a sense, it is quite analogous to translation between languages, wherein the information conveyed in one language must be preserved, but the language in which it is conveyed is changed. Signal conversion occurs at the interface.

> Signal conversion occurs at the interface.

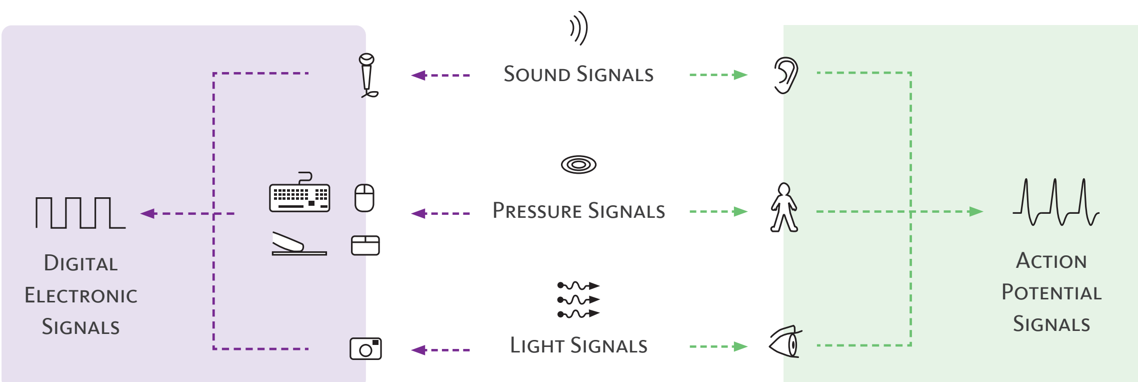
### Input devices

Input devices are the parts of the system that handle information coming in. External signals are converted to internal signals while preserving the information they carry. This ensures that the system gets accurate information about its surroundings. Each device is dedicated to converting a specific type of external signal into the required internal signal.

**THE INPUT DEVICE CONVERTS A SPECIFIC FORM OF EXTERNAL SIGNAL INTO THE INTERNAL SIGNAL OF THE SYSTEM.**

Computer parts responsible for input are called input devices and are often referred to as sensors. Each device is dedicated to the task of converting one form of signal into electronic digital signals. Humans, similarly, have sensory organs which are responsible for accepting input and converting one form of external signal into internal action potential signals.

Sound, pressure and light are commonly converted external signals by both computers and humans. Computers have a microphone and camera to convert sound and light, respectively, while humans have ears and eyes. Computers have various input devices that are capable of converting pressure depending on the type of computer; these are the keyboard, mouse, touchpad and touchscreen. Humans have the skin as a sensory organ capable of converting pressure signals into action potentials.




Sound, pressure and light are commonly converted external signals by both computers and humans. The computer has various input devices, such as the microphone, camera, keyboard, mouse, touchpad and touchscreen, to convert sound, light and pressure into internal signals. In comparison, the human body has the ear, eye and skin, respectively.

In addition, many smartphones are equipped with input devices or sensors that can detect position and motion. Some of the most common sensors used for this purpose include:

1. **Accelerometers** are used to detect linear motion in any of the 3 dimensions: up or down, left or right and forward or backward.
2. **Gyroscopes** are used to detect rotational motion in any one of the 3 dimensions: rotating clockwise or counterclockwise, rolling left or right and pitching up or down.
3. **Magnetometers** are used for magnetic fields. They are used to detect the orientation of the smartphone relative to the Earth's magnetic field.
4. **GPS** are sensors that use satellite signals to determine the location of its computer or device on Earth.

These four sensors work together to determine speed, direction, orientation, position and location, allowing the smartphone to track its own movement and its relative position in Space. This makes them useful in a variety of applications, such as games, fitness trackers, maps and navigation.

In comparison, while humans do not have sensory organs that can detect the Earth's magnetic field or GPS satellite signals, their vestibular system provides highly accurate information about linear movement, rotational movement, orientation and position in space. We can easily tell if we are moving and in which direction. We can also easily tell if we are leaning in any direction relative to the ground.

Furthermore, humans have additional senses. The skin not only provides pressure information, but it also provides information about temperature and pain. In addition, humans are capable of detecting chemical signals through taste and smell.

## OUTPUT DEVICES

In contrast to input devices, output devices are responsible for responding to the outside world. Responses, or output, can be in the form of a signal or an action, each of which requires a dedicated output device. In either case, an internal electronic digital signal is converted to an external signal or an external action. This is similar to the case of humans in whom an internal action potential is converted to an external signal, such as the sound of speech, or an external action, such as the movement of a limb.

THE OUTPUT DEVICE CONVERTS THE INTERNAL SIGNAL OF THE SYSTEM TO A SPECIFIC FORM OF EXTERNAL SIGNAL OR ACTION.

Computers mainly output two forms of signals: light from the screen display and sound from the speaker. In addition, the handheld computers, namely the tablet and smartphone, can also vibrate, emitting what is called a haptic signal or pressure that can be felt by whoever is holding the computer.

Output devices that respond by performing an action are called actuators; they are capable of moving. Actuators are part of other machines or devices that can be connected to a computer or are controlled by an embedded computer. The printer is a machine that connects to a computer and contains an actuator. The printer's printhead is an actuator that moves across the paper, printing line by line. Other examples of actuators include motors and valves, which are used in various machines such as home appliances, vehicles, manufacturing equipment and robots.

THE ACTUATOR IS AN OUTPUT DEVICE CAPABLE OF PERFORMING AN ACTION.

## Hardware & Software

The human computer interface is not just a physical one: it is also a logical one. The hardware component of the interface includes the various physical components, the input and output devices, which are responsible for transmitting and receiving signals. The hardware of the computer handles the physical layer of the interface, which are the signals it is transmitting and receiving.

> The physical layer is the signals; it is handled by the hardware.

The software component of the interface determines the information (logic) to be carried by the signals. The software component includes the applications and other programs that determine how the computer behaves and what information it shows the user. The content displayed on the screen is determined by software. What happens when a user clicks a button is determined by software. Everything a computer does is determined by software.

So when we interact with computers, we are interacting with both hardware and software. At the human computer interface, an exchange of signals carrying information is occurring, with the signals being the physical layer and the information being the logical layer.

> The logical layer is the information in the signals; it is handled by the software.

### Example 3.1

When two humans have a conversation, they are interacting. This interaction occurs in both physical and logical layers. In the physical layer, there is an exchange of sound signals, which are sensed by the ears and emitted by the mouth. However, the content of the conversation, the actual information represented by these sound signals, is what forms the logical layer. The logical layer contains the meaning carried by the sound signals.

## User Interface

The user interface (UI) is a term that refers to the software component of the human computer interface, also referred to as the software interface. The UI is responsible for translating user input into instructions that the computer can understand and execute, as well as displaying the results of those instructions for the user.

**THE USER INTERFACE IS THE SOFTWARE COMPONENT OF THE HUMAN COMPUTER INTERFACE.**

There are several types of user interfaces that evolved over time; these include in chronological order:

1. **Command-Line Interface** (CLI): A CLI requires users to type commands into a text-based interface to perform functions. CLI interfaces are commonly used in programming and system administration but are challenging for users unfamiliar with programming or technical jargon. Examples include Windows Command Prompt and macOS Terminal.

```
Last login: Wed Apr 13 15:35:52 on ttys001
computer-name:~ username$
```

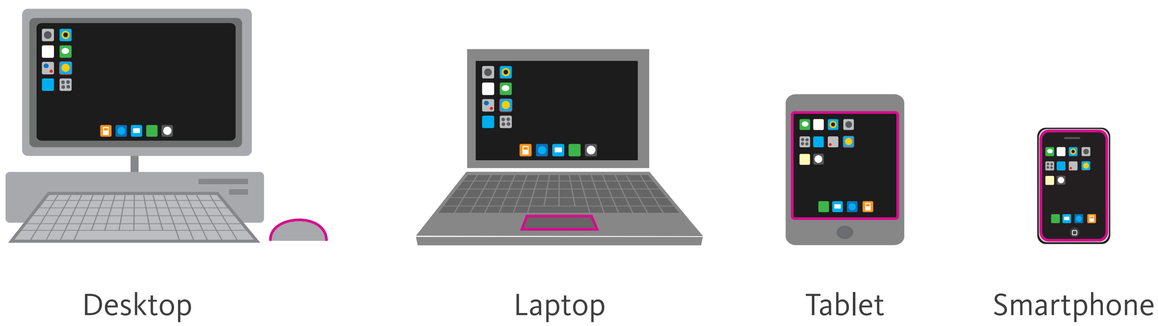
The Terminal window on macOS showing a standard output typical display. The first line indicates the date and time of the last login. The second line displays the name of the computer (in this case, "computer-name") followed by the username (in this case, "username") and a dollar sign ($), indicating that the terminal is ready to accept commands.

2. **Graphical User Interface** (GUI): A GUI uses graphical elements such as windows, icons and menus to represent different functions and commands. The GUI made computer systems more accessible and user-friendly by reducing the need for users to remember complex commands and by providing a more intuitive and visual interface. Examples include Windows and macOS.

### *Graphical Elements of the GUI*

The GUI has three common graphical elements: windows, icons and menus. A window is a separate area of the screen containing content related to a specific task or application. More than one window can be opened, allowing users to interact with multiple applications or tasks simultaneously. In addition, icons and menus allow users to select applications, files and commands.

> THE GRAPHICAL USER INTERFACE USES WINDOWS, ICONS AND MENUS TO REPRESENT DIFFERENT FUNCTIONS AND COMMANDS.




Desktops and laptops have pointing devices, the mouse and touchpad, respectively. These hardware components are used to control a software component displayed on the screen, the pointer. Tablets and smartphones have touchscreens, a technology that allows pointing to be replaced by touching.

There are two main types of GUIs that are distinguished based on the technology they incorporate to select and control graphical elements: the WIMP and touchscreen interfaces.

***WIMP Interface***: The acronym WIMP stands for windows, icons, menus and pointer. A pointer is a graphical element, typically in the form of an arrow or hand, used to interact with the interface by selecting icons and items in menus and windows. It is controlled by a mouse or touchpad, both of which are input devices known as pointing devices.

***Touchscreen Interface***: With the rise of mobile devices, the touchscreen interface became popular, allowing users to interact with devices directly by touching the screen using natural gestures such as swiping, tapping, or pinching, making it easy for users to learn and use. Touchscreens kept the windows, icons and menus but eliminated the need for pointing devices, such as the mouse or keyboard.

3. **Voice User Interface** (VUI): Most recently, the (VUI) has become more prevalent, allowing users to interact with devices using natural language and voice commands. VUIs are becoming increasingly popular in smart home devices and personal computers, such as Apple's Siri, Amazon's Alexa, Google's Assistant and Microsoft's Cortana.

### Career Tips

Human Computer Interaction is a multidisciplinary field that combines principles from computer science, cognitive psychology, design and other fields to study how people interact with computers and how to best design user interfaces.

## Usability

Usability is a design feature that refers to how easy it is for users to interact with a computer in order to complete a particular task. The evolution of the user interface and its related hardware components reflects an evolution in usability.

> **Usability is the ease with which a device can be used to complete a particular task.**

The touchscreen interface is often considered highly intuitive; users can simply touch the screen to interact with it, which many people find easy to learn and use regardless of their knowledge and skills with computers.

The WIMP interface requires a mouse or touchpad, which may require some learning and practice for users who are unfamiliar with the technology. However, once users become familiar with the interface, it can be quite efficient and effective for many computing tasks.

The CLI requires more learning and memorization of specific commands on the part of the user. As a result, it is generally only usable by IT professionals or those with advanced computing skills.

The VUI is an interface that has emerged in recent years. It allows users to interact with computers through speech. While it may seem a step up in usability, this interface is still in its early stages of development.

Overall, the usability of a user interface is an important consideration in designing effective and efficient human-computer interaction. The evolution of hardware components and the user interface itself has reflected a trend towards more intuitive and easy-to-use interfaces, which has allowed a wide range of people to use computers.

## SUMMARY

A system is a set of interconnected components that work together as a whole, serving a particular purpose. Humans and computers are examples of information processing systems.

A signal is a message carried in physical form. In an information processing system, signals ensure communication and coordination between the parts of the system. Every system has its own specific type of internal signal, e.g. humans have action potentials, and computers have digital electronic signals.

The interface is the boundary at which a system interacts with other systems and the external world. Signals must be converted between external signals and internal signals. This conversion occurs at the interface and must preserve the integrity of the information carried by the signals.

The interface of a system comprises both input and output devices. Input devices convert external signals into internal signals. Output devices convert internal signals into external signals. Each device is dedicated to a specific type of external signal.

Some output is in the form of action. The output device that performs this action is referred to as an actuator.

Information exists in two layers: the physical and virtual layers. The physical layer contains the actual signals and is handled by the hardware. The virtual layer is the information represented in the signals and is handled by the software.

The user interface is the software component of the human computer interface. This interface has evolved from a command-line interface to a graphical user interface and, more recently, a voice user interface.

The graphical user interface uses windows, icons and menus to represent different functions and commands. The WIMP interface and touchscreen interface are both types of graphical user interfaces. They both have windows, icons and menus, but the WIMP interface includes either a mouse or touchpad, which the touchscreen interface replaces with a touchscreen.

Usability is the ease with which a device can be used to complete a particular task.

A directory is a logical construct used by the filing system to organize files in a computer. A directory is used to hold files and, in many cases, other directories, referred to as subdirectories. The term directory is synonymous with “folder”.

## Activity 3.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F | STATEMENT |
|---|---|
| T | A system is a set of interconnected components that work together to achieve a particular task or purpose. |
| F | Humans and computers do not share any common features as systems. |
| F | Signal conversion at the interface does not need to preserve the integrity of the message. |
| T | Input devices convert external signals into internal signals while preserving the information they carry. |
| T | Both humans and computers are capable of converting sound, pressure and light signals. |
| F | Actuators are only used in computers and not in other machines or devices. |
| F | The human-computer interface only includes hardware components. |
| F | Command-Line Interface (CLI) is the most user-friendly type of interface. |
| T | The Graphical User Interface (GUI) made computers more accessible by providing a more intuitive and visual interface. |
| F | The Voice User Interface (VUI) is a highly developed and mature interface technology. |

## Activity 3.2

Answer the questions below.

1. What do computers and humans have in common as systems?
   a) They both use digital electronic signals.
   b) They are both capable of information processing.
   c) They both use action potentials.

2. What is the purpose of an interface in an information processing system?
   a) To store information
   b) To create new signals
   c) To convert external signals into internal signals

3. Which form of energy typically acts as a signal in computers?
   a) Thermal energy
   b) Electrical potential energy
   c) Kinetic energy

4. In the context of information processing systems, what is a signal?
   a) A physical form of a message
   b) A method of storing data
   c) A type of energy conversion

5. Signals going into an information processing system are referred to as:
   a) Output
   b) Input
   c) Interface

## Activity 3.3

Answer the questions below.

1. What is the primary function of input devices?
   a) Converting internal signals to external signals
   b) Converting external signals to internal signals
   c) Storing signals

2. Which two devices are responsible for converting sound signals?
   a) Microphone - Keyboard
   b) Mouse - Speaker
   c) Microphone - Speaker

3. What type of sensor is used to detect linear motion in smartphones?
   a) Gyroscope
   b) Accelerometer
   c) Magnetometer

4. What type of output device is responsible for performing an action?
   a) Sensor
   b) Actuator
   c) Transducer

5. Which of the following is an example of an actuator in a printer?
   a) Ink cartridge
   b) Paper tray
   c) Printhead

## Activity 3.4

Answer the questions below.

1. The human-computer interface consists of:
   a) Both input and output devices
   b) Only input devices
   c) Only output devices

2. Which type of interface uses elements such as windows, icons and menus?
   a) Command-Line Interface
   b) Voice User Interface
   c) Graphical User Interface

3. Which interface allows users to interact with devices using natural gestures?
   a) WIMP Interface
   b) Touchscreen Interface
   c) CLI Interface

4. The term “usability” refers to:
   a) How easy it is for users to interact with a computer
   b) The amount of memory a computer has
   c) The processing power of a computer

5. Which interface is considered the most user-friendly for non-technical users?
   a) Command-Line Interface (CLI)
   b) Graphical User Interface (GUI)
   c) Voice User Interface (VUI)

## Activity 3.5

Answer the questions below about signal conversion at the interface.

1. What is the purpose of signal conversion at the interface of an information processing system?

   The purpose of signal conversion at the interface is to transform external signals from the outside world into internal signals while preserving the integrity of the message. This conversion allows the system to use and respond to the information it receives from the external environment.

2. Why is preserving the integrity of the message important during signal conversion at the interface?

   Preserving the integrity of the message during signal conversion at the interface is crucial because it ensures that the information conveyed by the signal remains accurate and meaningful when it is transformed into the internal signals used by the information processing system. This allows the system to correctly interpret and respond to the input it receives from the external environment.

## Activity 3.6

Answer the questions below about humans and computers.

1. State the similarities that exist between humans and computers.

   They are both systems.

   They both can process information.

   They both can sense and respond to their environment.

   They are both capable of signal conversion at their interface.

   They both use electrical potential energy for internal signaling.

2. "The ability to gather information from the environment adds functionality to a system and enhances its chances of success."
   Comment and give an example.

   The more information a system can gather from its environment, the better its chances of navigating through its environment to find sources of energy and shelter, as well as to avoid obstacles and threats. Each sense adds a new set of information. Therefore, adding senses or input devices adds more information. The sense of smell can help someone avoid a fire before seeing it. The sensing of Earth's magnetic field helps birds migrate to warmer climates. Having a smartphone with more sensors makes it more useful.

## Activity 3.7

Answer the questions below about input and output devices.

1. What does the term “actuator” refer to in technology? Provide an example.

   An actuator is an output device that converts internal digital electronic signals to movement. An example is the motor that drives the turntable of a microwave oven.

2. Identify and list the input devices on your personal computer.

   Windows: 1) Right-click on “Start”, 2) select “Power User Menu”, 3) select “Device Manager” and then search the listed hardware categories.

   macOS: 1) Click on the Apple logo, 2) select “About This Mac”, 3) select “System Report” or “More Info” and then search the listed hardware categories.

   Androids & iPhones: A third-party app, such as “Sensors Multitool” or “Sensors Toolbox” will provide device-sensor information.

## Directories

Any information stored in a computer is stored in the form of a file. All files are organized and managed by the computer's filing system. The filing system uses the idea of a folder to organize files. A folder acts as a virtual container used to hold files and, in many cases, other folders, referred to as subfolders.

A more technical term for folder is directory. To restate using this term, the filing system organizes files under directories. A directory is used to hold files and, in many cases, other directories, referred to as subdirectories. This organization of directories within other directories is referred to as a hierarchical structure.

In this example, the directory structure shows the root directory and two levels of subdirectories.

In addition, four example files are shown that appear in green with their file extensions.

```
root directory /
               |--- Applications
               |      |--- Microsoft Word
               |               |--- Word.exe
               |               |--- Readme.txt
               |--- Documents
                      |--- My Documents
                      |        |--- Project A.docx
                      |        |--- Project B.docx
                      |--- Shared Documents
```

In the GUI, the folder icon is a graphical element of the user interface used to represent a directory. It is a visual abstraction of the more complex filing system managed by the computer. It is a simplification that allows a wide range of users to use the computer without having to deal with complex filing systems.

### Notes

Abstraction is a fundamental concept in many fields, including computer science, sciences, mathematics and philosophy. It allows us to focus on the essential characteristics of a concept or system without getting bogged down in irrelevant details and is, therefore, a key aspect of problem-solving, innovation and creativity.

## Activity 3.8

Use the command line interface to create a file in a specific folder. Then, find and add content to this file through the graphical user interface. Compare the two interfaces.

1. In the CLI, create a text file in the Documents directory and name it “My File” (see instructions). Add content, then save and close the file.
2. In the GUI, open “My File”, add additional content, then save and close the file.
3. Compare the two interfaces and their usability.

Instructions for macOS

1. Open Terminal, and list the directories under the root directory using the ‘ls’ command.

```
ls
```

2. Change directory using the ‘cd’ command followed by the name of the destination directory, which in this case is ‘Documents’.

```
cd ‘Documents’
```

3. List the contents of the directory sorted in alphabetical order using the ‘ls | sort’ command.

```
ls | sort
```

4. Create a text file using the ‘touch’ command followed by the name of the file and its extension.

```
touch ‘My File.txt’
```

5. Open the text file using the ‘nano’ command followed by the name of the file and its extension.

```
nano ‘My File.txt’
```

6. Add text by simply typing the text, then save and close using ‘Ctrl+X’ and then ‘Y’ followed by ‘Enter’.
7. List the contents of the directory again to check that the file was created.

```
ls | sort
```

Instructions for Windows

1. Open Command Prompt, and list the directories under the root directory using the ‘dir’ command.

```
dir
```

2. Change directory using the ‘cd’ command followed by the name of the destination directory, which in this case is ‘Documents’.

```
cd “Documents”
```

3. List the contents of the directory sorted in alphabetical order using the ‘dir’ command with the ‘/O:N’ option.

```
dir /O:N
```

4. Create a text file using the ‘notepad’ command followed by the name of the file and its extension to open a file in the Notepad app.

```
notepad “My File.txt”
```

5. Add text by simply typing the text, then save and close using ‘Ctrl+S’ then ‘Ctrl+W’.
6. List the contents of the directory again to check that the file was created.

```
dir /O:N
```

## Activity 3.9

Use the command line interface and the graphical user interface to create a backup folder that contains copies of your original files. Compare the two interfaces.

1. In the CLI, create a new directory under Documents. Name the new directory My Backups and move a copy of the file you created in Activity 3.8 into this directory.
2. In the GUI, create another directory under Documents. Name the new directory My Backups 2 and move a copy of the file you created in Activity 3.8 into this directory.
3. Compare the two interfaces and their usability.

Instructions for macOS

1. Open Terminal, and navigate from the root directory to the ‘Documents’ directory.

```
cd ‘Documents’
```

2. Make a new directory using the ‘mkdir’ command. Name the directory ‘My Backups’.

```
mkdir ‘My Backups’
```

3. Copy the file ‘My File.txt’ using the ‘cp’ command. Name the copy ‘Copy My File.txt’.

```
cp ‘My File.txt’ ‘Copy My File.txt’
```

4. Move the copied file to the new directory ‘My Backups’.

```
mv ‘Copy My File.txt’ ‘My Backups’
```

5. Change directory using the ‘cd’ command to the ‘My Backups’ directory.

```
cd ‘My Backups’
```

6. List the contents of the directory to check that the file is in it.

```
ls
```

Instructions for Windows

1. Open Command Prompt, and navigate from the root directory to the ‘Documents’ directory. Replace {your_username} with your actual username.

```
cd C:\Users{your_username}\Documents
```

2. Make a new directory using the ‘mkdir’ command. Name the directory ‘My Backups’.

```
mkdir “My Backups”
```

3. Copy the file ‘My File.txt’ using the ‘cp’ command. Name the copy ‘Copy My File.txt’.

```
copy “My File.txt” “Copy My File.txt”
```

4. Move the copied file to the new directory ‘My Backups’.

```
move ‘Copy My File.txt’ ‘My Backups’
```

5. Change directory using the ‘cd’ command to the ‘My Backups’ directory.

```
cd ‘My Backups’
```

6. List the contents of the directory to check that the file is in it.

```
dir
```

# Lesson 4: Behind the Interface

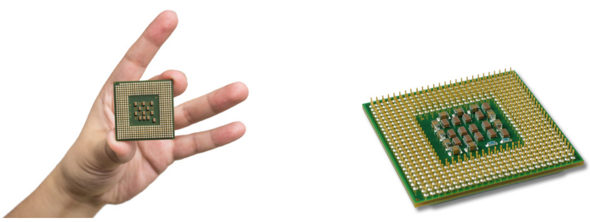
At the interface of an information processing system, only the input and output are visible to the user, whereas the steps that occur inside the system are not. This is because the interface is designed as an abstraction, a simplification of the complex internal workings of the system. Such internal workings are the topic of interest in this lesson.

## The Processor

In order for a system to generate a response, it must be able to process the information it has gathered. The system must be able to change input to output. This is the task of the processor. The processor is that part of the system dedicated to changing input to output, thereby allowing the system to respond.

> The processor is the part of the system that changes input to output.

Just as the user interface is both hardware and software, in the context of processing, we can broadly state that the processor is the hardware through which the signals move and are operated on, while the stored programs are the software that determines the order in which these operations happen and the components of the processor that will perform them.



A typical desktop CPU package measures about 4-5 square centimeters. However, the size of the actual processor chip inside the package is much smaller, typically only a few square millimeters or less.

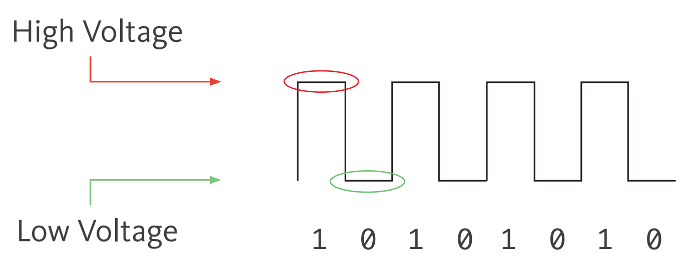
## Digital Electronic Signals

Inside the computer, information is carried as electrical signals. These signals travel through wires arranged in groups of 64 or 32 parallel wires, depending on the computer, with each wire carrying 1 signal or 1 bit of information. A computer with a 64-bit architecture uses 64 parallel wires, while a computer with a 32-bit architecture uses 32 parallel wires, allowing for 64 bits and 32 bits of information to be transmitted simultaneously, respectively.

Each bit is a pulse of electrical potential energy, whose intensity is measured in volts. Therefore, signals can be distinguished based on the intensity of energy they have, into either high voltage or low voltage. To simplify this, a high voltage signal is labeled '1', and a low voltage signal is labeled '0', even though it does not actually have zero volts of energy. Since a signal can only be one of two options, high or low, 1 or 0, we call this type of signal 'binary', which means 'two'. Electrical signals that are binary in nature are referred to as digital electronic signals.

**Digital electronic signals are binary in nature, being either high voltage (1) or low voltage (0).**

A sequence of 8 bits is referred to as a 'byte', whereas the term 'word' is used to refer to the maximum number of bits that can be transmitted simultaneously. In a 32-bit processor, a 'word' is 32 bits long, whereas in a 64-bit processor, a 'word' is 64 bits long.




This is an 8-bit sequence of signals with alternating high and low voltages. The byte can be represented in binary as a series of alternating 1's and 0's: `10101010`.

## Logic Circuits

A logic circuit is a collection of electronic components, mainly transistors, connected by wires in specific configurations. The configuration of transistors determines the operation the logic circuit can perform. Changing the configuration changes the operation. Therefore, a logic circuit that performs addition operations would have a different configuration of transistors from a circuit that performs multiplication operations.

**A logic circuit is a collection of electronic components designed to perform a particular operation.**

Processing occurs when digital electronic signals pass through a logic circuit. For example, the adder is a logic circuit that performs an addition operation. It receives two input signals, adds them and then outputs one signal, their sum. There are other logic circuits in the processor, such as the multiplier, comparator and multiplexer.

| Logic Circuit | Purpose |
|---|---|
| Adder | Adds 2 inputs |
| Multiplier | Multiplies 2 inputs |
| Shifter | Shifts the bits of 1 input by 1 or more places |
| Comparator | Compares 2 inputs as equal, less or greater than |
| Multiplexer | Selects 1 from several inputs |

**The operation a logic circuit performs is determined by the configuration of its components.**

## EXAMPLE 4.1

Let's assume there is a small processor with a 4-bit architecture instead of the more common 64 bits. This hypothetical processor has 4 logic circuits: the adder, multiplier, shifter and comparator.

Let's also assume that the processor receives two 4-bit signals as input, `0110` and `0001`. If these two inputs pass through the adder, the output would be their sum, `0111`.


Two input signals, `0110` and `0001`, pass through an adder logic circuit. The circuit performs the addition operation and generates an output which is the sum, `0111`. Note that addition follows the same rules in binary as it does in decimal.

```
  0 1 1 0
+ 0 0 0 1
---------
  0 1 1 1
```

Now, let's pass the same 4-bit signals, `0110` and `0001`, through the multiplier, the output would be their product, `0110`. This happens because the logic circuits are arranged in a manner that performs a multiplication operation rather than an addition.


The same two input signals, `0110` and `0001`, pass through a multiplier logic circuit. The circuit performs the multiplication operation and generates an output which is the product `0110`. Note that multiplication also follows the same rules in binary as it does in decimal. In this case, any number multiplied by 1 is itself.

```
  0 1 1 0
× 0 0 0 1
---------
  0 1 1 0
```

On the other hand, the shifter logic circuit would handle the two inputs differently. The first input, 0110, would be considered the signal on which it would perform the shift operation. The second input, 0001, would be the number of places that would be shifted, which in this case is 1 place to the left.


The same two input signals, 0110 and 0001, pass through a shifter logic circuit. The logic circuit performs the shift operation on the first input by shifting its bits by the number of places indicated by the second input, which in this case is 1. The left-most bit is dropped, and a 0 is added as the right-most bit.

| | | | | |
|---|---|---|---|---|
| In | 0 | 1 | 1 | 0 |
| | 1 | 1 | 0 | |
| Out | 1 | 1 | 0 | 0 |

The comparator accepts two signals but can output 1 of 3 possible outputs: 1) greater than, 2) less than or 3) equal to. It will generate an output signal based on the comparison of the first signal with the second signal. In our small 4-bit processor, we will assign a specific sequence of bits that will be interpreted as greater than, less than or equal to, 1111, 1001 or 0000, respectively.


The same two input signals, 0110 and 0001, pass through a comparator logic circuit. The logic circuit compares them. Since the first signal is greater than the second, 0110 > 0001, the comparator will output the signal for greater than, 1111.

| | | | | |
|---|---|---|---|---|
| > | 1 | 1 | 1 | 1 |
| = | 1 | 0 | 0 | 1 |
| < | 0 | 0 | 0 | 0 |

## ALU

The arithmetic logic unit (ALU) is that part of the processor responsible for executing instructions. It contains all the logic circuits needed to perform the operations required for executing instructions.

> THE ALU IS THAT PART OF THE PROCESSOR RESPONSIBLE FOR PERFORMING OPERATIONS ON DATA.

An instruction typically specifies the operation that needs to be performed. However, for an operation to function, it requires operands or information to work on. The instruction must provide this information either as data or as an address. Data can be provided immediately in the instruction itself, or the instruction can provide a source address at which the data can be found.

## EXAMPLE 4.2

The instruction to add two numbers must specify that the operation to be performed is an addition. It must also specify the two numbers that are to be added, the operands. Let's assume that the numbers to be added are 2 and 3. In this case, the instruction in human-readable language would be:

```
ADD 2,3
```

**Notes**

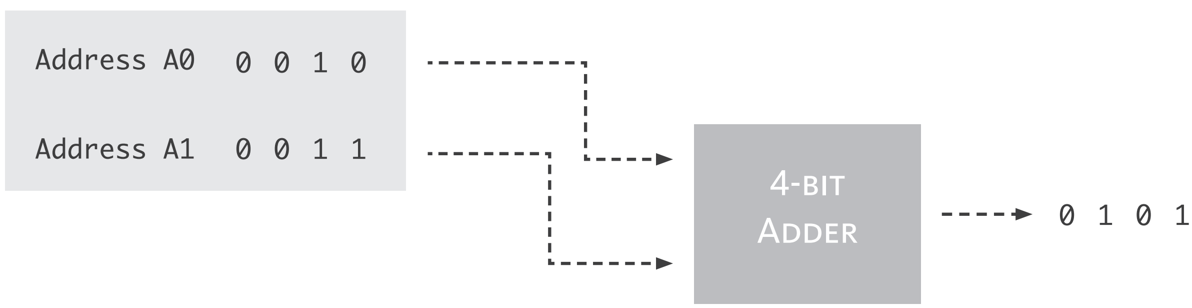
Learning the concepts that underpin programming is more critical than learning the programming language itself. Pseudocode is used for that purpose. Written in English, it allows the student to focus on thought processes rather than syntax and semantics.


The instruction specifies that the input signals should be 2 and 3, which in binary are represented as `0010` and `0011`, respectively. The output from the adder logic circuit would be 5, `0101`.

```
    1
  0 0 1 0
+ 0 0 1 1
---------
  0 1 0 1
```

Alternatively, the numbers 2 and 3 may already be in the computer and are located at Address A0 and Address A1, respectively. In this case, the instruction would provide the processor with the addresses from where to obtain this data.

```
ADD A0,A1
```

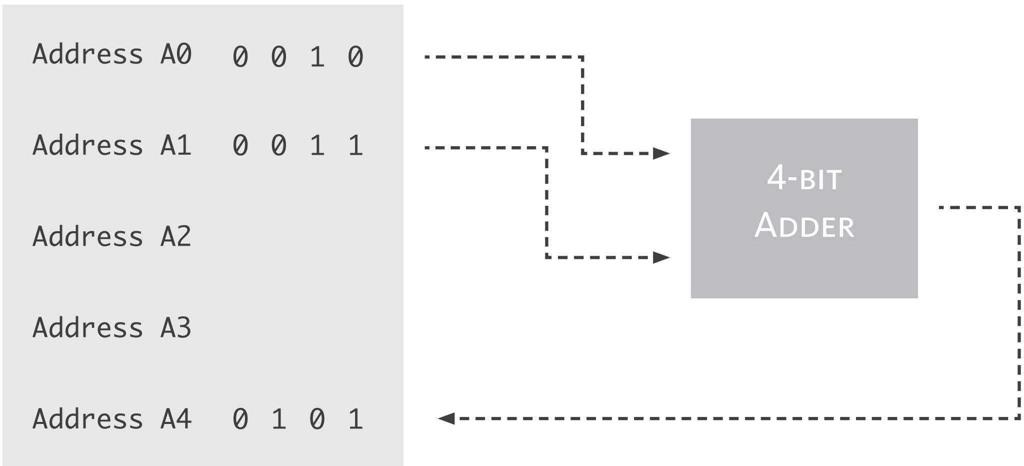



In this case, the instruction provides the data by referring the processor to the source addresses at which the data can be found, Address A0 and Address A1. The data is fetched and fed through the adder logic circuit.

```
    1
  0 0 1 0
+ 0 0 1 1
---------
  0 1 0 1
```

However, we still need to figure out where the ALU should send the output of its logic circuits. Just like the ALU has a source for its information, it must also have a destination. Therefore, the instruction must specify, in addition to a source address, a destination address - the address to which the output should be sent.

```
ADD A4: A0,A1
```




The instruction specifies the destination address first, followed by a colon. The two source addresses, Address A0 and Address A1 follow the colon. The processor provides the data from the two source addresses to the adder logic circuit and then delivers the output to the specified address, Address A4.

```
    1
  0 0 1 0
+ 0 0 1 1
---------
  0 1 0 1
```

Since every instruction must be processed by a logic circuit and the ALU can fit only so many logic circuits, that means that the number of instructions one ALU can execute is limited. In fact, despite the many tasks we perform using our computers, from preparing essays to playing games and watching videos, all these tasks are carried out using a limited set of instructions.

**THE INSTRUCTION SET OF A PROCESSOR IS THE SET OF ALL INSTRUCTIONS IT CAN PERFORM.**

When designing a processor, engineers first determine the set of instructions the processor needs to execute. Then, they determine the operations needed to execute each instruction and select the logic circuits that perform each operation. Finally, they organize the logic circuits in such a way that allows the operations to be performed in the correct order.

## Control

The control unit, CU, is an integral part of the processor. The CU is a collection of logic circuits whose main purpose is to direct the flow of signals through the processor and other parts of the computer. It does so by: 1) decoding instructions to identify the operations that need to be performed, 2) generating control signals to activate the logic circuits that perform these operations and 3) coordinating the operation of the different components of the computer.

In order for the CU to accomplish its purpose, it must be able to identify the operation that needs to be performed; therefore, every instruction in a processor's instruction set must have a unique operation code, opcode. The CU uses this opcode to determine the logic circuits it must activate.

> The control unit directs the flow of signals through the processor and computer.

## Example 4.3

Going back to our 4-bit processor and its four logic circuits, we can create a binary sequence to represent each of the four operation codes required.

| | |
|---|---|
| Add | 0 0 0 1 |
| Multiply | 0 0 1 0 |
| Shift | 0 0 1 1 |
| Compare | 0 1 0 0 |

Let's revisit the Add instruction that delivers two immediate operands for processing, namely 2 and 3, or `0010` and `0011`, respectively. In its human-readable form, it reads:

```
ADD 2,3
```

In machine language, using binary signals, this instruction is 12 bits in length and reads:

```
0001 0010 0011
```

Opcode for the Add operation

Data for the Add operation

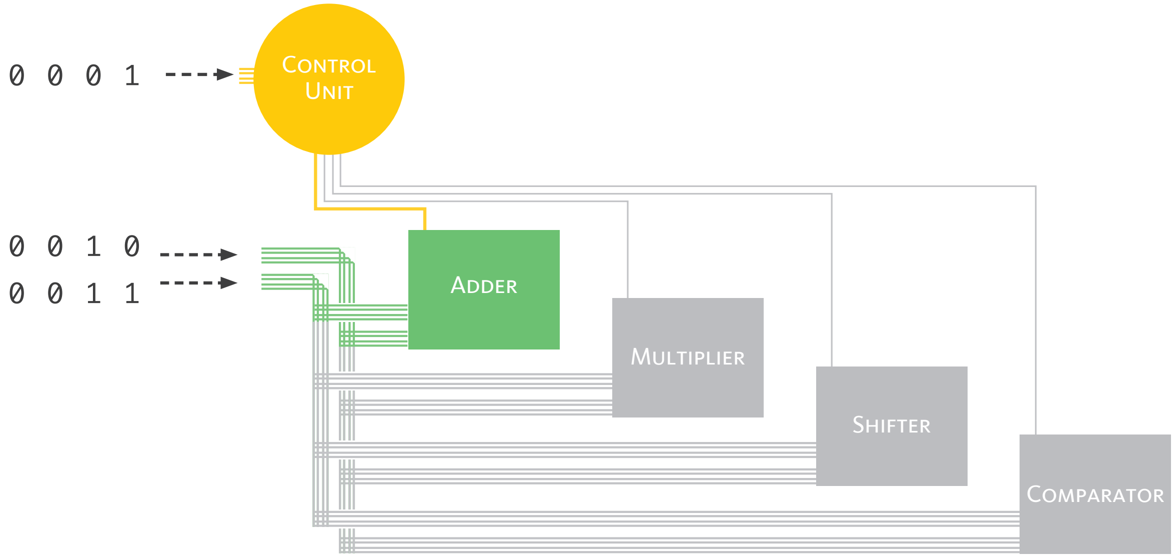
On a side note, this means for our hypothetical small computer to work, 4 bits is not a large enough architecture, and we need to upsize by adding 8 more lines to handle 12 bits instead of 4 bits.

When the instruction is fetched, it is split into its 3 fields: the 4-bit opcode and two 4-bit sequences that represent data. The opcode is sent to the CU for decoding, and, accordingly, a control signal is sent to activate the appropriate logic circuit to which the data is delivered.

**MACHINE LANGUAGE IS IN THE FORM OF ELECTRONIC SIGNALS AND IS WRITTEN FOR HUMANS AS ZEROS AND ONES.**

## Notes

Information exists in two main layers: the logical layer and the physical layer. The logical layer makes sense to us and is represented in the form of human-readable instructions and data. The physical layer is the signals and the logic circuits that process them. The physical layer is the implementation of the logical layer.

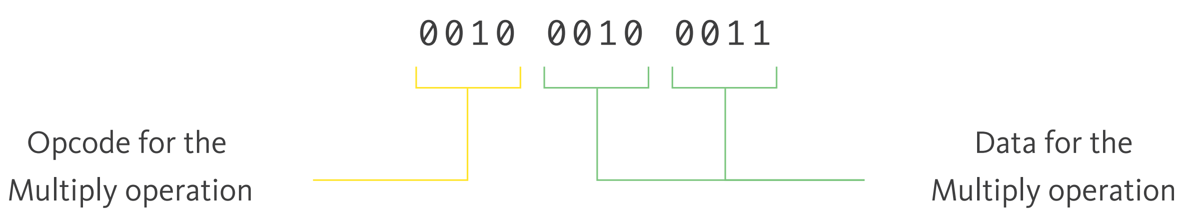



The CU has a control line to each of the four logic circuits it must manage. Once the instruction is fetched, it is split. The opcode `0001` is sent to the CU that interprets it as an Add operation, and then proceeds to send an activation control signal to the adder logic circuit. The activated adder can now receive the two items of data, `0010` and `0011`.

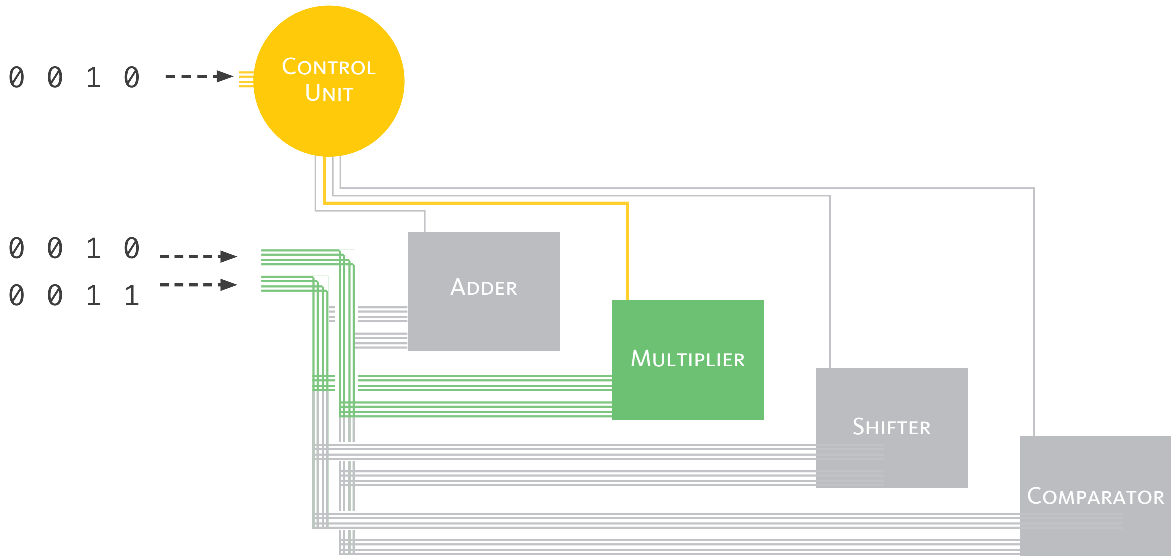
Suppose the Add instruction is changed to a Multiply instruction keeping the same immediate operands, namely 2 and 3, or `0010` and `0011`, respectively. In its human-readable form, the instruction reads:

```
MULTIPLY 2,3
```

The opcode for the Multiply operation is `0010`. In machine language, the 12-bit binary instruction reads:



As with every instruction, the instruction is fetched, then it is split into its respective fields. The leftmost field is sent to the CU for decoding, a process that identifies and activates the required components of the computer. The remaining two fields are treated as operands and sent to the activated logic circuit for processing.




In this example, the instruction is for Multiply operation whose opcode is `0010`. The 12-bit binary instruction is split. The 4-bit opcode is sent to the CU, which then activates the multiplier logic circuit by sending a control signal. The activated logic circuit can then receive the two items of data, `0010` and `0011`.

**IN MACHINE LANGUAGE, THE INSTRUCTION IS DIVIDED INTO FIELDS, EACH REPRESENTING A PIECE OF INFORMATION.**

The previous hypothetical instructions contained immediate operands, which are data immediately delivered to the ALU for processing. However, not all instructions necessarily contain immediate data. In many cases, instructions involve source addresses that indicate the locations where the data is stored. One of the components that hold such data waiting to be processed is the register file.

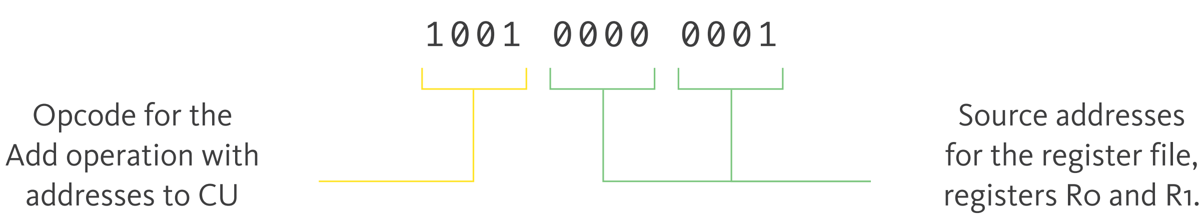
A register is a small and fast memory component within the processor. A processor typically has several registers that are grouped into one working unit referred to as the register file.

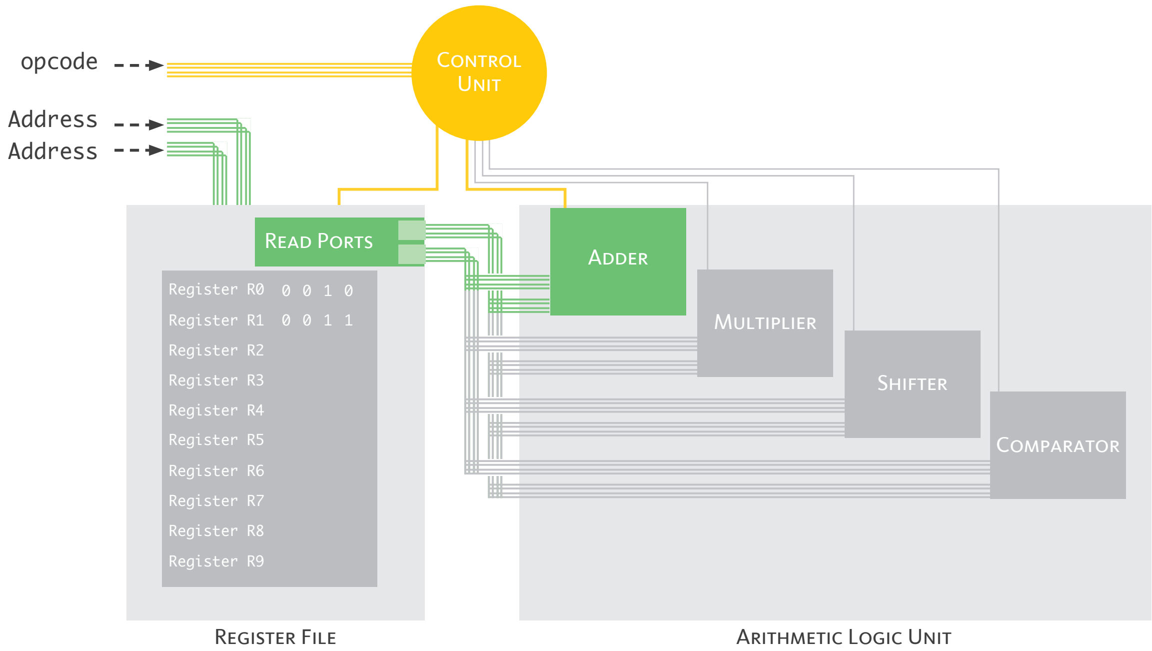
> THE REGISTER FILE IS A GROUP OF SMALL AND FAST MEMORY COMPONENTS LOCATED IN CLOSE PROXIMITY TO THE ALU.

Like other components of the processor, the register file is managed by the CU. For instructions that contain source addresses rather than immediate data, the opcode is different. This allows the CU to distinguish between the need to activate the register file and not. When the CU decodes the opcode and identifies that the operation requires the services of the register file, it treats the following two fields as addresses.

A fundamental concept in computing is that any sequence of signals is interpreted based on the context in which it is presented for execution. So if the opcode is for adding immediate data, the following fields are interpreted as numbers. Whereas, if the opcode is for adding the data at source addresses, the following fields are interpreted as addresses.

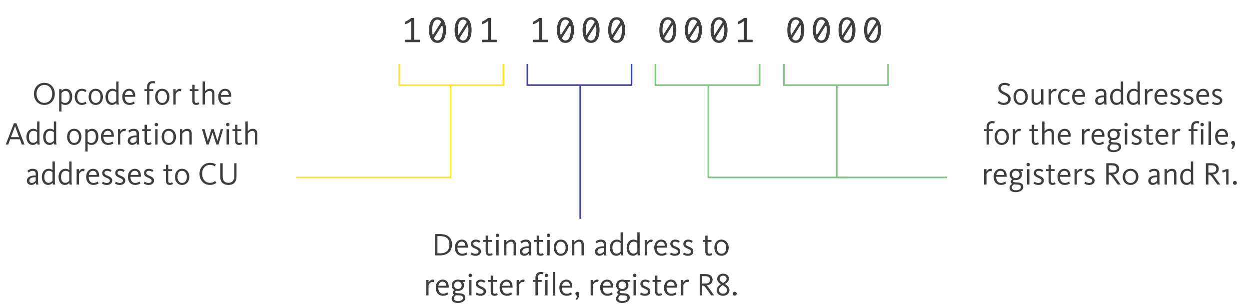
> THE INTERPRETATION OF ANY SIGNAL DEPENDS ON THE CONTEXT WITHIN WHICH IT IS BEING EXECUTED.






The instruction is split into 3 fields: the opcode and the two source addresses. The opcode is sent to the control unit, where it is decoded. The CU signals the register file to accept the two addresses as source addresses and to load their contents to the read ports. The adder can then read the data and perform the addition operation.

So far, in our hypothetical processor, we have not considered that data must go somewhere after processing. Once an operation like addition is performed, the result needs to be stored for further use or future instructions. To specify where the output should be stored, the instruction includes a destination address field. This field points to a specific register where the result of the operation will be held.

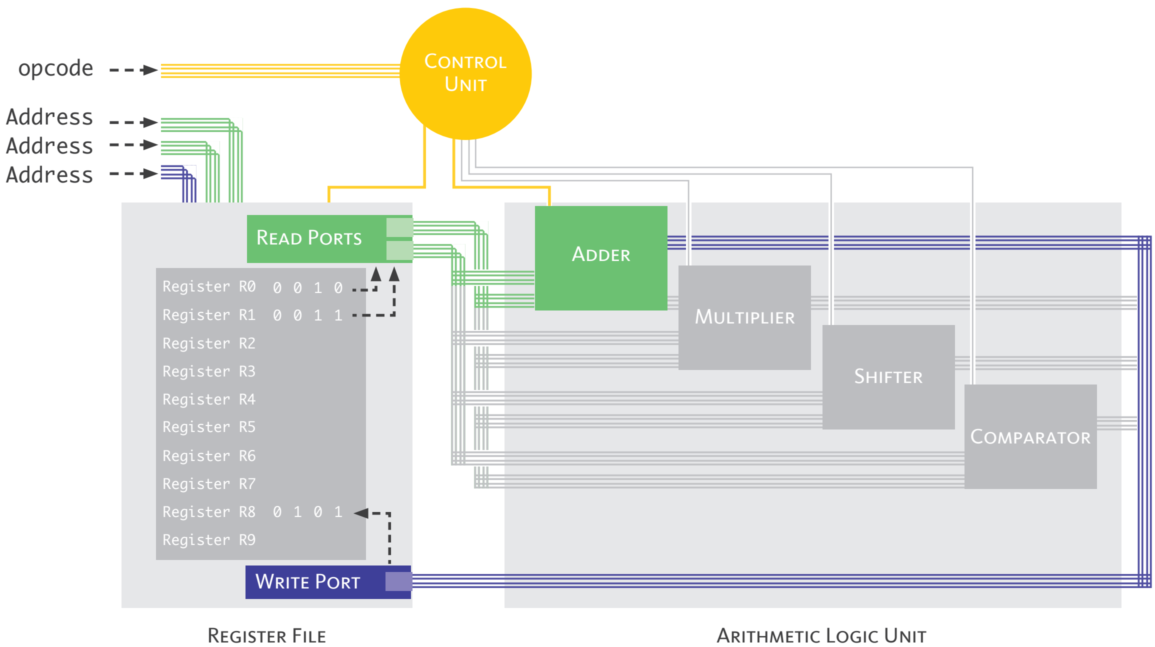


This 16-bit binary sequence, `1001 1000 0001 0000`, means: "add the contents of registers R0 and R1 and place the sum in register R8." In human-readable form, it is transcribed as:

```
ADD R8: R0,R1
```

The destination address, like the source address, typically represents a register number that identifies a specific register within the register file. After decoding the instruction, the control unit signals the register file to write the result into the register specified by the destination address.

By having a designated destination address, the processor ensures that the result is properly stored and can be accessed by subsequent instructions or utilized in further computations.

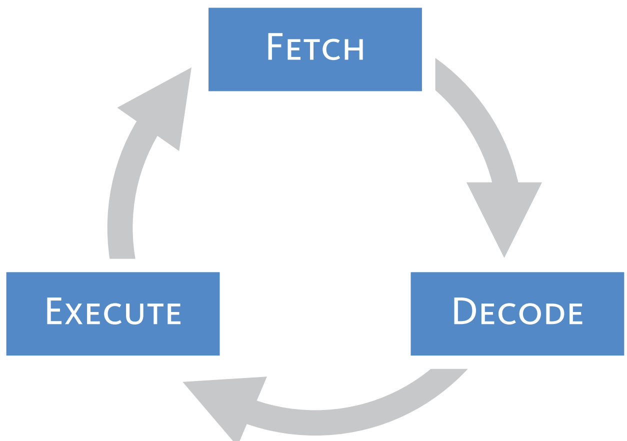



The instruction is split into 4 fields: an opcode, two source addresses and a destination address. The CU signals both the register file and the Adder in the ALU. The register file places the contents of the registers indicated by the source addresses in the read ports. The Adder reads the data, performs the operation and sends the sum back to the register file, which is placed in the register indicated by the destination address.

It is important to note that this lesson reveals the interface between the logical layer, represented in human language, and the physical layer of machine language, represented in digital electronic signals and implemented by logic circuits.

## THE INSTRUCTION CYCLE

As long as a processor is working, it is processing instructions one after the other. It fetches an instruction, decodes it, executes it and then fetches the next instruction to repeat the process over and over again. This is referred to as the instruction cycle. Each step is referred to as a stage. Therefore, the instruction cycle is composed of three basic stages: fetch, decode and execute.




THE INSTRUCTION CYCLE IS A SERIES OF REPEATED STAGES: FETCH, DECODE AND EXECUTE.

## Summary

*While this lesson presented a hypothetical processor, the concepts presented apply to real-life processors.*

A computer processes signals referred to as digital electronic signals. These signals are binary in nature in that they can only be one of two values, either high voltage (1) or low voltage (0). Digital electronic signals constitute what is referred to as machine language.

The processor is that part of the computer system that changes input into output. The processor contains logic circuits, which are responsible for processing. Each logic circuit is a collection of electronic components that have a particular arrangement designed to perform a particular operation. The operation performed by a logic circuit is determined by the configuration of its components.

Examples of logic circuits are: adder, multiplier, shifter, comparator and multiplexer.

The processor, also known as the central processing unit (CPU), has three main parts: arithmetic logic unit (ALU), control unit (CU) and register file.

The ALU is responsible for performing operations that include: addition, subtraction, multiplication, division, shifting and comparisons.

The CU coordinates and controls the activities of the processor. It fetches instructions, decodes them and generates the necessary control signals, which ensures that instructions are executed in the correct sequence.

The register file is a group of small and fast memory components called registers. It is located in close proximity to the ALU and is used to hold data and addresses needed in the execution of instructions.

The instruction is split into fields that represent the operation code (opcode), data and/or addresses at which data can be found. The leftmost field is the opcode. The interpretation of the remaining fields depends on the opcode.

The instruction set of a computer is the set of all instructions its processor is designed to perform.

The instruction cycle is a series of stages: fetch, decode and execute. The cycle repeats itself with every instruction.

## Activity 4.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F | Statement |
|---|---|
| F | The processor is not involved in changing input to output. |
| T | The main units of a processor are the arithmetic logic circuits, control unit and register file. |
| T | A digital electronic signal is binary in nature. |
| T | In a 64-bit processor, a 'word' refers to a sequence of 64 bits. |
| T | The transistor is an important component of a logic circuit. |
| F | The configuration of components in a logic circuit does not affect the operation it performs. |
| T | The ALU is responsible for performing arithmetic operations. |
| F | The CU is responsible for sending control signals to input and output devices. |
| T | The opcode determines how the remaining signals in the instruction will be interpreted. |
| T | Fetch, decode and execute are the three stages of the instruction cycle. |

## Activity 4.2

Complete the sentences below.

1. Changing input to output is the responsibility of the ________ .
   a) memory
   b) processor
   c) network adapter

2. Performing arithmetic operations is the responsibility of the ________ .
   a) ALU
   b) CU
   c) Register file

3. Directing the flow of data is the responsibility of the ________ .
   a) ALU
   b) CU
   c) Register file

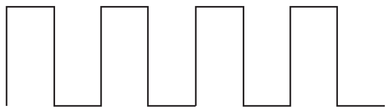
4. Which of the following is not true about the register file?
   a) The register file is a short-term memory unit.
   b) The register file is a very fast memory unit.
   c) The register file is located away from the processor.

5. Signals going into an information processing system are referred to as:
   a) decode - fetch - execute
   b) fetch - decode - execute
   c) execute - decode - fetch

## Activity 4.3

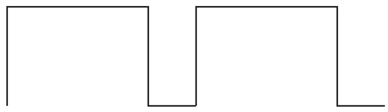
Study the digital electronic signals. Encode each in binary form using ones (1) and zeros (0).

1.



1 0 1 0 1 0 1 0

2.

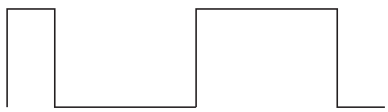


1 1 1 0 1 1 1 0

3.

1 0 0 0 0 0 1 0

4.



1 0 0 0 1 1 1 0

## Activity 4.4

Perform an addition operation on the 4 pairs of binary numbers.

Note that binary addition is similar to decimal addition, except that 1+1 = 10, which requires the 1 to carry forward.

1.

$$\begin{array}{rcccc} & 0 & 0 & 1 & 0 \\ + & 0 & 0 & 0 & 1 \\ \hline & 0 & 0 & 1 & 1 \end{array}$$

2.

$$\begin{array}{rcccc} & 0 & 1 & 0 & 1 \\ + & 1 & 0 & 0 & 0 \\ \hline & 1 & 1 & 0 & 1 \end{array}$$

3.

$$\begin{array}{rcccc} & 1 & 1 & 0 & 0 \\ + & 0 & 0 & 1 & 0 \\ \hline & 1 & 1 & 1 & 0 \end{array}$$

4.

$$\begin{array}{rcccc} & & & 1 & \\ & 0 & 0 & 0 & 1 \\ + & 0 & 0 & 0 & 1 \\ \hline & 0 & 0 & 1 & 0 \end{array}$$

5.

$$\begin{array}{rcccc} & 1 & & 1 & \\ & 0 & 1 & 0 & 1 \\ + & 0 & 1 & 0 & 1 \\ \hline & 1 & 0 & 1 & 0 \end{array}$$

6.

$$\begin{array}{rcccc} & & 1 & 1 & \\ & 0 & 0 & 1 & 1 \\ + & 0 & 0 & 0 & 1 \\ \hline & 0 & 1 & 0 & 0 \end{array}$$

## Activity 4.5

Answer the questions below.

1. Which instruction adds two immediates and places the sum in register R8?
   a) `ADD R8: R1,R3`
   b) `MULTIPLY R8: 4,5`
   c) `ADD R8: 4,5`

2. Which instruction multiplies data fetched from source addresses and places the product at a destination address?
   a) `MULTIPLY R5: R4,R0`
   b) `MULTIPLY R8: 4,5`
   c) `ADD R8: R1,R3`

3. Which instruction is missing a destination address?
   a) `ADD R8: 4,5`
   b) `ADD 2,3`
   c) `ADD R3: R1,R0`

4. Which instruction is missing a source address?
   a) `MULTIPLY R6: 0,7`
   b) `MULTIPLY R3: R1`
   c) `MULTIPLY R4: R1,R3`

5. Which instruction is missing both source and destination addresses?
   a) `ADD R8: R1,R3`
   b) `MULTIPLY R8: 4,5`
   c) `ADD : 4,5`

## Activity 4.5

Use the information provided in the tables to translate between human language and the machine of a hypothetical processor to answer the questions.

**Opcodes**

| | |
|---|---|
| Add (immediate) | 0001 |
| Add (address) | 1001 |
| Multiply (immediate) | 0010 |
| Multiply (address) | 1010 |

**Numbers**

| | | | |
|---|---|---|---|
| 0 = 0000 | 4 = 0100 | 8 = 1000 | 12 = 1100 |
| 1 = 0001 | 5 = 0101 | 9 = 1001 | 13 = 1101 |
| 2 = 0010 | 6 = 0110 | 10 = 1010 | 14 = 1110 |
| 3 = 0011 | 7 = 0111 | 11 = 1011 | 15 = 1111 |

**Register Numbers**

Register R0 (0000)
Register R1 (0001)
Register R2 (0010)
Register R3 (0011)
Register R4 (0100)
Register R5 (0101)
Register R6 (0110)
Register R7 (0111)

1. Which 4-bit binary sequence is designated as the opcode for the operation of adding immediate data?
   a) 1001
   b) 0001
   c) 1101

2. Which 4-bit binary sequence represents register R4?
   a) 0001
   b) 0111
   c) 0100

3. Which 4-bit binary sequence represents the number 15?
   a) 1001
   b) 0101
   c) 1111

4. Identify the human-readable form of this instruction `1001 0111 0000 0010`.
   a) `ADD R4: R1,R2`
   b) `MULTIPLY R7: 0,2`
   c) `ADD R7: R0,R2`

5. Identify the human-readable form of this instruction `0010 0010 0110 0001`.
   a) `MULTIPLY R2: 6,1`
   b) `MULTIPLY R2: R6,R1`
   c) `ADD R2: 6,1`

6. Identify the machine language form of this instruction `ADD R4: R1,R2`.
   a) `1001 0001 0010 0100`
   b) `1001 0100 0001 0010`
   c) `0001 0100 0001 0010`

7. Identify the machine language form of this instruction `MULTIPLY R7: 6,7`.
   a) `0010 0110 0111 0111`
   b) `1010 0111 0110 0111`
   c) `0010 0111 0110 0111`

8. Write the instruction `1010 0110 0100 0101` in human-readable form.

   `MULTIPLY R6: R4,R5`

9. Write the instruction `ADD R3: 4,5` in machine language.

   `0001 0011 0100 0101`

## Activity 4.6

Use the information provided in the translation tables of Activity 4.5 to answer the questions.

1. In how many ways can the 4-bit binary sequence `1010` be interpreted? Explain.

   It can be interpreted in 3 ways: as an opcode, number and register.

   If it is the first field in an instruction, it is interpreted as the opcode for multiply using addresses. If it is the second field, it is interpreted as the register number for the destination register. If it is the third or fourth field, its interpretation will depend on the opcode. An opcode for immediate operations will result in interpreting it as a number, while an opcode for address-related operations will result in interpreting it as a register number.

2. What is the maximum number of integers that can be encoded using 4-bit binary sequences? Can we add more opcodes and registers to our hypothetical processor?

   The maximum number of integers that can be encoded is 16, which are the integers 0 to 15. The table in Activity 4.5 lists all the possible 4-bit binary sequences. Alternatively, we can calculate this number. Since there are 4 places and each place can hold one of two values, 0 or 1, we can use 2^4 = 16.

   Yes, we can add opcodes and registers each to a maximum of 16.

## Activity 4.7

Answer the questions below.

1. “Instructions provide a context in which signals are interpreted.” Is this statement correct? Explain.

   Yes, instructions do provide a context in which signals are interpreted.

   Since every instruction has an opcode and the opcode determines how the data is interpreted, then changing the instructions changes the way in which data is interpreted.

2. “I tried to open my Word document using the Excel application, but it didn’t open.” Explain why this statement is true. Does this apply to other applications and files?

   In essence, Word and Excel have different ‘instructions’ or ‘opcodes’ for interpreting and handling data, which is why you cannot open a Word document in Excel and expect it to be displayed correctly.

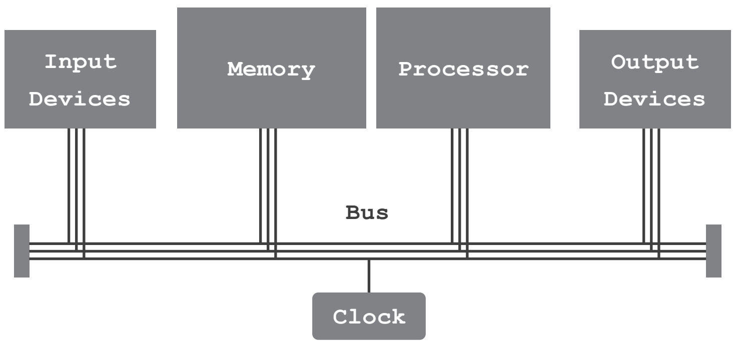
   Yes, this principle generally applies to all software applications and their respective files. Each application is specifically designed to work with certain file formats and structures.

# LESSON 5: DESIGN OPTIMIZATIONS

Many considerations are taken when designing a system, all in an effort to design the computer to be fast, dependable, energy efficient, easy to use and affordable. Optimizations target design features, such as performance and dependability. This lesson will focus on the strategies used to optimize performance.

## ARCHITECTURE

All computer systems contain four basic components: input devices, output devices, a processor and a memory. Input devices and output devices allow the exchange of signals with the outside world. The processor handles the conversion of input to output in accordance with stored programs. The memory is tasked with holding information. All information, whether data or instructions, are held in memory.




The basic architecture of a computer includes a processor, memory components, input devices and output devices. In addition, the bus acts as a communication network connecting these components and a clock to ensure all activities are synchronized.

THE COMPONENTS OF ANY COMPUTER SYSTEM: A PROCESSOR, A MEMORY, INPUT DEVICES, OUTPUT DEVICES, A BUS AND A CLOCK.

## The Bus System

The bus, in computing, is a shared communication pathway that allows the components of a computer system to transmit signals to each other. These signals are divided into 3 groups, each carrying one type of information: data, addresses and control signals. Each of these is transmitted on a dedicated set of conducting lines.

| | |
|---|---|
| **Data bus** | The set of lines that transmit the actual data being transferred between components. It is bidirectional, meaning it can transmit data in both directions. |
| **Address bus** | The set of lines that transmit the addresses used to specify the source or destination of data. |
| **Control bus** | The set of lines that transmit bus control signals that are used to coordinate and control the operations of the components within the computer system. |

THE BUS IS A SHARED COMMUNICATION PATHWAY THAT CONNECTS ALL THE COMPONENTS OF A COMPUTER SYSTEM.

In a computer system, each component is connected to the bus and assigned an address, a number that uniquely identifies it. The address allows the components to be identified and accessed by the CPU or other components in the system.

When a component, such as the CPU, wants to communicate with another specific component, such as memory or an I/O device, it places the target component's address on the address bus. This address is then transmitted over the bus to indicate the source or destination of the data transfer.

### Example 5.1

Let's say the CPU is busy processing data for streaming a video that is being viewed on the web browser. With the mouse, the user moves the pointer and selects the window for a Word document essay. The screen switches to the essay window, and the video stops playing. What happened inside the computer?

| | |
|---|---|
| **1. User** | The user moves the mouse and clicks the left button. This generates signals inside the mouse, which are data signals. |
| **2. Mouse** | The mouse prepares the data for transmission and places it on the data bus. It places the CPU address on the address bus. It sends an interrupt signal to the control bus. |
| **3. Bus** | The bus, based on the address and control signals, delivers the data to the CPU. |
| **4. CPU** | The CPU interrupts the process it is working on, streaming the video, in order to process the incoming data. Based on the new data, the CPU switches the active screen from the web browser to the word processor. |

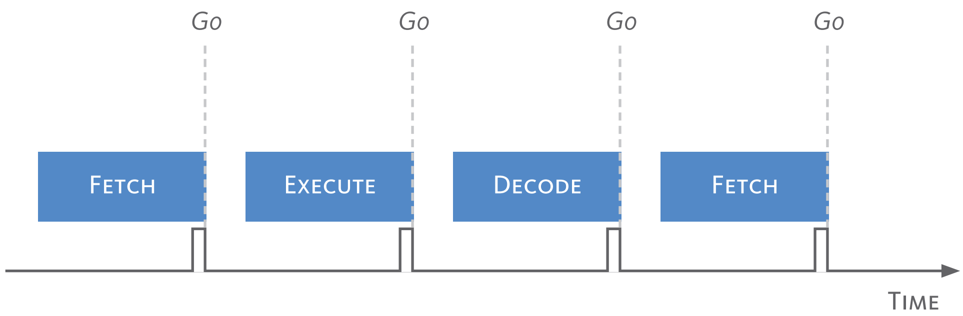
### Synchronization

The computer system is a busy place with signals traveling at light speed between the various components while the processor executes one instruction cycle after the other.

Activities occur at different rates. Some components are slower than others, while some instructions take less time to execute than others. This would result in both delays and in errors if left unchecked. Fast operations would pile up, waiting their turn while a slow instruction is being executed, which causes delays and slows performance. Not to mention the errors that can occur if two components try to send data at the same time.

In a synchronized system, all the components operate according to a common time reference, which is typically provided by a time signal generated by an internal clock and distributed to all the components.

The clock has a quartz crystal that oscillates back and forth, vibrates very quickly. Each oscillation is referred to as a clock cycle. With each clock cycle, it generates a timing signal that is placed on the control bus and sent out to all components.




*Clock Signal*

This timing signal is like a "Go" signal; it tells all the components to move their signals to the next stage. This way, all signals move at the same time and don't "bump" into each other.

Inside the CPU, the clock signals the move from one stage in the instruction cycle to the next. Outside the CPU, the clock signals the transmission of data and addresses on the bus between components. The clock signal is one of the signals transmitted on the control bus.

**THE CLOCK GENERATES A TIMING SIGNAL THAT SYNCHRONIZES THE TRANSFER OF ALL SIGNALS THROUGHOUT THE COMPUTER.**

The processor operates very quickly, mostly due to the high clock frequency. The clock frequency is the number of clock cycles or oscillations that occur per second. For example, if a clock generates 2 billion clock cycles every second, that means its clock frequency is 2 billion cycles per second, or 2 GHz.

| | | |
|---|---|---|
| 2 | Billion | Cycles per second |
| ↓ | ↓ | ↓ |
| 2 | Giga | Hertz |
| ↓ | ↓ | ↓ |
| 2 | G | Hz |

Since the clock determines the rate at which the computer operates, then adjusting the clock frequency affects the overall performance of the computer. The clock frequency can be used as a measure of performance, with higher clock frequencies indicating faster performance.

**The clock frequency is one measure of computer performance.**

## Example 5.2

Let's assume there are two hypothetical CPUs, CPU $_A$ and CPU $_B$. We will make two assumptions for our CPUs, each CPU has 1 processor, and each processor executes only 1 stage of the instruction every clock cycle. If the clock frequency of CPU $_A$ is 6 GHz and that of CPU $_B$ is 3 GHz, then we can say that CPU $_A$ is faster than CPU $_B$.

$$\frac{6 \text{ x } 10^9 \text{ clock cycles}}{1 \text{ second}} \text{ x } \frac{1 \text{ instruction}}{3 \text{ clock cycles}} = 2 \text{ x } 10^9 \text{ instructions / second}$$

We can also calculate the instruction count per second and compare the two CPUs. If both CPUs are executing 1 stage of the instruction cycle per clock cycle and there are 3 stages, then it will take 3 clock cycles to complete 1 instruction. So CPU $_A$ that runs at 6 GHz, or 6 billion clock cycles per second, will be capable of executing 2 billion instructions per second.

On the other hand, CPU $_B$ is running at 3 GHz or 3 billion clock cycles per second. This is half the speed and is reflected in the calculation, which reveals an instruction count of 1 billion instructions per second.

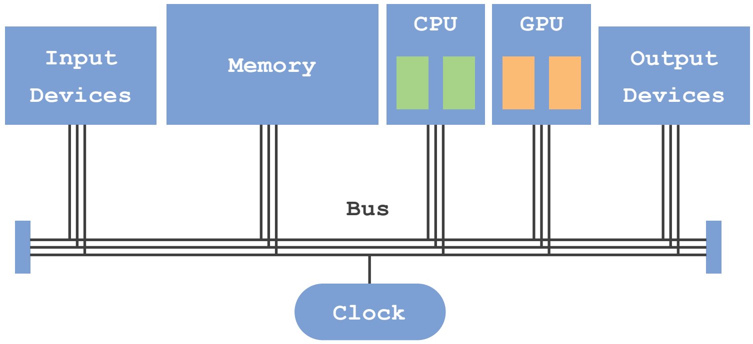
$$\frac{3 \times 10^9 \text{ clock cycles}}{1 \text{ second}} \times \frac{1 \text{ instruction}}{3 \text{ clock cycles}} = 1 \times 10^9 \text{ instructions / second}$$

## Parallelism

Another strategy for optimizing performance is referred to as parallelism, which means performing tasks simultaneously at the same time. One way this strategy manifests in the design of the computer is in multi-core CPUs, which are CPUs that have more than 1 processor, each referred to as a core.

Another way this strategy is implemented is in the introduction of the graphic processing unit, GPU. The GPU is a processing unit that focuses only on executing instructions that pertain to graphics. It does not concern itself with running the whole of the computer and its apps. In modern-day computers, particularly those used for games, it is common to have both a CPU and a GPU, and each one of these processing units has multiple cores. This is referred to as parallel architecture.

**PARALLELISM REFERS TO THE ABILITY TO PERFORM TASKS SIMULTANEOUSLY.**




A parallel architecture that includes a GPU operating in parallel with a CPU. The CPU attends to the operation of the whole computer, while the GPU focuses on processing graphics-related instructions.

It is important to note that while a dual-core CPU has two processors, it is faster than a single-core CPU. However, it does not necessarily mean that it is twice as fast. The overall performance of a computer is not limited to clock frequency; it also depends on the task required.

### EXAMPLE 5.3

Let's assume CPU $_A$ has a single-core, while CPU $_B$ has a dual-core, and both CPUs are operating at the same clock frequency of 2 GHz. Let's also assume that there is a Program A and that the time it takes for CPU $_A$ to execute Program A is 10 seconds.

If CPU B is tasked with executing Program A, its operating system will have to decide whether it can divide Program A into two parts. If it can, then CPU B can execute both parts simultaneously, each on a core. This would shorten the time to execute Program A from 10 seconds to 5 seconds (assuming both parts are equal).

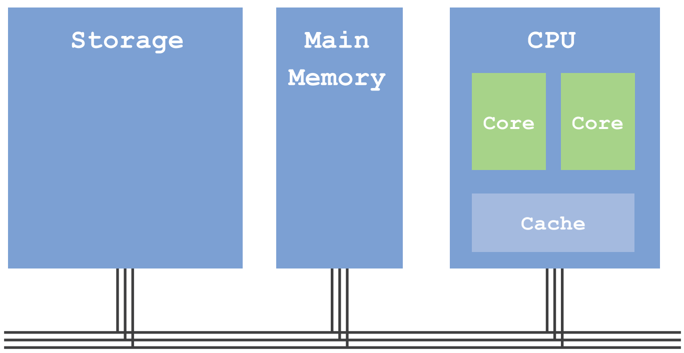
If the operating system cannot divide Program A into two parts, then only one core out of the two will be used, and the execution time remains at 10 seconds.

## PROXIMITY

Proximity is another strategy used to optimize computer design. The closer components are together, the better the communication and the faster the transfer of information between them one such implementation of this strategy is the on-chip cache.

Understanding this concept requires an understanding of the memory system and its relation to the processor. Memory is composed of several components that work together as one system, the memory system. The components of this system, in order of size, are: storage, main memory, cache and register file.

**THE COMPONENTS OF THE MEMORY SYSTEM ARE: STORAGE, MAIN MEMORY, CACHE AND REGISTER FILE.**




The components of the memory system are the storage, main memory, cache and register file. The register file is not illustrated as it is considered part of the core.

The processor depends on memory to supply it with the information it needs. The execution of tasks requires both instructions and data to be delivered promptly from memory to the processor. The overall performance of the computer depends on memory matching its performance with that of the processor. Memory must keep up with the processor; otherwise, the computer's performance will slow down.

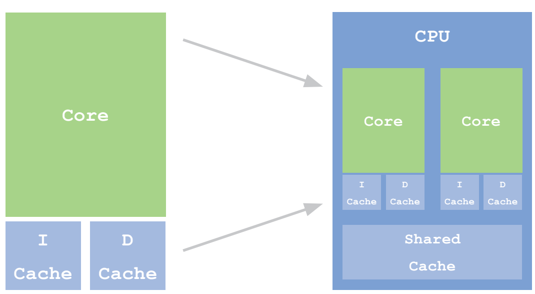
Since close means fast, keeping components, such as the register file and cache, on the same chip as the processor is one way to optimize the performance of the computer. The processor does not have to wait for information to be transmitted on the bus from storage or from main memory. The information it needs is in the registers, and if not there, the information is in the cache. In either case, the information is close by, and the processor does not have to wait.

**PROXIMITY OPTIMIZES PERFORMANCE BY PLACING COMPONENTS CLOSE TO THE PROCESSOR.**

## Prediction

Of the optimization strategies, prediction is by far the most ingenious. The ability to predict what comes next greatly improves performance. One such implementation is the separate instruction cache and data cache design.

In this design, the cache is divided into two caches: an instruction cache and a data cache. Having a mixed cache requires the processor to sift through large amounts of information in search of the next instruction. However, we know that instructions follow each other, and therefore, we can predict that if instruction 1 is executed, it will be followed by instruction 2. So, why not arrange all the instructions in order and separate them from the clutter of data?




Each core has two separate caches: one for instructions and the other for data, labeled I-Cache and D-Cache, respectively. The CPU also has a shared cache for all cores on the chip.

Predicting the next instruction allows us to arrange instructions in order and separate them from data. This makes finding the next instruction easy and fast.

> PREDICTION OPTIMIZES PERFORMANCE BY PLACING THE MOST LIKELY INFORMATION TO BE USED CLOSEST TO THE PROCESSOR.

Another implementation of prediction is prefetching. When the processor searches for data, it starts by checking the closest memory component to it: the data cache. If it doesn't find the desired data there, it proceeds to check the next component, the shared cache, the main memory, and finally, storage. What's interesting is that when the processor does find the desired data, it doesn't stop at fetching just that particular data item. Instead, it also fetches the adjacent data items. The rationale behind this approach is that if one data item is needed, there is a high chance that the next adjacent data item will be needed soon as well.

By prefetching data that is predicted to be used soon, the processor increases its chances of having the required information available nearby. This proactive data fetching significantly improves the performance of the computer system.

### Notes

The principle of locality is fundamental to the operation of the computer. While not explicitly stated as a principle, the lesson clearly identifies both temporal and spatial locality as concepts.

## Summary

The components of any computer system are: a processor, a memory, input devices, output devices, a bus and a clock.

The bus is a shared communication pathway that connects all the components of a computer system. The bus is divided into the address bus that transmits source and destination addresses, the control bus that transmits control signals, including the clock timing signal, and the data bus that transmits the actual data.

The clock generates a timing signal that synchronizes the transfer of all signals throughout the computer. Synchronization is a strategy to optimize performance by ensuring that all signals travel without congestion or delay.

The clock frequency is one measure of computer performance. Its unit of measurement is the Hertz. Other measures of performance are the instruction count per second and the program execution time.

Parallel processing refers to the ability to perform tasks simultaneously. Parallelism is a strategy that is implemented in the form of multi-core processors and in the use of graphics processing units that work in parallel with central processing units.

The components of the memory system are: storage, main memory, cache and register file.

Proximity optimizes performance by placing components close to the processor. The closer components are, the faster the exchange of signals. The inclusion of the cache on the same chip as the processor is one implementation of the proximity concept.

Prediction optimizes performance by placing the most likely information to be used closest to the processor. The ability to predict what comes next significantly improves performance. The separation of the instruction and data caches is one implementation of the strategy of prediction. The other is referred to as prefetching, where the processor fetches not only the requested data item but also the adjacent data items. If one data item is needed, there is a high chance that the next adjacent data item will also be needed soon.

## Activity 5.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F | Statement |
|---|---|
| T | The bus is a component of the computer system. |
| T | The clock sends out a timing signal to synchronize all components in the computer system. |
| F | Synchronization is not essential for the proper functioning of the computer.. |
| F | The data bus and the address bus are the only two bus lines. |
| F | The execution time is not a measure of performance. |
| T | Performing tasks simultaneously is referred to as parallelism. |
| T | The CPU and GPU work in parallel. |
| T | Placing the cache on the same chip as the processor is one implementation of the proximity principle. |
| T | Prediction is a fundamental concept in computing. |
| F | Prediction does not improve performance. |

## Activity 5.2

Complete the sentences below.

1. The ________ is a system of lines that connects all computer components.
   a) bus
   b) processor
   c) cache

2. The clock cycle is the time duration between two ________ .
   a) frequencies
   b) instructions
   c) timing signals

3. The ________ is a measure of computer performance.
   a) instruction count
   b) execution time
   c) mean time to failure

4. The unit of clock frequency is the ________ .
   a) Second
   b) Giga
   c) Hertz

5. The unit of clock cycle is the ________ .
   a) Second
   b) Meter
   c) Hertz

## Activity 5.3

Answer the questions below.

1. The clock is an implementation of which concept?
   a) synchronization
   b) parallelism
   c) prediction

2. The cache is an implementation of which concept?
   a) parallelism
   b) proximity
   c) synchronization

3. The separate I-cache and D-cache is an implementation of which concept?
   a) parallelism
   b) prediction
   c) synchronization

4. The multi-core processor design is an implementation of which concept?
   a) parallelism
   b) proximity
   c) prediction

5. The strategy of prefetching is an implementation of which concept?
   a) prediction
   b) synchronization
   c) proximity

## Activity 5.4

Answer the questions below.

1. Assuming the CPUs have the same number of cores, which CPU is the fastest?
   a) $CPU_A$ : 3.5 GHz
   b) $CPU_B$ : 3.1 GHz
   c) $CPU_C$ : 2.7 GHz

2. Assuming the CPUs operate at the same clock frequency, which CPU is the fastest?
   a) $CPU_A$ : dual core
   b) $CPU_B$ : octa core
   c) $CPU_C$ : quad core

3. Where is the first place the processor will look for instructions?
   a) D-cache
   b) I-cache
   c) Shared cache

4. Where is the first place the processor will look for data?
   a) Main memory
   b) Shared cache
   c) D-cache

5. If the processor doesn't find data in main memory, where will it look next?
   a) Storage
   b) Shared cache
   c) Register file

## Activity 5.5

Answer the questions below.

1. Write the name and specifications of your processor.

   Name: Intel Core i5

   Cores: Quad-Core

   Clock: 3 GHz

2. List, in order, the memory components the processor checks when looking for data. Assume it has already checked the register file.

   D-cache, shared cache, main memory, storage

3. What is prefetching, and why does it work?

   Prefetching is a process by which the processor retrieves the data it is looking for and then fetches it along with adjacent data items. This is a strategy to improve performance. It is effective because if one data item is needed, there is a high chance that the next adjacent data item will be needed soon as well.

## Activity 5.6

Measure your computer's execution time. Type the program below, presented in Python. The program tasks the CPU with adding a long list of numbers and measures the time it takes the CPU to complete the task.

| Code | Explanation |
|---|---|
| `import time` | Imports the time module. |
| `def cpu_intensive_task():`<br>`        result = 0`<br>`    for i in range (1, 1000000):`<br>`        result = result + i`<br>`    return result` | Defines a function by the specified name. The function adds all the numbers in range between 1 and 1,000,000. |
| `start_time = time.time()` | Starts the timer. |
| `result = cpu_intensive_task()` | Runs the function. |
| `end_time = time.time()` | Stops the timer. |
| `execution_time = end_time - start_time` | Calculates the execution time. |
| `print ("Result:", result)` | Prints the sum. |
| `print ("Execution Time:", execution_time, "seconds")` | Prints the execution time. |

1. Run the program for different ranges of numbers and document the execution time for each.

Execution time for adding 1 to 1,000,000 : ____________________

Execution time for adding 1 to 10,000,000 : ____________________

Execution time for adding 1 to 100,000,000 : ____________________

________________________________________

________________________________________

# 2

# Computers & Beyond

# Lesson 1: Human-Computer Interaction

Human-Computer Interaction (HCI) is a multidisciplinary field that focuses on the study of the design of computers and, in particular, the interaction that occurs between humans and computers.

Initially, HCI focused only on computers, but this has changed over the years to currently include almost all forms of information technologies. Understanding how humans interact with computers and other technologies has greatly changed our lives. It has made computers and technology more user-friendly and intuitive, which has greatly broadened the range of people who can comfortably use these technologies.

## Human-Computer Interface

A system's interface is the boundary across which it interacts with the outside world. The system receives input and delivers output across its interface, which is physically determined by the system's input and output devices.

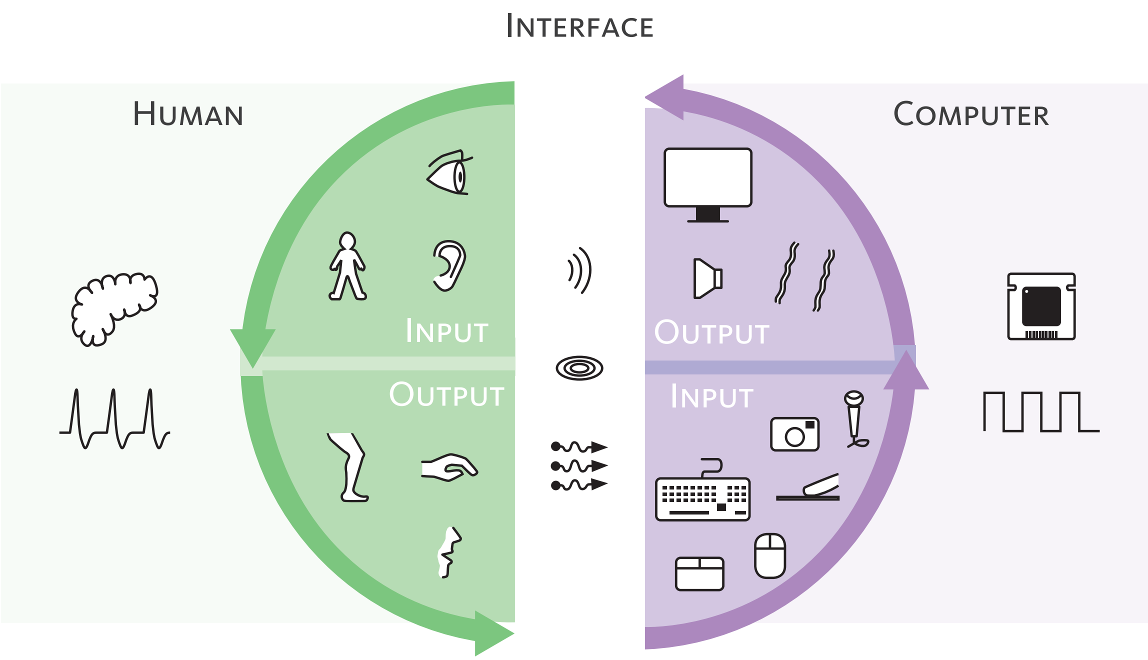
Two systems can interact if their interfaces are compatible and allow them to exchange information. Humans have designed computer interfaces to be compatible with their own. This allows humans to exchange information with computers and to interact with them. So when a human uses a computer, the boundary where the two systems meet and exchange signals is referred to as the human-computer interface.

> THE HUMAN-COMPUTER INTERFACE IS THE BOUNDARY ACROSS WHICH THE TWO SYSTEMS INTERACT.

## INPUT/OUTPUT LOOP

The input/output (I/O) loop is fundamental to human-computer interaction. The interface between the two systems allows signals to be passed back and forth. The output of one system becomes the input of the other, which in turn generates an output. This process repeats itself, creating an I/O loop across the interface. The I/O loop perpetuates until one of two conditions arises: 1) the task is completed, or 2) the loop is interrupted for some reason.

**HUMAN-COMPUTER INTERACTION IS A LOOP OF INPUTS AND OUTPUTS BEING EXCHANGED BY THE TWO SYSTEMS.**




The human-computer interface is established by the input and output components of each system. The user receives a signal emitted from the computer. The sensory organs convert it to action potentials that the human can process and then accordingly respond. The human's response, or output, then becomes the computer's input. The incoming signal emitted by the human is converted to electronic digital signals, which the computer uses to process and respond, creating a loop of inputs and outputs.

### EXAMPLE 3.1

When a user enters text into a word processor, the keyboard sends the input to the computer, which processes the input and displays the output on the screen. The user can then see the output and make changes, which are then sent back to the computer. The process repeats in a loop until the desired result is achieved, which is completing the essay, or the loop is interrupted by the user being called for dinner.

## PERIPHERALS

Peripheral devices are input and output devices that can be connected to a computer, extending the human-computer interface. They augment an existing functionality or add a new one that the computer could not previously perform. Peripheral devices can act as input, output or as both input and output devices (I/O devices).

> PERIPHERAL DEVICES EXPAND THE HUMAN-COMPUTER INTERFACE AND PROVIDE ADDITIONAL FUNCTIONALITIES.

### EXAMPLE 3.2

The audio headset is a peripheral output device. It connects to a computer and provides the user with privacy and clarity of sound. In some cases, the headset has a built-in microphone, making it an I/O device that can provide both input and output of sound. This device enhances an existing functionality by bringing the input and output devices closer to the mouth and ears of the user. This new interface provides a more direct and clear transmission of sound signals.

### Example 3.3

Another example of a peripheral device is the projector, a peripheral output device commonly used in classrooms. This technology outputs light to a much wider area than the computer screen, allowing for a much larger audience of viewers, making the interface not just with one person but several. The smartboard is another classroom peripheral device. It can both display content and allow users to manipulate it directly; it is an I/O device.

There are several peripheral devices that connect with a computer and extend its interface. Some devices connect through wired connections, while others have wireless connections.

| | |
|---|---|
| **Input Devices** | Camera - Webcam - Barcode reader - Scanner - Game controller - Microphone |
| **Output Devices** | Printer - 3D Printer - Plotter - Projector - Earphones - Speakers - Television |
| **I/O Devices** | Headset - Smartboard |

By extending the human-computer interface through peripheral devices, users can experience more interactions with computers, allowing them to perform a wide range of tasks across various contexts, such as work, education, entertainment and communication.

It is important to note that each of these devices has its own software, programs that contain the instructions that allow the peripheral device to exchange information with the computer. While this software may not be directly visible in the user interface, it does perform critical work behind the scenes. Without the software, the hardware will not work.

## Smart Machines

Embedded computers have impacted our world significantly, with processors and other digital components embedded in almost every machine, ranging from home appliances to automobiles.

Embedded computers allow us to enhance the machine's interface as well as provide additional functions that can be performed using the machine. Therefore, smart machines have an interface that provides an enhanced level of functionality and usability, ensuring that the machine performs its intended tasks to the satisfaction of its user.

**The Human-Machine Interface is the interface across which humans interact with machines.**

### The Smartphone

The first cellphone was released in 1973 by Motorola, the DynaTAC 8000X. It was large, weighing about 1 kg and costing over $3,900. As a device, it was purely a phone and could only make and receive calls.

This changed in 1996 with the release of the Nokia 9000 Communicator, the first phone to have an embedded computer. This cellphone provided internet capability, with web browsing and emailing as functionalities. However, it was not until 2007 that the Apple iPhone became the first smartphone to achieve success in the market.

Nowadays, smartphones are no longer just cellphones used for the sole purpose of making calls. Instead, smartphones have become general-purpose machines capable of performing a wide range of tasks. It is now considered a personal computer and not just a smartphone.

### The Cash Register

A cash register is a common example of a machine that contains an embedded computer. The human-machine interface of the register comprises:

| **Input Devices** | |
|---|---|
| **Barcode scanner** | It is used to scan the barcode on items. The barcode contains information like the name of the product and its price, which is input into the system. |
| **Keyboard** | It is used to input information, such as the price of items and method of payment. |

| **Output Devices** | |
|---|---|
| **Drawer** | It is used to store cash. An actuator moves and releases the drawer, allowing it to spring forward and open. |
| **Printer** | It is used to print receipts for customers. There are two actuators for this output device: 1) a motor to turn the role of paper and 2) a printhead that moves across printing line by line. |
| **Screen** | It is used to show the transaction information, such as the price of items, total cost and change due. |

Unlike personal computers that are intended to have a general purpose, the cash register is designed as a machine that serves a specific purpose. Therefore, its interface must be optimized to allow humans to perform the required tasks that serve that purpose. As such, the extent of the interaction between human and machine is limited to the tasks that need to occur at a point of sale, such as identifying items and prices, receiving payments and printing receipts.

## Robots

A robot is a machine capable of performing a variety of tasks autonomously or under varying degrees of control by an operator. A robot must interact with the world and humans; as such, it typically consists of input devices in the form of sensors, output devices in the form of actuators that provide movement across joints and a body that holds everything together. The design of the robot and its interface is dictated by the purpose it must serve.

> ROBOTS ARE MACHINES CAPABLE OF PERFORMING TASKS AUTONOMOUSLY OR WITH SOME HUMAN INTERVENTION.

Robots can be categorized depending on their degree of autonomy, mobility and purpose.

- Autonomy: refers to the degree to which a robot can act independently. An autonomous robot can make decisions on its own without human intervention. A semi-autonomous robot can make most decisions on its own. In comparison, a non-autonomous robot relies on a human operator to control its actions. Therefore, the embedded computer in each of these three cases has varying degrees of decision-making capabilities that can vary from making all decisions on its own to simply assisting in coordinating input and output, leaving the decisions to the human operator.
- Mobility: refers to a robot's capability of moving and changing location. Some robots are stationary, such as the robotic arms used in car factories, while other robots are mobile. Mobile robots have varying degrees of mobility. Some can only negotiate a smooth, flat surface, such as robotic vacuum cleaners. In contrast others, such as some advanced humanoid robots, can walk outdoors on uneven terrain, run, jump and even do somersaults.
- Purpose: refers to the task or set of tasks a robot is intended to perform. Some tasks are menial and highly repetitive, such as those required in

manufacturing. While other tasks require entering difficult or dangerous locations, such as exploring underwater caves or finding people trapped by fire. Still, other robots, humanoid robots, are designed to serve the purpose of satisfying the engineering achievement of creating machines very much in our own likeness.

Robots have brought about both benefits and concerns. On the one hand, robots have taken over jobs that are unpleasant or even unsafe for humans. Menial and repetitive tasks tend to make jobs unpleasant, and robots have been designed to perform such tasks as:

1. Cleaning floors and windows.
2. Sorting and organizing items in warehouses.
3. Tightening screws or fastening components on assembly lines.
4. Packaging products in factories.

Robots have also taken over jobs and tasks that are dangerous, thereby keeping humans out of harm's way.

1. Handling hazardous materials or chemicals in environments such as nuclear power plants or chemical facilities.
2. Conducting underwater inspections or repairs in deep-sea environments.
3. Performing tasks in areas with high levels of radiation or toxic substances.
4. Exploring hostile or extreme environments like volcanoes, disaster sites and even other planets.
5. Conducting military operations, such as bomb disposal or surveillance in conflict zones.

However, robots have also raised several concerns. One concern is the displacement of humans from their jobs due to the increasing advancement and capability of robots, which can now perform tasks traditionally done by humans. According to a study by the McKinsey Global Institute, up to 800 million jobs could be lost to automation by 2030. This is equivalent to 14% of the global workforce.

Other concerns involve ethical implications. As robots become more independent, will they behave ethically and responsibly? Will they prioritize the best interests of humans or potentially cause them harm? Will they adhere to the ethical standards we hold ourselves to?

In 1942, Isaac Asimov, a science fiction author, introduced the three laws of robotics.

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given to it by human beings, except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Although these laws were introduced in science fiction publications, they are based on the ethical considerations that humans are concerned about.

Overall, it is important to carefully balance the benefits of robotics while addressing these concerns as we move towards a more automated future—a future in which humans interact with computers, machines, and robots.

## SUMMARY

Human-Computer Interaction is a multidisciplinary field that focuses on the study of the design of computers and its impact on the interaction that occurs between humans and computers. HCI include computers and other information technologies, particularly those that have an embedded computer.

The Human-Computer Interface is the boundary across which the two systems interact. For the computer, this includes all its input and output devices. For the human, this includes the sensory organs and hands. Interaction occurs in an input/output loop that results across the interface.

The interface of the computer can be expanded by installing peripheral devices. These devices are input and output devices that augment an existing functionality or add a new one. Peripherals enhance human-computer interaction.

Embedded computers have made a wide variety of machines and electronic devices smart. Introducing embedded computers enhances the human-machine interface by adding new functionalities and improving usability.

Robots are machines capable of performing tasks autonomously or with some human intervention. They are categorized depending on their autonomy, mobility and purpose. Robots bring the benefits of taking over menial and repetitive jobs, as well as dangerous ones. The concerns are that robots are taking over too many jobs leaving people jobless, and that they may not in the future act ethically and in our best interest.


### Career Tips

Mechatronics is an interdisciplinary field that combines mechanical engineering, electrical engineering, and computer science. Mechatronic systems can be found in various applications, such as manufacturing, transportation, healthcare and robotics.

## Activity 1.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F | STATEMENT |
|---|---|
| T | Humans interact with computers across an interface. |
| F | Interaction occurs in one direction, from human to computer. |
| T | The I/O loop is a cycle of exchanges of signals. |
| F | The computer's interface cannot be expanded. |
| T | Peripheral devices are input and output devices that can expand the interface. |
| T | Embedded computers make machines smart. |
| T | The smartphone used to be a cellphone with an embedded computer; now, it is a general-purpose computer. |
| F | All robots perform tasks autonomously. |
| T | Taking over menial jobs is considered a benefit of robots. |
| F | Isaac Asimov was a scientist. |

## Activity 1.2

Answer the questions below.

1. Which component is not part of the interface?
   a) Processor
   b) Mouse
   c) Keyboard

2. Which device acts as a peripheral device for a tablet?
   a) Touchscreen
   b) Memory
   c) Digital pen

3. Which device acts as a peripheral device for a laptop?
   a) Screen
   b) Printer
   c) Keyboard

4. Which device acts as a peripheral device for a smartphone?
   a) Wireless earphones
   b) Touchscreen
   c) Processor

5. Which device acts as an I/O device?
   a) Smartboard
   b) Keyboard
   c) Whiteboard

## Activity 1.3

Complete the sentences below.

1. The purpose of an interface in a machine is _____ .
   a) to control the machine's operations
   b) to provide functionality and usability
   c) to connect the machine to the internet

2. For a microwave oven, an embedded computer adds the functionality of ____ .
   a) cooking food
   b) heating beverages
   c) programmable cooking modes

3. For an air conditioner, an embedded computer adds the functionality of ____ .
   a) automatically adjusting room temperature
   b) blowing hot air to warm up a room
   c) blowing cold air to cool down a room

4. An example of a modern human-machine interface is ____ .
   a) dials and knobs
   b) switches and buttons
   c) graphical user interface

5. The first to have an embedded computer was the ____ .
   a) Nokia 9000 Communicator
   b) Motorola DynaTAC 8000X
   c) Apple iPhone

## Activity 1.4

Answer the questions, and complete the sentences below.

1. Which of the following is an example of a task that a robot can perform in a factory?
   a) Conducting underwater inspections
   b) Exploring hostile environments
   c) Tightening screws on an assembly line

2. Which category of robot can make decisions on its own without human intervention?
   a) Semi-autonomous robot
   b) Non-autonomous robot
   c) Autonomous robot

3. A robotic vacuum cleaner that does not require human control is best described as ____ .
   a) stationary and non-autonomous
   b) stationary and autonomous
   c) mobile and autonomous

4. A robotic arm on an assembly line that requires some human control is best described as ____ .
   a) stationary and non-autonomous
   b) stationary and semi-autonomous
   c) mobile and autonomous

5. What is one potential benefit of using robots in hazardous environments?
   a) Improved job opportunity
   b) Improved safety
   c) Improved cost savings

## Activity 1.5

Answer the questions below.

1. Which types of computers do you use? List the devices that constitute each computer's interface.

   Desktop: screen, keyboard, mouse, speaker, microphone, camera

   Laptop: screen, keyboard, touchpad, speaker, microphone, camera

   Tablet / smartphone: touchscreen, speaker, microphone, camera

2. What peripheral devices have you added to your computer's interface?

   Wireless / wired earphone +\- microphone, webcam, printer, scanner

3. Of the peripheral devices you use, what functionality does each device provide you? How do you find that beneficial?

   The printer allows me to print homework from the comfort of my home.

   Earphones allow me to listen at higher volumes without bothering others.

## Activity 1.6

Write your answer(s) on the lines provided below.

1. List smart machines found at home or school.

   Smart (interactive) board

   Smart thermostat or air conditioning unit

   Smart TV

2. Most modern elevators have embedded computers that perform various tasks to ensure the safe and smooth operation of the elevator. Describe the user interface of a modern elevator.

   Outside the elevator, there is a screen to show the floor number at which the cabin is currently at, as well as call buttons to call for the cabin. Inside the elevator, there is also a screen to indicate the current floor, a control panel of floor numbers and a speaker to sound arrival at the requested floor.

3. "A smartphone is a computer, not a cellphone with an embedded computer." Is this statement true? Explain.

   This statement is mostly true as it depends on the main use or purpose of the phone. In general, the smartphone is mainly used for its general-purpose capabilities: navigating the web and social media, communicating via text or audiovisual messages, reviewing and completing assignments, attending online classes, playing games, listening to music, watching videos, etc.

## Activity 1.7

Write your answer(s) on the lines provided below.

1. List three tasks robotic arms perform in factories.

   Load and unload items

   Weld and solder

   Insert and tighten screws

2. Investigate the rovers sent to explore the surface of Mars. Choose one and list three of the various functions it can perform based on its sensors and actuators.

   Curiosity landed on Mars in August 2012.

   It has six wheel actuators that can move across rocky terrain.

   It has sensors to monitor temperature, pressure, wind speed and radiation.

   It has a robotic arm actuator to collect samples.

3. "Cars that can drive themselves are robots." Is this statement true? Explain. .

   If we define robots as machines capable of performing tasks autonomously or under varying degrees of control, then self-driving cars can be considered robots. These cars have sensors, actuators and onboard computers to perceive the environment, make decisions and control their movements. They can operate without direct human intervention.

# Lesson 2: Computer Interaction

Computers interact with each other by sending and receiving information just like their internal components. Information exchange follows similar principles as those that apply inside the computer. It, therefore, follows that there must be signals, a medium through which signals will travel, a way to organize these transmissions and addresses that indicate source and destination of transmitted information.

## Networks

A network is a system of interconnected computers and devices that are capable of exchanging information. Networks vary depending on their size, area of coverage, and the technology used to transmit information. The smallest network may consist of two devices in one room, while the largest network is the internet a system of communication that consists of billions of devices and spans the entire globe.

The method of information transfer depends on the technology used. In some cases, transmission occurs through wired connections where data is transmitted as electrical or light signals across wires and cables. Conversely, there is wireless transmission, which uses radio waves to send signals through the air.

A NETWORK IS A SYSTEM OF INTERCONNECTED COMPUTERS AND DEVICES CAPABLE OF EXCHANGING INFORMATION.

### Personal Area Network

The smallest network is a personal area network, PAN. It typically consists of a personal computer connected to peripherals and other devices in its vicinity. Any personal computer can be used to establish a PAN.

Typically, the PAN is centered around an individual and the personal computer being used. Connections in a PAN can be wired, wireless or a combination of both.

A PERSONAL AREA NETWORK IS A NETWORK CENTERED AROUND AN INDIVIDUAL.

### Example 2.1

Sarah sat in her school's library. She had some homework she needed to finish and print. She was using her tablet, which was connected to a digital pen, her earbuds and the library's printer. Sarah's devices were connected in PAN through wireless connections.

These short-range wireless connections are based on Bluetooth technology. Each device has a Bluetooth antenna capable of sending and receiving radio wave signals. Most Bluetooth antennas are only capable of a range of about 10 meters, they are referred to as Class 2 Bluetooth antennas and are a good choice for short-range communications. On the other hand, Class 1 Bluetooth antennas are capable of transmissions up to a range of 100 meters.

### Example 2.2

The librarian also had her own PAN. But in her case, the PAN was wired rather than wireless. Her desktop was connected to a printer, scanner, and a smartboard by cables. This allowed her desktop to exchange information with devices in its vicinity.

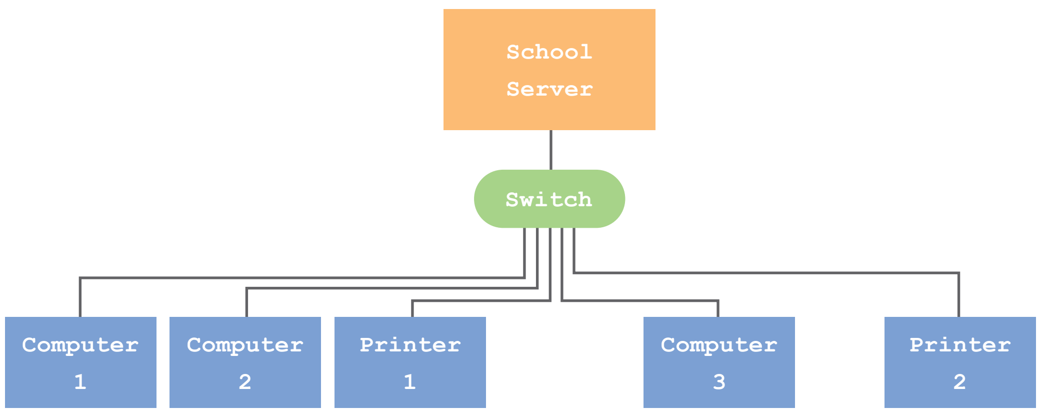
## Local Area Network

A local area network, or LAN, is larger than a PAN and covers a wider area such as a home, office building, or school campus. A LAN is more extensive than a PAN and serves a larger group of users.

## Example 2.3

The librarian can use her desktop to connect to other computers at school. This includes the school server which holds the school's database and information management system. By having access to the server, the librarian gains information about the inventory of books in the library, which books have been checked out and by who, as well as overdue to books. This information is shared with the school principal and teachers. Even students can login to view their list of checked out and overdue books.

Our hypothetical school LAN consists of a central server and all the computers connected to it at school. Connections are established by wires and cables, specifically Ethernet cables. This is a wired LAN, since all its connections are in the form of cables.




The figure shows the school as one local area network that consists of 3 computers, 2 printers and a server all of which connect and communicate via one switch.

Switches are networking devices that connect multiple computers and other devices within a network. They receive information from one device and forward it to the appropriate destination within a LAN.

> A SWITCH IS A NETWORK DEVICE THAT CONNECTS MULTIPLE COMPUTERS AND DEVICES WITHIN A LAN.

## Wide Area Network

Wide area networks, WAN, are much larger. They vary in size of coverage from city-wide to as wide as our whole planet. The internet is the largest and most widely known example of a WAN. It is a global network of interconnected smaller networks that spans across the entire planet.

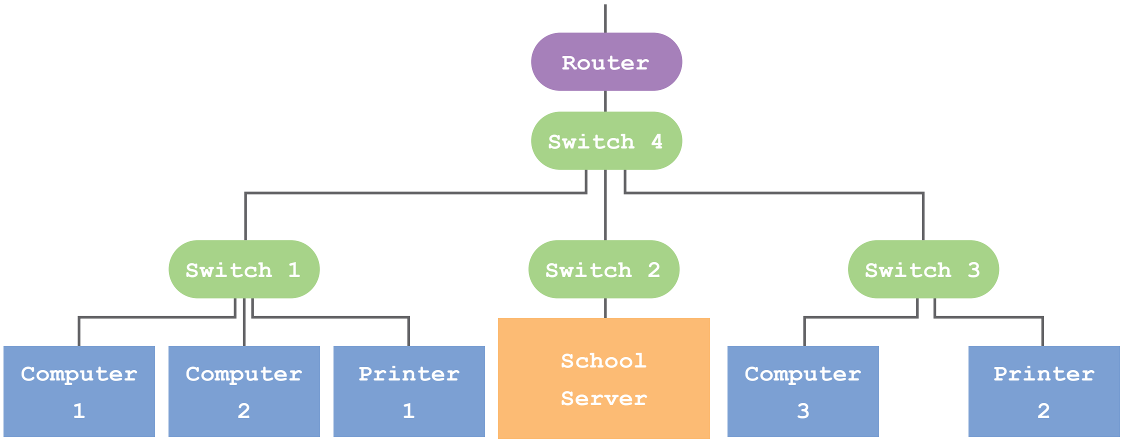
While a switch is capable of handle connections inside a LAN, it cannot perform the tasks necessary to handle connections outside its LAN. For such tasks, a router is needed. Routers act as gateways to the internet and its traffic controllers. When information has to pass outside a LAN, the router identify its destination and the best path to get there.

> A ROUTER IS A NETWORK DEVICE THAT ACTS AS A GATEWAY TO THE INTERNET AND AS ITS TRAFFIC CONTROLLER.

## Example 2.4

In the previous example, we assumed a small school. A larger school would have many more computers, printers and other devices. Therefore, instead of having a large LAN, it is more appropriate to divide the school into smaller LANs.

Each section of the school would have its own switch and form its own LAN. To connect these smaller LANs, another switch would be needed forming one larger LAN. The addition of a router provides the school with access to the internet.




In this figure, the whole school campus is depicted as 1 larger LAN consisting of 3 smaller LANs. Each smaller LAN is established by a switch. All three switches are connected by Switch 4. The router provides connectivity to the internet.

The internet is considered to be the widest network. The backbone of the internet is a complex series of submarine fiber optic cables that connect all the continents and transmit information in the form of light signals. As of early 2023, there are 552 active and planned submarine fiber optic cables used for the internet. These cables span over 1.4 million kilometers and carry more than 95% of the world's international internet traffic.

**The INTERNET is a complex system of networks working together to provide communication across the globe.**

But the internet does not solely rely on fiber optic cables for signal transmission, it also employs satellite and telephone communication networks, such as land lines and the cellphone grid.

## Adapters

The three common forms of energy used to as communication signals are electrical, light and radio waves. Electrical signals and light signals are used to transmit information in wires, typically copper wires for electrical signals and fiber optic cables for light, while radio signals are used to transmit information through the air. Each type of signal requires a specific type of hardware to handle its conversion but in general, communication channels are broadly categorized as wired and wireless.

> THERE ARE TWO BROAD CATEGORIES OF COMMUNICATION CHANNELS WIRED AND WIRELESS.

Since the signals of communication traveling in a network are different from the electronic digital signals of the computer, signals must be converted. This is the task of the network adapter, a hardware component of the computer interface. The job of the adapter is to convert the internal signals of the computer to the external signals of the network and vice versa. As such, the adapter is considered an I/O device.

> EACH ADAPTER CONVERTS INTERNAL SIGNALS TO A SPECIFIC TYPE OF EXTERNAL SIGNAL INTENDED FOR TRANSMISSION.

### Wired Adapters

Each of the adapters converts electronic digital signals into a different type of transmission signal. They typically connect via a port.

1. **Land line Modems** convert between the electronic digital signals of the

computer and the continuous (analog) electrical signals used to transfer information through telephone land lines. They connect LANs to the internet.

2. **Ethernet network interface card, NIC,** converts between the electronic digital signals of the computer and another electronic digital signals used to transfer information through long cables. It also connects devices in LANs and the connect LANs to the internet.
3. **Fiber optic NIC** typically includes a fiber optic transceiver, which converts the electrical signals used by the device into optical signals that can be transmitted over the fiber optic cable. It is commonly used in data centers where high-speed data transmission and reliability are critical.

**Wireless Adapters**

The following three adapters convert between the electronic digital signals of the computer and radio waves, which are used to transfer information through the air. They capture and emit signals using an antenna. One main difference between them is their range of transmission.

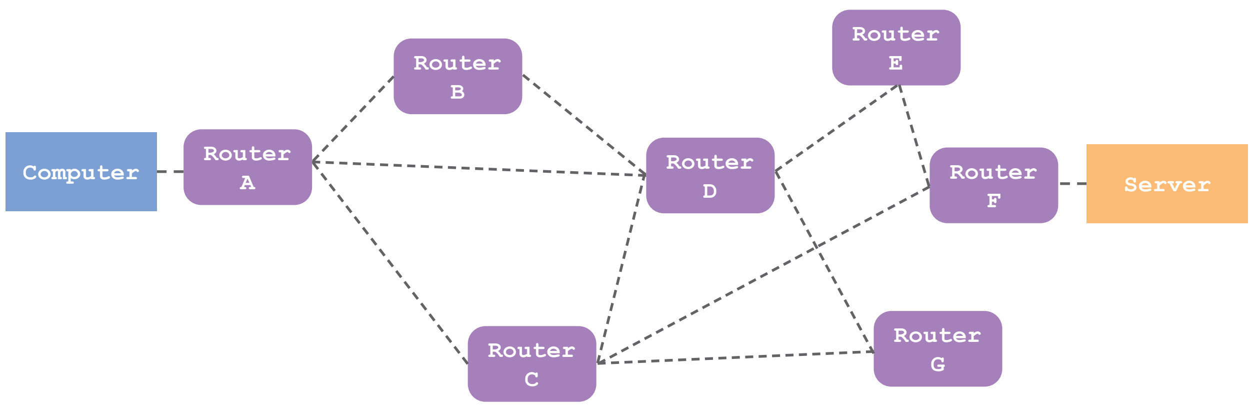
1. **Cellular Modem** is included as a component in smartphones and some tablets and is designed to transmit signals over several kilometers and is used to connect to the internet via cellular networks.
2. **WiFi NIC** can transmit signals up to 45 meters indoors and up to 150 meters outdoors. It is typically included as an internal component in most personal computers or can be externally connected to one. It is used to connect to LANs and the internet.
3. **Bluetooth Adapter** can transmit special type of radio wave signal, referred to as frequency-hopping spread spectrum (FHSS) used for short distance of upto 10 meters indoors and is used to connect to nearby devices. This allows the computer to established a personal area network between devices in close proximity such as Bluetooth-enabled earphones, headphones, speakers, television sets and other personal computers.

## Best Route

The internet is a busy place. Once the signal leaves the source computer, depending on the distance it must travel, it will likely have to pass through several routers before arriving at its destination. The path the signal takes from source to destination is determined by routers.

### Example 2.4

A computer sends a request to a server. Router A is acting as the computer's gateway to the internet. Therefore, Router A needs to know the best route to send the signal to the server.




One method of determining the best route for delivering a signal to its destination is to count the number of routers the signal must pass through before arriving. This is referred to as the hop count. Every router keeps a routing database of all the possible routes and their metrics, such as the hop count.

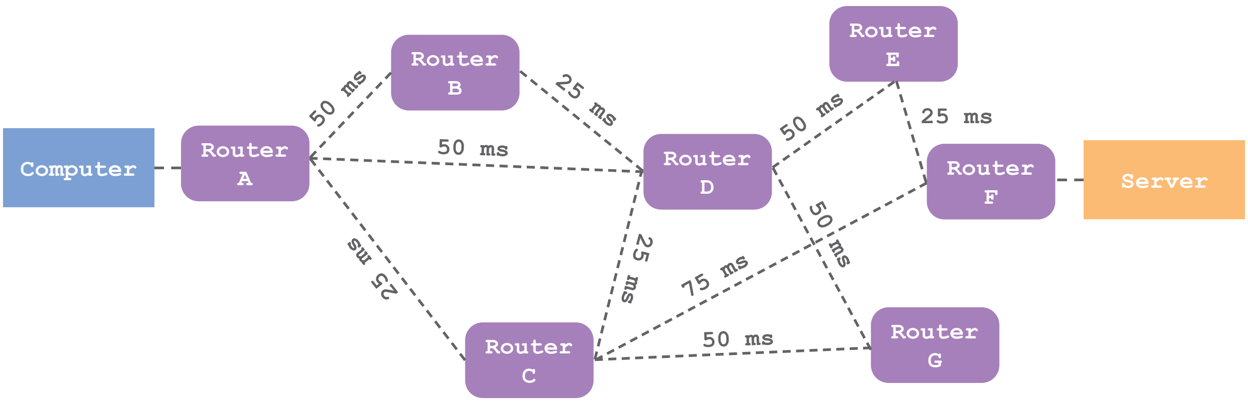
Based on this metric, Router A would take route option 3 because it has the fewest hops from source to destination.

| Route Option | Hop 1 | Hop 2 | Hop 3 | Hop 4 | Hop Count |
|---|---|---|---|---|---|
| 1 | B | D | E | F | 4 |
| 2 | D | E | F | | 3 |
| 3 | C | F | | | 2 |

Another metric routers can use for determining the best paths is latency, or the time it takes for a signal to travel from one point in the network to another. One common method for measuring latency is the 'ping', which measures the round-trip time for a signal to travel from the source to the destination and back.

In some network configurations, routers may ping their neighboring routers. These latency measurements are then shared with other routers, which use them to choose the best route for sending signals.

A long ping indicates either that the distance between the source and destination is long or that there are delays in processing the signal, due to either heavy traffic or a slow router.




The figure shows the latency time in milliseconds between routers. Router A would require such information to determine the best route to its destination, the server. Total latency is estimated as the sum of all intermediary latencies.

| Route Option | Hop 1 | Hop 2 | Hop 3 | Hop 4 | Total Latency |
|---|---|---|---|---|---|
| 1 | B: 50 | D: 25 | E: 50 | F: 25 | 150 ms |
| 2 | D: 50 | E: 50 | F: 25 | | 125 ms |
| 3 | C: 25 | F: 75 | | | 100 ms |

According to the database information, route option 3 remains the best route to the destination. This route not only has the fewest hops, 2, but also has the shortest latency, 100 ms.

> **The best path is determined using metrics such as hop count and latency.**

What if Router C becomes congested with many requests to route signals through? What if the latency between Router A and Router C becomes 100 ms? In such a case, route option 3 would not seem to be so desirable having a total latency of 175 ms, the highest among all three options. Instead, route option 2 becomes the best route to the destination.

Some cases are not as clear. What if the latency between Router A and Router C becomes 50 ms? Now route options 2 and 3 have the same total latency, 125 ms.

In general, routers differ based on the decision-making program used to determine the best route. Some programs place greater importance on hop count, while others favor latency, and still others may use yet another metric.

## Trust

When transmitting signals across the internet, or between any two devices, trust must be established. The user must be able to trust that the signals are transmitted without any unintentional error and without any intentional change or theft. The transmission must maintain the integrity and security of the message it is carrying.

> The user trusts that the transmission will maintain the integrity and security of the message.

## Errors

Errors in data during transmission can happen for many reasons, which, for the most part, are generally related to the phenomena of attenuation and interference.

Loss of strength, or attenuation, can occur with any form of signal in any medium if that signal is required to travel great distances. The longer the distance, the more likely it is that the signal will lose its strength, whether it is light in a fiber optic cable, electricity in a copper wire, or radio waves traveling through air.

> Errors in data transmission happen generally due to attenuation or interference.

Another common cause of errors is interference. There are several reasons for interference, such as storms, buildings, neighboring wires, and cosmic and solar radiation.

Storms and severe weather can interfere with and disrupt radio waves, while obstacles such as buildings can block or reflect radio waves, creating multiple paths of waves that interfere with each other.

Nearby electrical wires or electronic devices can generate electromagnetic radiation that interferes with the signals, particularly electrical signals in copper wires.

Other forms of energy, such as cosmic and solar radiation, can be problematic for satellites in low earth orbit and other devices, such as computers and routers located at high altitudes.

## Protection

There are methods used to protect against errors and decrease the risk of their occurrence. For signal attenuation, repeaters are placed at a certain distance along the path of propagation of the signal, whether it is light in fiber-optic cables, electricity in copper wires, or radio waves traveling through the air. Repeaters are devices that capture, amplify, and retransmit the same signal, so signals whose energy is weakening become restrengthened.

Shielding works to protect against interference in copper wires and satellites. For copper wires, adding an insulating layer around the wires helps shield the signals inside from outer, neighboring interference. As for satellites, various types of shielding and design techniques are used to protect the internal electronic circuits from cosmic, and to some extent, solar radiation.

Redundancy in data means maintaining additional copies of the original data.

A simple and common strategy used to protect from complete loss of information is redundancy, a term which means having more of the same. When any transmission occurs, whether it is between components of a computer or between computers, only a copy of the original information is sent. The original information, or file, stays put. It is copied, and that copy is the signal that is transmitted. If the transmitted information is lost or disrupted, the original information is still safe and can be recopied and retransmitted.

## Detection

When a signal is transmitted, the receiving computer sends back an “All is Good” signal to the transmitting computer, which basically means that “I have received your message.” This is referred to as an acknowledgement. If the acknowledgement is not received, the sender will assume that the message was lost and will transmit another copy.

If the message does arrive, then it must be checked for errors. There are several methods of error checking, one common method is a checksum. A checksum is a value calculated from the data that is sent along with the data itself. The receiver can calculate its own checksum from the received data and compare it to the one that was sent. If they match, the data is correct; if not, an error occurred.

**The acknowledgement and the checksum are two methods of detecting errors in data transmission.**

## Example 2.5

The checksum works at the byte level. Let's assume that the message "Hello" is going to be sent. Every character in the alphabet has an 8-bit binary code to represent it. This byte actually represents the number assigned to the character.

| | H | e | l | l | o | |
|---|---|---|---|---|---|---|
| | 0100 1000 | 0110 0101 | 0110 1100 | 0110 1100 | 0110 1111 | |
| | 72 | 101 | 108 | 108 | 111 | |
| **Checksum** | | | **72 + 101 + 108 + 108 + 111 =** | | | **500** |

In this case, the values for each of the 5 characters in the word "Hello" are added. The sum, referred to as the checksum, is sent along with the message. The computer on the receiving end will repeat the checksum calculation on the message it received. If the receiver's checksum is equal to the sender's checksum then the signals are all correct and no errors have occurred during transmission. The message the receiver has is the same as the one sent by the sender.

## Encryption

Just like we expect to maintain the integrity of a message during transmission, we also expect that the transmission will be secure. The message should arrive error-free and it should not be amenable to hacking. Hacking means gaining unauthorized access to a computer, network or transmission, with the intention of changing, stealing or destroying data.

> **Hacking is gaining unauthorized access to a computer, network or transmission, to change, steal or destroy information.**

One way to keep a message secure from hackers is to encrypt it. Encryption changes the message in such a way as only the sender and receiver can read it. So even if hackers gained access to the message they would not know what to do with it because they cannot read it.

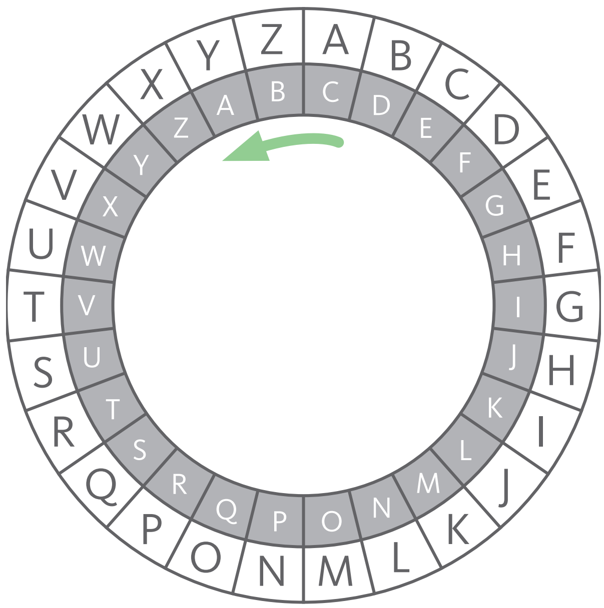
> **Encryption is the process of rewriting a message in a way that can only be read by a user who has the key.**

Encryption is a process that transforms a message from a readable form, plaintext, into an unreadable form, ciphertext. The transformation is based on a set of steps, an algorithm, that determines how the message will be changed. The encryption key determines by how much the message will be changed.

## EXAMPLE 2.6

The Caesar cipher is a simple encryption algorithm that dates back to the time of the Roman Empire. It is named after Julius Caesar, who is said to have used it to protect the confidentiality of his military messages. This allowed Caesar to send messages across the span of the empire with the security that should a message fall into the wrong hands it could not be read.

The Caesar cipher works by shifting each letter in the plaintext by a fixed number of positions in the alphabet. A Caesar cipher disk, is a tool used to encrypt and decrypt messages using the Caesar cipher. It consists of a circular disk or wheel with the alphabet printed on it, typically in two concentric rings. The outer ring represents the plaintext alphabet, while the inner ring represents the corresponding ciphertext alphabet after applying a specific shift value.




A Caesar cipher disk whose inner ring is turned 2 positions counterclockwise. 'A' on the outer ring corresponds to 'C' on the inner ring, 'B' corresponds to 'D', 'C' corresponds to 'E', and so on. To encrypt a message replace each letter of the plaintext on the outer ring with the corresponding letter of the ciphertext on the inner ring. In this case, the word "HELLO" would be encrypted as "JGNNQ".

The encryption algorithm would be to shift the inner wheel in a certain direction by a certain number of positions. The key would be the direction and number of positions to shift, which, in the case of this example, is 2 positions counterclockwise.

Encryption algorithms and keys are far more elaborate when considering computer-to-computer data transmission. Every communication application would have its own encryption algorithms and keys that it uses to ensure that all messages and information sent by it are secure.

Although this presents a very simplified view of computer-to-computer interaction, or data transmission, it does emphasize the importance of establishing trust by ensuring the integrity and security of transmitted information.

## Career Tips

Cybersecurity is a rapidly evolving field that encompasses various aspects of protecting computer systems, networks, and data from unauthorized access, attacks, and damage.

Ethical hackers, also known as ‘white hat’ hackers are professionals who are authorized to identify vulnerabilities and weaknesses in computer systems, networks, or applications with the goal of strengthening their security measures.

## Summary

A network is a system of interconnected computers and devices capable of exchanging information. Networks vary depending on size, area of coverage, and the technology used to transmit information.

The smallest network is a personal area network, PAN, which is a network centered around an individual. Any personal computer can be used to set up a PAN. It can be wired, wireless or a combination of both.

A local area network, LAN, is larger that a PAN and covers a wider area and serves a larger number of users. A switch is a network device that connects multiple computers and devices within a LAN.

Wide area networks, WAN, vary in size and coverage. They consist of multiple LANs and can provide coverage for a city. The largest WAN is the internet. The internet is a complex system of networks working together to provide communication across the globe. A router is a network device that acts as a gateway to the internet and as its traffic controller.

The adapter is a computer component that converts the internal signals of the computer to the external signals of the network and vice versa. Each adapter converts internal signals to a specific type of external signal intended for transmission. There are two broad categories of communication channels wired and wireless. Wired adapters are: land line modems, ethernet NICs, and fiber optic NICs. Wireless adapters are: cellular modems, WiFi NICs, and Bluetooth adapters.

Routers are tasked with identifying the best route between source and destination. The best path is determined using metrics such as hop count and latency.

Transmission must maintain the integrity and security of the message. Errors in data transmission happen generally due to attenuation or interference. To protect against signal attenuation, repeaters are used to amplify, and retransmit signals whose energy is weakening. To protect against interference, shielding is used around copper wires and on satellites.

The acknowledgement and the checksum are two methods of detecting errors in data transmission. If the acknowledgement is not received, the sender will assume that the message was lost and will retransmit. If the receiver's checksum is not the same as that of the sender, then there is an error in the message.

Encryption is the process of rewriting a message in a way that can only be read by a user who has the key. Every encryption requires an encryption algorithm and key to transform the message from readable to unreadable forms.

## Activity 2.1

Read the following statements and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F | STATEMENT |
|---|---|
| F | A network does not act as a system. |
| T | The smallest network is a personal area network. |
| T | The internet is a complex system of networks. |
| T | The adapter is considered an I/O device. |
| F | Only two forms of energy are used to convey signals: light and electricity. |
| T | Network interface cards and modems are different types of adapters. |
| F | The best route is determined based only on the hop count. |
| F | Errors in data transmission happen only due to attenuation. |
| T | The checksum is one method to detect errors. |
| F | A key is not required to encrypt a message. |

## Activity 2.2

Complete the sentences below.

1. A network is _____ .
   a) a device that stores information
   b) a technology used for sending signals
   c) a system of interconnected computers and devices capable of exchanging information

2. The internet transmits information ____ .
   a) only through wired connections
   b) only through wireless transmissions
   c) using a combination of both wired and wireless transmissions.

3. The primary function of a switch in a network is to ____ .
   a) serve as a power supply.
   b) receive and forward signals within a LAN
   c) connect a LAN to the internet

4. The device used to handle connections outside a LAN is ____ .
   a) a server
   b) a switch
   c) a router

5. The backbone of the internet is formed by ____ .
   a) a complex series of submarine fiber optic cables
   b) satellite and telephone communication networks
   c) routers and switches

## Activity 2.3

Answer the questions below.

1. What are the three common forms of energy used for communication signals?
   a) Magnetic, sound and light
   b) Electrical, light and radio waves
   c) Heat, mechanical and electrical

2. What are the two broad categories of communication channels?
   a) Internal and external
   b) Visible and invisible
   c) Wired and wireless

3. What is the primary function of a network adapter?
   a) To convert between internal computer signals and external network signals.
   b) To convert AC power to DC power.
   c) To increase the speed of the computer processor.

4. Land line Modems convert electronic digital signals to what kind of signals?
   a) Radio waves
   b) Continuous (analog) electrical signals
   c) Frequency-hopping spread spectrum signals

5. Bluetooth Adapters can transmit a special type of radio wave signal referred to as?
   a) Frequency-hopping spread spectrum (FHSS)
   b) Continuous wave spectrum (CWS)
   c) Modulated wave spectrum (MWS)

## Activity 2.4

Complete the sentences below.

1. The role of a router in internet connectivity is _____ .
   a) to send signals directly from the source computer to the destination server.
   b) to store data from the source computer.
   c) to determine the path the signal takes from source to destination.

2. Every router keeps ____ .
   a) a record of all the users in the network.
   b) a routing database of all the possible routes and their metrics.
   c) a list of all the signals it has processed.

3. In networking, a ping is used for ____ .
   a) measuring the round-trip time for a signal to travel from the source to the destination and back
   b) counting the number of routers the signal has passed through
   c) identifying faults in the network

4. In networking, latency is the ____ .
   a) time it takes for a signal to go and come back
   b) distance between source and destination
   c) time it takes for a signal to arrive at its destination

5. In general, routing algorithms use ____ .
   a) different metrics
   b) only hop count as a metric
   c) the same metrics

## Activity 2.5

Answer the questions below.

1. What can cause signal attenuation during transmission?
   a) The strength of the device sending the signal
   b) The distance the signal has to travel
   c) The user's trust in the transmission system

2. What are some common causes of interference during signal transmission?
   a) Storms and neighboring wires
   b) Rainbows and moonlight
   c) Animals and plants

3. What is the purpose of a repeater in signal transmission?
   a) To create a copy of the signal
   b) To capture, amplify, and retransmit the same signal
   c) To protect the signal from interference

4. What is the purpose of an acknowledgement in signal transmission?
   a) To inform the sender that the message was sent
   b) To inform the sender that the message was received
   c) To inform the sender that the message was encrypted

5. What is encryption used for in signal transmission?
   a) To increase the speed of transmission
   b) To reduce errors in transmission
   c) To protect the data from hackers

## Activity 2.6

Answer the questions below.

1. Give an example of a PAN you use in your daily life.

   Answers will vary. Example: My smartphone connected to my Bluetooth headphones and smartwatch.

2. Give an example of a LAN you use in your daily life.

   Answers will vary. Example: My computer science classroom has a switch that connects the desktops to each other and to the school server allowing them to share resources.

3. How do routers function in a network?

   Routers act as gateways to the internet and control its traffic. When information has to pass outside a LAN, the router identifies its destination and the best path to get there.

## Activity 2.7

Answer the questions below.

1. How is the function of a Fiber optic NIC different from an Ethernet NIC?

A Fiber optic NIC converts the electrical signals used by the device into optical signals that can be transmitted over the fiber optic cable, while an Ethernet NIC converts between the electronic digital signals of the computer and another electronic digital signals used to transfer information through long cables.

2. Why are Fiber optic NICs commonly used in data centers?

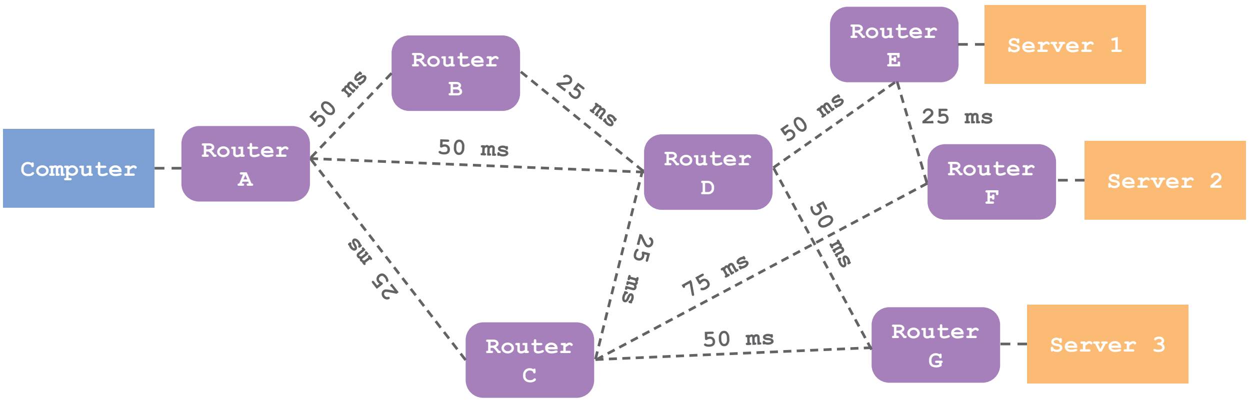
Fiber optic NICs are commonly used in data centers because they offer high-speed data transmission and reliability, which are critical in data center. environments

3. Examine your personal computer and a desktop computer at school. List the various network adapters these computers have.

The answers will vary widely depending on the computers being examined.

## Activity 2.8

Use the network diagram to answer the questions below.




1. Complete the routing database below between Computer and Server 1.

| Route Option | Hop 1 | Hop 2 | Hop 3 | Hop 4 | Total Latency |
|---|---|---|---|---|---|
| 1 | B: 50 | D: 25 | E: 50 | | 125 ms |
| 2 | D: 50 | E: 50 | | | 100 ms |
| 3 | C: 25 | D: 25 | E: 50 | | 100 ms |
| 4 | C: 25 | F: 75 | E: 25 | | 125 ms |
| 5 | C: 25 | G: 50 | D: 50 | E: 50 | 175 ms |

2. Complete the routing database below between Computer and Server 2.

| Route Option | Hop 1 | Hop 2 | Hop 3 | Hop 4 | Total Latency |
|---|---|---|---|---|---|
| 6 | B: 50 | D: 25 | E: 50 | F: 25 | 150 ms |
| 7 | D: 50 | E: 50 | F: 25 | | 125 ms |
| 8 | C: 25 | D: 25 | E: 50 | F: 25 | 125 ms |
| 9 | C: 25 | F: 75 | | | 100 ms |

3. Complete the routing database below between Computer and Server 3.

| Route Option | Hop 1 | Hop 2 | Hop 3 | Hop 4 | Total Latency |
|---|---|---|---|---|---|
| 10 | B: 50 | D: 25 | G: 50 | | 125 ms |
| 11 | D: 50 | G: 50 | | | 100 ms |
| 12 | C: 25 | G: 50 | | | 75 ms |

4. Based only on hop counts, list the best route to each server. Use latency to resolve a tie.

Computer to Server 1: Route Option 2

Computer to Server 2: Route Option 9

Computer to Server 3: Route Option 11 and 12, based on latency 12 wins.

5. Based only on latency, list the best route to each server. Use hop count to resolve a tie.

Computer to Server 1: Route Option 2 and 3, based on hop count 2 wins.

Computer to Server 2: Route Option 9

Computer to Server 3: Route Option 12

6. Use Command Prompt or Terminal to measure ping for the web address, www.google.com, and any other of your choosing.

ping <web address>

## Activity 2.9

Answer the questions below. Use an ASCII table to identify the decimal number that represents each character. Note that capital and small characters have different ASCII values and that a "space" has a value.

1. You send a message, "Hi there", to your friend. The checksum at the recipient's end is 704. Did the message arrive intact without error?

   The checksum, based on ASCII values, is 745.

   So there is an error in the message.

2. You send a message, "I finished homework", to your friend. The checksum at the recipient's end is 1855. Did the message arrive intact without error?

   The checksum, based on ASCII values, is 1855.

   So there is no error in the message.

3. A transaction code is sent between two banks, "AA103XT47". The checksum at the recipient's end is 551. Did the message arrive intact without error?

   The checksum, based on ASCII values, is 557.

   So there is no error in the message.

## Activity 2.10

The program below, in Python, is intended to accept a message from the user and to calculate the checksum. Use this program to check your answers for Activity 2.9.

| Code | Explanation |
|---|---|
| `def calculate_checksum(message):` | Defines the function that calculates the checksum by adding the ASCII values. |
| `    checksum = 0` | Defines the checksum as a variable and sets it to zero. |
| `    for char in message:` | Starts a loop that takes a character from the message each time. |
| `        ascii_value = ord(char)` | Converts character to its ASCII value. |
| `        checksum = checksum + ascii_value` | Updates checksum by adding the new ASCIi value to the current checksum value. |
| `    return checksum` | Repeats the loop. |
| `message = input('Enter message: ')` | Gets a string input from the user and assigns it to "message". |
| `checksum = calculate_checksum(message)` | Calculates the checksum using the function "calculate_checksum". |
| `print (f'The checksum for message is: {checksum}')` | Prints the checksum. |

```python
def calculate_checksum(message):

    checksum = 0

    for char in message:

        ascii_value = ord(char)

        checksum = checksum + ascii_value

    return checksum

message = input('Enter message: ')

checksum = calculate_checksum(message)

print (f'The checksum for message is: {checksum}')
```

### Notes

The intention of this activity is to familiarize students with coding and the concept of algorithms and iteration. The intention is not to teach Python or any other coding language. Students should, however, appreciate and understand each step in the line of code.

## Activity 2.11

Use a Caesar cipher wheel to answer the questions below. You may choose to construct one in class.

1. Encrypt the message "Hi there". The key = counterclockwise 2.

   Fg rfcpc

2. Encrypt the message "Beware of hackers". The key = clockwise 5.

   Gjbfwj tk mfhpjwx

3. Encrypt the message "Beware of hackers". The key = counterclockwise 10.

   Rumqhu ev xqsauhi

## Activity 2.12

The program below, in Python, is intended to accept a message from the user and to calculate the checksum. Use this program to check your answers for Activity 2.9. Note that this program only performs a clockwise shift. To perform a counterclockwise shift subtract the shift value instead of adding it.

| | |
|---|---|
| `def caesar_cipher_encrypt(plaintext,shift):` | Defines the function that changes original plaintext to ciphertext by a specified shift. |
| `    Ciphertext = “ ”` | Defines the ciphertext as an empty string. |
| `    for char in plaintext:` | Starts a loop that takes a character from the message each time. |
| `        if char.isalpha():` | Checks if the character is alphabetic. |
| `            if char.isupper():` | Checks if the character is upper case. |
| `                encrypted_char = chr((ord(char) - 65 + shift) % 26 + 65)` | If the character is uppercase, it calculates the ASCII value of the encrypted character by subtracting 65 (the ASCII value of ‘A’) from the ASCII value of the current character. The shift value is added, and the modulo 26 operation is applied to ensure the result stays within the range of the alphabet. Finally, 65 is added back to obtain the ASCII value of the encrypted character. The chr() function converts the ASCII value back to a character. |
| `            else:` | |
| `                encrypted_char = chr((ord(char) - 97 + shift) % 26 + 97)` | If the character is not uppercase, the same process is applied, but with 97 (the ASCII value of ‘a’) as the base value. |
| `            ciphertext = ciphertext + encrypted_char` | The encrypted character is added to the ciphertext string. |
| `        else:` | |
| `            ciphertext = ciphertext + char` | If the character is not alphabetic it is added as is to the ciphertext string. |
| `    return ciphertext` | Repeats the loop. |
| `plaintext = input(“Enter message: ”)` | Gets a string input from the user and assigns it to “message”. |
| `shift = int(input(“Enter shift value: ”))` | |
| `encrypted_message = caesar_cipher_encrypt (plaintext,shift)` | Calculates the checksum using the function “calculate_checksum”. |
| `print (“Encrypted message: ”, encrypted_messsage)` | Prints the checksum. |

# Lesson 3: Clouds & Things

The basic model for a computer was a device that provided the four basic functions of input, output, memory and processing. However, this model has gradually evolved, and the four basic functions decoupled and dispersed across a wide geographical area. No longer do these four basic functions have to occur on one device.

The internet brought about two advances in computing models, the cloud and the internet of things.

> THE INTERNET BROUGHT ABOUT THE CLOUD AND THE INTERNET OF THINGS.

## The Cloud

The term "cloud" refers to a system of servers that are accessed over the Internet. The term originated from the cloud-shaped symbol often used in diagrams to represent the internet.



The ability to connect to the internet and share information with other computers brought about possibilities. What if these computers could store a whole lot more of your information? What if these computers could provide additional computational power and could process information that could not be performed by your computer?

This additional storage and processing power would greatly enhance the functionality of your computer. Storage does not need to be on your device, it can be anywhere as long as you can access it whenever you want and from wherever you want.

STORAGE CAN BE OFFERED AS A CLOUD-BASED SERVICE.

### EXAMPLE 3.1

Mario likes taking photos. All the photos he takes are stored in the drive of his smartphone. But the more photos he takes the less space he has available. What will happen when one day the drive is full? In such a case, Mario needs more storage space. He has three options:

1. **Memory Card**: If Mario's smartphone has a slot for a memory card, he could consider buying one to expand the storage. However, not all smartphones have this option, and even if his does, there's a limit to how much a memory card can store. Moreover, if Mario loses or damages his phone, he could potentially lose all his photos.

2. **External Drive**: External drives like flash drives, hard disk drives (HDD), or solid state drives (SSD) are also viable options. They provide ample storage and can be a good choice if Mario prefers having physical control over his data. The downside is they can be lost or damaged. Additionally, accessing or sharing the data might be inconvenient as Mario will have to carry the external drive around.

3. **Cloud-Based Storage Service**: Google Drive and Apple iCloud are examples of services that provide additional storage space for users and allow them to access files from any device that has internet access. They also automatically backup data, so if Mario loses his phone, his photos will still be safe. The downside is that if he doesn't have an internet connection, accessing his photos could be difficult. Also, storing data on the cloud means trusting the provider with the privacy and security of his data.

| Storage Option | Advantages | Disadvantages | Maximum Storage Capacity | Approximate Price per GB |
|---|---|---|---|---|
| **Memory Card**<br><br>(microSD card) | Portable and expandable storage.<br>Immediate access to data. | Limited by phone compatibility and capacity.<br>Risk of physical damage or loss. | Up to 1TB | Around \$0.20 - \$0.50 |
| **External Drive**<br><br>(HDD, SSD) | Large storage capacity.<br>Data under physical control. | Risk of physical damage or loss.<br>Not as portable or convenient as other options. | HDD: Up to 16TB<br><br>SSD: Up to 8TB | HDD: Around \$0.02 - \$0.03,<br><br>SSD: Around \$0.10 - \$0.20 |
| **Cloud Service**<br><br>(Google Drive, Apple iCloud) | Accessible from any device with internet.<br>Automatic backup and syncing. | Requires internet access.<br>Recurring cost for extra storage.<br>Dependence on service provider for data security. | For individual users, up to 2TB | Around \$0.025 - \$0.05 |

As for processing, that too can be provided as a cloud-based service. If you are connected to a much larger and more powerful computer then you could perform far more complex tasks than what your computer could perform alone.

> **PROCESSING CAN BE OFFERED AS A CLOUD-BASED SERVICE.**

## EXAMPLE 3.2

Liam loves playing video games. He has a basic laptop that he uses for schoolwork, but it's not powerful enough to run the latest high-definition games that require powerful processors and high-end graphics cards.

Google Stadia is a cloud-based gaming service. It allows Liam to play his favorite games directly from the cloud. The games are run on powerful servers, and the video is streamed to Liam's laptop. His interactions, like button presses and mouse clicks, are sent back to the server allowing him to play more complex games than what he could on his computer.

> A DATA CENTER IS A FACILITY THAT HOUSES SERVERS WORKING AS AN INTEGRATED SYSTEM SERVING THE REQUESTS OF MANY USERS.

## Data Centers

A datacenter is a facility that contains a great number of servers. The number of servers in a data center varies from tens to thousands depending on the size of the facility. It is important to realize that a data center is not simply a room filled with servers, it is a system of integrated components, such as servers, switches, power units, cooling systems and security systems, that come together to provide services such as the cloud-based services discussed in the previous section.

### Servers

The heart of a data center is the server, a powerful computer designed to handle large amounts of data and run applications that serve other computers and their users across a network.

Servers are typically mounted on top of each other in metal frames called racks. Racks not only house the servers but also the other hardware components of the data center, such as the cables and switches that are used to connect servers together. In fact, a data center is one big, complex network consisting of servers connected by cables and switches.

Servers, like any other computer, have a basic architecture that also consists of the four basic components: input, output, memory and processing.

Although at certain terminals in a datacenter, there are keyboards, mice and screens for data center administrators to take care of the internal workings of the servers, a server's main input and output component is a network adapter. This is because the majority of a server's input comes from the internet or the network on which it serves. Remember that the server is serving users by performing the tasks they request. As such, the server's input comes from the users and its output must be sent back to the users. So, whether it is input or output, the majority is being transmitted across a network, making the network adapter the most important I/O device for a server.

As for memory and processing components, servers also have central processing units (CPU), graphics processing units (GPU), caches, main memory and storage components.

### Performance

A data center is designed to respond to the requests of many users concurrently. Therefore, the performance of a server is measured by its throughput, which is defined as the number of tasks it can complete in 1 second.

A single server can handle thousands to tens of thousands of requests per second under regular conditions. Across an entire data center, the cumulative throughput could be in the millions or even billions of requests per second.

> **Throughput of a server is the number of requests processed per second.**

Throughput is dependent on the amount of data a server can receive in order to process, the more data it can process in one second the higher is its throughput. The amount of data that can be sent or received is referred to as the bandwidth which is measured as bits per second (bps), and ranges from megabits (Mbps) to gigabits (Gbps) and even to terabits (Tbps) per second.

Adequate bandwidth is critical to ensure that data can be transmitted rapidly, enabling high-speed communication between servers and users or between different parts of the network.

Both throughput and bandwidth are interdependent. A server can have high throughput, but if the bandwidth is insufficient, data delivery to the server would be too slow, and the server won't be able to process tasks at its full capacity. On the other hand, a network with high bandwidth won't be fully utilized if the server's throughput is low.

BANDWIDTH IS THE AMOUNT OF BITS TRANSMITTED PER SECOND AND CAN LIMIT THE PERFORMANCE OF A DATA CENTER.

The performance of a server, its throughput, is enhanced by implementing parallel processing, which is the execution of multiple tasks at the same time. This method has already be implemented in the multi-core architecture of a CPU. Having multiple cores enhances the performance of a CPU by allowing it to process multiple tasks in parallel.

In the context of a data center, servers are arranged in clusters. This internal network organization allows the implementation of parallel processing at the level of the server. Multiple interconnected servers in a cluster can work simultaneously to perform multiple tasks for multiple users.

SERVERS ARE ARRANGED IN CLUSTERS THAT ALLOW THE PARALLEL PROCESSING OF TASKS.

**Notes**

Concurrent processing is processing by rapidly switching between tasks. Parallel processing is when processing of multiple tasks occurs simultaneously. Both enhance performance.

Data centers have two bandwidths, the internal and external bandwidths. This is because servers communicate with one another inside the data center via an internal network, and they also communicate with their clients via the network that connects them to the outside world.

**Data centers have both an internal and external bandwidths.**

If servers in a data center are working together then the internal bandwidth on the network interconnecting them must be high enough to not limit their performance. This is often quite high, with connections between servers and switches operating between 10 Gbps to 100 Gbps.

The external bandwidth of a data center, which is the total capacity for data transfer between the data center and the outside world, can also vary widely. Smaller data centers might have external bandwidth in the range of 1 to 10 Gbps, while large data centers might have external bandwidth in the order of Tbps.

From the perspective of the user, the most important measure of performance is the response time, the time it takes to receive a response. The response time factors in travel time through the internet to the data center, processing time at the data center, travel time back through the internet and any delays or wait time that occurs in between.

**Response time is the time it takes for a user to receive a response from a server to a request.**

## Dependability

Regardless of response time, throughput and bandwidth, users depend on data centers to run without any faults or failures. Users do not appreciate errors nor the failure of a server to respond. The dependability of a system such as a data center, refers to the extent to which it can operate without fault or failure.

> **Dependability is the extent to which a system will operate without fault or failure.**

Common reasons for faults or failures include over-heating, power outages, security breaches and disasters such as fire.

1. **Over-heating**: Servers generate lots of heat due to the high volume of computational work they have to perform. Therefore, cooling systems such as Computer Room Air Conditioning (CRAC) units, Computer Room Air Handler (CRAH) units, and liquid cooling systems are essential to maintaining the proper operating temperature and preventing hardware failure.

2. **Power outage**: Power disruptions can cause significant issues for data centers, including hardware damage and loss of data. Uninterrupted Power Supply (UPS) systems and backup generators are used to maintain power to the data center during outages.

3. **Security Breach**: Physical and cyber security breaches can lead to data loss, service disruption, and significant harm to the data center's reputation. Physical and cyber security measures are implemented to protect the data center.

4. **Disasters**: Natural disasters and accidents like fires or floods can cause catastrophic damage to data centers. Fire suppression systems, flood defenses, and disaster recovery protocols including off-site data backups and redundant systems are crucial for protecting the data center and ensuring uninterrupted service.

## The Internet of Things

While the cloud has taken processing and storage capabilities beyond the confines of personal computers, the internet of things (IoT) extends input and output functionalities to a wide array of devices beyond traditional computers.

Computers are not the only devices connected to a network. A network may also connect other things. The term "things" refers to devices that can link to a network and either provide input or generate output forming an internet of things. IoT is a system of interrelated devices that are connected to a network and can send or receive data without the need for human interaction. These devices can include anything from small sensors to large appliances.

> The IoT is a system of interrelated devices connected to a network and transmit data.

In the context of the IoT, a personal area network, PAN, can be created to connect a variety of smart devices to a central hub, often a smartphone or a personal computer.

### Example 3.3

Sam lives in a smart home. Using a central hub, a small computer designed for the purpose of smart home management, Sam can control and automate his home particularly because his home is full of smart things connected by a network to the hub.

Here are some of the smart things Sam has: smart watch, smart coffee machine, smart refrigerator, smart air conditioner, smart curtains, smart cameras and smart lights.

At 6:30 am, Sam's smartphone alarm, synced with his sleep cycle via his smartwatch, wakes him up at the lightest point in his sleep, making waking up less of a struggle. The moment he switches off the alarm, his smart coffee machine in the kitchen starts brewing his morning coffee.

After getting ready, Sam walks into the kitchen, where his smart refrigerator has already identified that he's running low on milk and eggs. The fridge automatically adds these items to Sam's grocery list on his smartphone, and it even suggests a healthy breakfast recipe based on the items currently available.

As Sam leaves his house for work, he doesn't need to worry about whether he has switched off all the lights or the AC. His smart home hub, which is connected to all the IoT devices in the house, knows when he's leaving due to his smartphone's GPS. It automatically turns off unnecessary lights and adjusts the thermostat to an energy-saving mode.

During the day, Sam's home security system keeps an eye on the house. The smart cameras and sensors monitor for any unexpected activity. If the cameras detect movement, Sam instantly receives a notification on his phone. He can watch a live feed and, if necessary, sound an alarm or contact the authorities, all from his phone.

Once Sam heads back home, his smart home hub detects his smartphone's location and begins to prepare for his arrival. The thermostat adjusts to a comfortable temperature, and the lights turn on as he walks in. As he steps into the house, his smart home hub greets him and briefs him about his schedule for the next day, the weather forecast, and any unread emails or messages.

At night, Sam can use his voice to control his home. A simple command like, "Good night, Home," can cause the lights to dim, the doors to lock, the alarm system to activate, and his favorite relaxing playlist to start playing softly in the background.

In this manner, each device in Sam's smart home plays a role in automating routine tasks, enhancing his comfort, saving energy, and providing him with peace of mind about his home's security. All these devices form a network around Sam, working together to make his life easier and more efficient.

An IoT is not restricted by distance, as long as their is network coverage various sensors and smart devices can connect to a central computer. Areas as wide as farms and cities can be covered by IoT. Here are two examples.

**The IoT is not restricted by distance as long as there is network coverage.**

### Example 3.4

Meet Farmer Maya, an innovative agriculturist using smart farming to optimize her operations. At dawn, Maya's weather station, equipped with sensors, provides a precise forecast, helping her plan the day's tasks. Her smart irrigation system, connected to soil moisture sensors, already knows which parts of her farm need water. These sensors feed data to the system, which autonomously controls the water flow, ensuring each crop gets the right amount of hydration while conserving water.

Maya has drones equipped with infrared cameras. When flown over her fields, they provide valuable data about crop health, identifying potential disease outbreaks or pest infestations before they become visible to the human eye.

Livestock wear connected devices that monitor their health and location. If a cow's activity level drops or it strays from the herd, Maya gets a notification on her phone.

At the end of the day, all data collected is sent to Maya's computer. Thanks to her smart farm, Maya ensures that her crops grow well, her livestock are healthy and that she uses her water resources efficiently.

### Example 3.5

Director Dave, is in charge of a city's waste management department and has implemented smart technology to optimize garbage collection.

To start the day, Dave receives an automated report generated by smart sensors installed in public garbage bins throughout the city. These sensors monitor the fill levels of the bins and report back when they're nearing capacity. This real-time data helps Dave's team plan their collection routes, ensuring they only visit bins that need to be emptied and reducing unnecessary trips.

Dave's garbage trucks are equipped with GPS and route optimization software. This allows the central system to direct each truck along the most efficient route, saving time and reducing fuel consumption.

The smart waste bins also have sensors that identify different types of waste for recycling. This helps the city to improve its recycling rate and reduces the volume of waste sent to landfills.

With this smart waste management system, Dave has made garbage collection more efficient, improved recycling rates, and made his city cleaner and more sustainable.

### Career Tips

IoT Solution Architects: These individuals design comprehensive IoT solutions, ensuring various components like devices, cloud services, and connectivity solutions work seamlessly together.

IoT Hardware Engineers: They are responsible for designing and developing the hardware aspects of IoT devices, such as sensors, circuits, and embedded systems.

## Summary

The internet brought about the cloud and the internet of things.

The cloud is a system of servers that can be accessed over the internet. The cloud offers additional storage and processing services.

Additional storage can be obtained by using memory cards, external drives and cloud-based storage services. The advantages of cloud-based storage are its accessibility from any device with internet access and automatic backups ensure the most recent information is kept secure. The disadvantages are its dependence on the internet and service providers for accessibility and security, as well as its potential recurring cost.

A data center is a facility that houses servers working as an integrated system serving the requests of many users. It is a system of integrated components, such as servers, switches, power units, cooling systems and security systems.

The server is the heart of a data center. It is a powerful computer designed to handle large amounts of data and run applications that serve other computers and their users across a network.

Throughput is a measure of server performance and is defined as the number of requests processed per second. Throughput is dependent on bandwidth which is the amount of data that can be sent or received. Data centers have both an internal and external bandwidths. Servers communicate with one another inside the data center and they also communicate with clients via an external network.

From the perspective of the user, the most important measure of performance is the response time, the time it takes to receive a response.

The performance of a server is enhanced by parallel processing. Servers are arranged in clusters, which allows the for parallel processing. Servers in a cluster work simultaneously to perform multiple tasks for multiple users.

Dependability is the extent to which a system will operate without fault or failure, which can be caused by over-heating, power outages, security breaches and disasters such as fire.

The IoT is a system of interrelated devices connected to a network and transmit data autonomously. The IoT extends input and output functionalities to a wide array of devices beyond traditional computers. All these devices are connected to a central hub.

The IoT made smart homes, smart farms and smart cities possible.

## Activity 3.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F | STATEMENT |
|---|---|
| F | The cloud does not depend on the internet. |
| T | Both storage and processing can be provided as cloud-based services. |
| T | A data center is a facility with an integrated system of servers. |
| T | Servers, like other computers, have the four basic components of input, output, memory and processing. |
| F | Throughput is a measure of dependability. |
| T | Bandwidth refers to the amount of data transmitted per second. |
| T | To improve performance servers are arranged in clusters that perform parallel processing. |
| T | Dependability is the extent to which a system will operate without fault or failure. |
| T | The IoT is a system of interrelated devices that are connected and can transmit data. |
| F | The IoT only includes devices in close proximity. |

## Activity 3.2

Answer the questions below.

1. What is the “cloud” in cloud computing?
   a) A weather system
   b) A system of servers accessed over the Internet
   c) A data storage device

2. Which of the following is not an advantage of cloud storage?
   a) Accessible from any device with internet
   b) Automatic backup and syncing
   c) Doesn’t require an internet connection

3. Which of the following tasks can be performed by a cloud-based service?
   a) Only storage
   b) Only processing
   c) Both storage and processing

4. Which additional storage option provides the greatest space?
   a) cloud-based storage
   b) memory card
   c) external drive

5. Which additional storage option depends on the internet?
   a) cloud-based storage
   b) memory card
   c) external drive

## Activity 3.3

Answer the questions below.

1. What is the main component of a data center?
   a) Cooling system
   b) Network switch
   c) Server

2. What does the term "throughput" refer to in the context of a server?
   a) The amount of data a server can store
   b) The number of tasks a server can complete in 1 second
   c) The speed at which a server can download data

3. What is "bandwidth" in the context of data center performance?
   a) The physical width of the cables connecting servers
   b) The number of servers in a data center
   c) The amount of data that can be sent or received per second

4. What is the measure of performance from the user's perspective?
   a) Throughput
   b) Bandwidth
   c) Response time

5. What does "dependability" in the context of a data center refer to?
   a) The amount of data a server can store
   b) The extent to which a system will operate without fault or failure
   c) The speed at which a server can download data

## Activity 3.4

Answer the questions below.

1. What does the term "Internet of Things" refer to?
   a) The network connecting all the computers in the world
   b) A network of interrelated devices that can transmit data without the need for human interaction
   c) A network of satellites providing global internet coverage

2. What can smart home devices do in the context of IoT?
   a) Automate routine tasks
   b) Enhance comfort
   c) Both a and b

3. What is a central hub in the context of IoT?
   a) A device that connects to other devices on a network and manages them
   b) The main server in a data center
   c) The main router in a network

4. How does smart farming help in conserving water?
   a) By watering all crops equally irrespective of their needs
   b) By using soil moisture sensors to autonomously control the water flow
   c) By only watering the crops during the night

5. How can a city's waste management department benefit from IoT?
   a) By having to manually check the fill levels of the bins
   b) By planning their collection routes based on real-time data about the fill levels of the bins
   c) By increasing the volume of waste sent to landfills

## Activity 3.5

Answer the questions below.

1. How has the internet contributed to the evolution of the basic model for a computer?

   The internet has allowed for the decoupling and geographical dispersal of the four basic computer functions: input, output, memory, and processing. This has given rise to new computing models such as cloud computing and the Internet of Things.

2. What are some potential advantages and disadvantages of using a cloud-based storage service?

   Advantages of cloud-based storage services include accessibility from any device with internet, automatic backup and syncing of data. Disadvantages include the requirement of an internet connection to access data, potential security and privacy concerns, and the recurring cost for extra storage.

3. How can cloud-based services enhance the functionality of your computer?

   Cloud-based services can provide additional storage and processing power beyond the capabilities of one's personal computer. This allows for more complex tasks to be performed and larger amounts of data to be stored than what could be achieved with the computer alone.

## Activity 3.6

Answer the questions below.

1. What are the main components of a data center?

   The main components of a data center are servers, switches, power units, cooling systems and security systems.

2. How can bandwidth limitations impact the performance of a data center?

   Bandwidth limitations can limit the amount of data that can be sent to and processed by the servers, reducing their throughput. If the bandwidth is insufficient, data delivery to the servers would be too slow, and the servers won't be able to process tasks at their full capacity.

3. What measures can be taken to ensure the dependability of a data center?

   Measures to ensure the dependability of a data center include cooling systems to prevent overheating, Uninterrupted Power Supply systems and backup generators to maintain power during outages, physical and cyber security measures to protect against security breaches, and fire suppression systems,

## Activity 3.7

Answer the questions below.

1. How do smart home devices improve a user's quality of life?

   Smart home devices improve a user's quality of life by automating routine tasks, enhancing comfort, providing security, and increasing efficiency. They can adjust to the user's habits and preferences, such as brewing coffee when the alarm goes off, or suggesting recipes based on available ingredients.

2. What is one way smart farming benefits the environment?

   Smart farming benefits the environment by optimizing water usage, with soil moisture sensors controlling the water flow to ensure each crop gets the right amount of hydration. This helps in conserving water.

3. How does a smart waste management system contribute to a cleaner and more sustainable city?

   A smart waste management system contributes to a cleaner and more sustainable city by optimizing garbage collection, reducing unnecessary trips, and saving time and fuel. Sensors in the bins monitor fill levels and enable collection only when necessary. They also identify different types of waste for recycling, improving recycling rates and reducing the volume of waste sent to landfills.

## Activity 3.8

Cloud Storage Activity: Sign up for a free cloud storage service such as Google Drive or Dropbox. Upload a document or picture from your computer, and then access and download it using a different device. This will demonstrate the concepts of cloud storage and accessibility from multiple devices.

## Activity 3.9

Exploring Data Center Virtual Tour: Many data centers offer virtual tours online (like Google data centers). Explore these tours to understand the scale, organization, and components of a data center, including servers, cooling systems, and security measures. Then discuss or write a report on your findings.

## Activity 3.10

IoT Smart Home Design: Divide students into groups and ask each group to design a smart home using IoT devices. Students can research different types of IoT devices, decide what their smart home would include, and explain how the devices would communicate with each other and the central hub. This project can be presented as a diagram, a report, or a presentation.

## Activity 3.11

Creating a Smart City Map: Divide students into groups and ask each group to design a "smart city" that utilizes IoT technology. Students can research different applications of IoT in urban environments (like smart traffic lights, waste management systems, air quality sensors) and incorporate these into their city. They can present their city map and explain how each IoT element contributes to the efficiency, sustainability, or livability of the city.

# LESSON 4: INFORMATION SYSTEMS

In previous lessons we have explored how humans interact with their computers and how these computer communicate with each other across networks. We have explored how functions such as input, output, storage and processing can occur over networks as in the case of the cloud and the IoT. In this lesson, we will explore how all these can be brought together, humans, computers, networks, cloud and IoT, to form larger systems to handle information.

## FUNCTIONS OF INFORMATION SYSTEMS

Every information system serves a particular purpose, such as managing a school or providing navigation services. Regardless of the purpose, all information systems collect data, store data, and process data in order to extract meaningful information and share it to help guide decision-making.

> AN INFORMATION SYSTEM PERFORMS FOUR PRIMARY FUNCTIONS: DATA COLLECTION, STORAGE, PROCESSING, AND SHARING.

An information system performs four primary functions: data collection, data storage, data processing, and information sharing.

**Collection**: An information system gathers data from various sources. This could be through direct user input or automated data collection from sensors as in the case of the IoT.

**Storage**: The collected data is stored mainly in the form of databases on servers making the data accessible over a network. Storage may be on a local server or it may be on server based in the cloud.

**Processing**: This is where the data is transformed into useful information. The system might organize the data, perform calculations, apply various algorithms, or create visualizations. Depending on where the data is stored, processing may occur at the level of a local server or in the cloud. The aim is to turn raw data into meaningful information that supports decision-making or other activities.

**Sharing**: Information systems are designed to distribute and share data among users. This sharing ensures that the necessary data is available when and where it's needed. Information systems have multiple users with whom information is shared. Access to the information may be restricted to ensure the security and confidentiality of data and information.

To better understand information systems let's explore two types: a management information system and a geographic information system.

## School Information Systems

A school information system is a system designed to collect, store, process and share information pertinent to all stakeholders in a school, from students and their parents to employees and their administrators.

### Example 4.1

We will explore a SIS from the perspectives of a student, parent, teacher, principal and the SIS administrator.

**Story 1**: Student - Sara

Sara is a student in seventh grade. Every morning, she checks her SIS mobile app to see her schedule for the day. After school, she checks for assignments her teachers have posted. Once she completes her homework, she uses the SIS to submit her assignments. When she finishes a big project or a test, she eagerly awaits the grade on the SIS - it's the first place her grades are posted.

In addition, Sara has a smart ID card. She uses it to swipe in and out of school that way the SIS automatically takes attendance. She also uses it to purchase food and drinks from the school's cafeteria.

**Story 2**: Parent - Ahmed

Every morning, Ahmed checks his children's timely arrival to school using his SIS mobile app. At work, he frequently checks their schedules and academic progress using his desktop. One day, he noticed a dip in his son's math scores. He messaged the teacher via SIS, and arranged a virtual meeting to discuss improvement strategies. School announcements are easily accessed on SIS, keeping Ahmed informed. Despite his hectic schedule, SIS enables Ahmed to stay closely connected with his children's education.

**Story 3**: Teacher - Mr. Johnson

Mr. Johnson starts his day by checking his class rosters on the SIS, noting any changes. Throughout the day, he uses the SIS to take attendance, input grades, and assign homework. He appreciates how he can quickly message students or parents through the SIS to discuss grades or concerns. At the end of the day, Mr. Johnson uses information from the SIS to monitor students' progress and customize his lesson plans accordingly.

**Story 4**: Principal - Mrs. Gomez

Mrs. Gomez begins her day by reviewing reports from the SIS, looking at attendance records, academic performance, and other key metrics. The SIS helps her identify patterns, such as if a particular class is struggling with attendance or a specific subject. She also uses the SIS to communicate with the school community, whether it's a reminder about an upcoming event or a critical update.

**Story 5**: System Administrator - Mr. Lee

Mr. Lee oversees the SIS, ensuring it runs smoothly. His morning starts with system checks and addressing any issues. Throughout the day, he manages user accounts, granting necessary access to new staff, or students. Mr. Lee also installs updates and troubleshoot issues. He plays a crucial role in training staff on how to use the SIS effectively.

## Geographic Information Systems

A Geographic Information System (GIS) is a type of information system that is specifically designed to collect, store, process, and share geographical data. GIS allows users to visualize data primarily in the form of maps. It also provides users with information about various locations on the map, such as names of streets, traffic congestion, photographs, and directions to go from one place to another.

### Example 4.2

Google Maps is a commonly used geographic information system. We will explore it from the perspectives of a user, data analyst, street view driver, 3D modeling specialist and software engineer.

**Story 1**: User - Sofia

Sofia uses Google Maps in her everyday life. She starts her day by checking the fastest route to her office, taking into account real-time traffic updates. During lunch, she uses Google Maps to find a new restaurant nearby. After work, she uses Street View to virtually explore a neighborhood where she's considering renting a new apartment. For Sofia, Google Maps and its GIS underpinnings make her daily life more convenient and informed.

**Story 2**: Data Analyst - Kelly

Kelly starts her day by pulling new satellite images from multiple sources. She uses GIS software to correct and enhance these images for clarity. By overlaying new images on old ones, she detects significant changes, like deforestation or urban growth. She updates Google Maps with the new data, helping to ensure that users have the latest information when planning routes or exploring areas.

**Story 3**: Street View Driver - Mario

Mario is responsible for capturing street-level images for Google Maps. His van, equipped with a multi-lens camera, captures 360-degree views of the streets he drives. He spends his day driving assigned routes, often in new cities or areas where significant changes have occurred. The images he captures are sent to Google for processing and integration into the Street View feature.

**Story 4**: 3D Modeling Specialist - Mei

Mei creates realistic 3D models of buildings and landmarks for Google Maps. She uses high-resolution aerial images and elevation data as her primary sources. Through GIS software, she's able to translate this data into detailed 3D models. When a user explores a city using the 3D feature in Google Maps, they're seeing Mei's work.

**Story 5**: Software Engineer - Lucas

Lucas works on the backend of Google Maps, making sure everything runs smoothly. His day often involves optimizing algorithms that calculate routes, improving the map rendering speed, or ensuring that the huge amount of data Google Maps uses is stored efficiently. Lucas' work doesn't always have a visible output on the Google Maps interface, but without it, the service wouldn't be nearly as fast or reliable.

## Components of Information Systems

Every information system despite its complexity is composed of four components: hardware, software, data and people. These four components are used to perform the functions of data collection, storage, processing and sharing.

> An information system has four primary components: hardware, software, data, and people.

**Hardware**: The physical components of an IS, including computers, network equipment, and IoT devices. Computers include all types personal computers as well as servers, whether they are local servers or part of a data center. Network equipment includes switches and routers. While IoT devices can be a very wide variety of devices that range from simple sensors to smart cars.

The school information system hardware includes all the personal computers of its users as well as the school server and the smart ID carried by students. The geographic information system includes a tremendous number of hardware that range from satellites to every smartphone running the Google Maps app. Also included in the Google Maps information system are the data centers that store and process its data, and share its information with all its users.

**Software**: The programs and applications used to process and manage information. These also include the databases that contain all the data. Some information systems consist of one application that performs all the functions, whereas others may consist of multiple applications each performing a particular task and collectively forming an information system.

In the case of the school information system, it consists of one application which all users access. Whereas the geographic information system consists of multiple applications, Google Maps accessible by the general public, as well as data analysis software, photo-editing software and 3D modeling software to name a few.

**Data**: The raw, unprocessed facts and figures without any added interpretation or analysis. Data is simply the input collected either from users entering it using the application or automatically collected by IoT devices.

*School Information Systems (SIS) Data*

Student Data: This includes data such as student names, IDs, grades, test scores, attendance, demographic data, and any other information related to individual students.

Class Data: Information about each class, including the teacher, room number, subject, schedule, and enrolled students.

Teacher Data: Details about teachers, such as their name, contact information, subjects taught, schedules, and any other relevant information.

Parent Data: Parent or guardian names, contact information, and their relation to students.

School Calendar: Important dates for the academic year, including holidays, start and end dates of terms, and dates of significant events.

Administrative Data: Information about school operations, including budgets, resource allocation, staff details, and other operational data.

*Geographic Information Systems (GIS) Data*

Satellite Imagery: Images captured by satellites that provide a top-down view of the Earth's surface.

Aerial Photographs: These are photographs taken from aircraft and drones, which can often provide more detail than satellite imagery.

Maps: Both digital and paper maps can serve as data for a GIS.

GPS Data: Data from Global Positioning System devices can provide very accurate spatial data.

Street View Data: Images and data collected by vehicles equipped with 360° cameras, like those used for Google's Street View feature.

Environmental Sensor Data: Data from sensors that measure environmental conditions like temperature, humidity, soil moisture, etc., often used in environmental or agricultural applications of GIS.

**People**: Include both the end users who interact with the information system and the people that perform the tasks that keep the system operational, such as technicians, system administrators and software engineers to name a few.

The school information system includes its users, the students, parents, teachers and principle, as well as system administrators and technicians that help keep the system operational. The geographic information system has a far greater number of users, namely any person who has the app, and many other people that work to keep the system running and updated, such as street car drivers, data analysts, 3D modeling specialists and data engineers.

Information systems play an important role in our personal and professional life and demonstrate the importance of information in the decision-making processes of our lives.

## SUMMARY

Information systems perform four primary functions: data collection, storage, processing and sharing of data and information.

The overall goal of information systems is to guide decision-making by collecting data, processing it to extract meaningful information, and to share both data and information with users in such a way as to assist them with making decisions.

Information systems have four primary components: hardware, software, data and people. The hardware includes all IoT devices, personal computers, servers and network hardware such as switches and routers. The software includes the programs and applications that are used to collect, store, process and share data and information. Data is a critical component of an information system and is collected in a database as input from IoT devices and users. It is processed to extract useful information. People are part of the information system. They act as users and sources of data input as well as professionals that ensure the information system operates properly.

There are many types of information systems and each has a purpose. A school information system collects, stores, processes and shares data and information with members of the school community to assist each in making decisions related to education and the educational process. A geographic information system collects, stores, processes and shares data and information with the public to assist people in making decisions related to places and travel.


### Career Tips

Software Engineers: These individuals design and build information systems. They are responsible for creating the software components that make up the system.

Information System Managers: They are responsible for overseeing the implementation of new systems, managing IT teams, and allocating resources.

## Activity 4.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F | STATEMENT |
|---|---|
| T | An information system is responsible for collecting data. |
| T | Data processing is used to extract meaningful information. |
| F | An information system does not share information with its users. |
| T | An information system may incorporate IoT devices. |
| F | Cloud storage is not an option for information systems. |
| T | Any device that collects data can be used as component of an information system. |
| F | An information system cannot include more than one software. |
| T | A database must remain secure. |
| F | People do not play an important role in an information system. |
| T | Information systems presents information to users to guide their decisions. |

## Activity 4.2

Answer the questions below.

1. What are the four primary functions of an information system?
   a) Data collection, data mining, data processing, information sharing
   b) Data collection, data storage, data processing, information sharing
   c) Data collection, data storage, data analysis, data sharing

2. What is the primary purpose of an information system?
   a) To guide decision-making
   b) To store data
   c) To collect data

3. Which of the following is a source of data for a school information system?
   a) teacher
   b) smart ID
   c) Both a and b

4. Which of the following is a source of data for a geographic information system?
   a) satellite
   b) map
   c) Both a and b

5. What is the primary way information systems help with decision-making?
   a) By collecting data from users
   b) By processing raw data into meaningful information
   c) By storing data in databases

## Activity 4.3

Answer the questions below.

1. What is the role of a system administrator in a school information system?
   a) To enter data
   b) To oversee the system and ensure it runs smoothly
   c) To analyze data

2. What is the role of a data analyst in a Geographic Information System like Google Maps?
   a) To drive the Street View vehicle
   b) To pull new satellite images and enhance them for clarity
   c) To create 3D models of buildings

3. What are the components of an information system?
   a) Hardware, software, information
   b) Software, data, IoT devices, cloud
   c) Hardware, software, data, people

4. What kind of devices does the 'hardware' component of an information system include?
   a) servers, personal computers
   b) IoT devices, routers
   c) Both a and b

5. What do Geographic Information Systems (GIS) primarily visualize data in the form of?
   a) Charts
   b) Maps
   c) Tables

## Activity 4.4

Answer the questions below.

1. What are the main functions of an information system?

   The main functions of an information system are to collect data, store data, process data into meaningful information, and share information.

2. What are some types of data stored in a school information system?

   A school information system stores various types of data including student data (names, IDs, grades, attendance), class data (teacher, room number, subject, schedule), teacher data, parent data, school calendar, and administrative data (budgets, resource allocation).

3. What are some types of data for a geographic information system?

   Satellite images, aerial photographs, maps, GPS data, Street view data, environmental sensor data

## Activity 4.5

Answer the questions below.

1. What are the main components of an information system?

   The main functions of an information system are to hardware, software, data, and people.

2. What is the role of IoT devices in information systems?

   IoT devices play a crucial role in information systems by automating data collection. These devices can collect data from their environment and send it to the system for processing and analysis.

3. What role do people play in an information system?

   People are both the end users who interact with the system and the professionals who keep the system operational. They contribute to data input and use the output information for various purposes. Professionals such as system administrators and software engineers ensure smooth operation and maintenance of the system.

## Activity 4.6

Interview an IT Professional: Students can prepare and conduct an interview with a school's IT administrator or a GIS specialist. They can ask about the everyday challenges, maintenance requirements, and the skills needed to operate and manage these systems.

## Activity 4.7

Build a Mock School Information System: Students can design and create a basic mock-up of a school information system. They can define what data should be collected (such as student names, grades, attendance), decide how to store it, define methods for processing the data, and consider how it should be shared with users. This activity will help them understand the practical aspects of an information system.

## Activity 4.8

Explore Google Maps: Students can explore the various features of Google Maps (such as Street View, satellite imagery, traffic updates, etc.) and report on how these features help in their daily life. This activity will help them understand how a Geographic Information System works.

## Activity 4.9

Discussion on Privacy and Security: Organize a discussion or a debate on the privacy and security concerns related to information systems. It will make students think critically about the ethical aspects of data collection, storage, and sharing.

## Activity 4.10

Design User Interface: Students can design the user interface for a new feature in the school information system or Google Maps. They can then present their design to the class, explaining how users would interact with the system via this interface. This activity encourages students to think about the user experience aspect of information systems.

# Appendix

# Glossary

| Term | Definition | Page |
|---|---|---|
| Accelerometer | A sensor in devices like smartphones that detects straight-line motion, such as moving up and down, forward and backward, or side to side. | p. 58 |
| Acknowledgement | A signal sent back to confirm that a message was received successfully. | p. 151 |
| Actuator | An output device that performs an action by converting signals into movement, like a motor in a printer or robot. | p. 59 |
| Action Potential | The internal electrical signal used by the human nervous system to transmit information between body parts. | p. 55 |
| Adder | A logic circuit that performs addition on two input signals and outputs their sum. | p. 78 |
| Address | A specific location in memory or registers where data is stored or retrieved. | p. 82 |
| Address Bus | A set of lines in the bus system that carry the addresses of where data should be read from or written to. | p. 103 |
| Administrative Data | Information about how a school or organization runs, such as budgets, staff details, or resources. | p. 198 |
| Aerial Photographs | Pictures taken from aircraft or drones that provide detailed views of the Earth's surface, often used in GIS. | p. 199 |
| Algorithm | A set of steps or rules a computer system follows to process data and solve a problem. | p. 193 |
| Application (App) | A stored program that performs a specific task, such as writing an essay, playing music, or browsing the web. | p. 20–21 |
| Architecture | The basic structure of a computer system, including the processor, memory, input/output devices, a bus, and a clock. | p. 102 |
| Arithmetic Logic Unit (ALU) | The part of the processor that performs calculations and logical operations on data. | p. 81 |
| Autonomous Robot | A robot that can make decisions and act on its own without human control. | p. 128 |
| Bandwidth | The amount of data that can be sent or received in one second across a network, usually measured in bits per second (bps). | p. 175 |
| Binary | A number system using only two values, 0 and 1, which computers use to represent information. | p. 77 |
| Bit | The smallest unit of information in computing, representing either a 0 or a 1. | p. 77 |

| | | |
|---|---|---|
| Bluetooth Adapter | A type of adapter that uses short-range radio waves to connect nearby devices, such as headphones, speakers, or keyboards. | p. 146 |
| Bus | A shared communication pathway that connects all components of a computer so they can send and receive signals. | p. 103 |
| Byte | A group of 8 bits that together can represent more complex information. | p. 77 |
| Cache | A small, high-speed memory placed close to the processor to store frequently used data and instructions for quick access. | p. 109 |
| Caesar Cipher | An old encryption method where each letter in a message is shifted by a fixed number of positions in the alphabet. | p. 154 |
| Capability | How powerful a computer is, based on the complexity of tasks it can handle and the speed of its calculations. | p. 34 |
| Checksum | A number calculated from data before transmission, used to check if the data arrived correctly without errors. | p. 152 |
| Class Data | Information about a class, including the teacher, subject, schedule, and enrolled students. | p. 198 |
| Clock | A part of the computer that sends out regular timing signals to keep all components working in sync. | p. 104 |
| Clock Cycle | One complete vibration of the clock's quartz crystal, which acts like a "go" signal for moving data to the next step. | p. 104 |
| Clock Frequency | The number of clock cycles per second; higher frequency usually means faster performance. | p. 106 |
| Cluster | A group of interconnected servers that work together at the same time to handle many user requests in parallel. | p. 175 |
| Command-Line Interface (CLI) | A type of user interface where the user types text commands to make the computer perform tasks. | p. 61 |
| Comparator | A logic circuit that compares two inputs and outputs whether one is greater than, less than, or equal to the other. | p. 80 |
| Communication | The exchange of information between two or more people, often using methods like messaging or email. | p. 18 |
| Computer | A technology designed to process information according to a stored program. | p. 10, 15 |
| Computational Power | The ability of a computer to process large amounts of information and perform calculations very quickly. | p. 34 |
| Concurrent Processing | When a computer quickly switches between tasks to give the impression that multiple tasks are running at the same time. | p. 175 |
| Control Bus | The part of the bus that carries control signals, including timing signals from the clock. | p. 103 |
| Control Unit (CU) | The part of the processor that directs signals, decodes instructions, and activates the correct logic circuits. | p. 84 |

| | | |
|---|---|---|
| Core | An individual processing unit inside a CPU or GPU; modern processors may have multiple cores working in parallel. | p. 107 |
| Data | Raw facts or numbers that can be processed to create meaningful information. | p. 15 |
| Data Bus | The part of the bus that carries the actual data being transferred between components. | p. 103 |
| Data Center | A large facility that houses many servers and related systems (cooling, power, security) to provide cloud-based services. | p. 173 |
| Database | An organized collection of data stored on servers that can be accessed and managed by software. | p. 192 |
| Dependability | How well a system, like a data center, can keep running without faults or failures. | p. 177 |
| Destination Address | The location where the result of an operation is stored after processing. | p. 83 |
| Digital Electronic Signal | The electrical signals used inside computers, which can only be high (1) or low (0). | p. 77 |
| Directory (Folder) | A virtual container used by the computer's filing system to organize files; it can also contain other directories (subdirectories). | p. 73 |
| Encryption | The process of changing a readable message into a secret code so that only authorized people can read it. | p. 153 |
| Environmental Sensor Data | Measurements collected by sensors about conditions like temperature, humidity, or soil moisture. | p. 199 |
| Error Detection | Techniques, like checksums and acknowledgements, used to check if data was transmitted correctly. | p. 151–152 |
| Ethernet NIC | A network interface card that connects computers using cables to exchange data. | p. 146 |
| External Bandwidth | The total data transfer capacity between a data center and the outside world (users, internet). | p. 176 |
| Field | A part of a machine-language instruction that carries specific information, such as the opcode or an address. | p. 87 |
| File | A unit of stored information, such as a document, photo, or program, that has a name and address. | p. 19 |
| Filename | The name given to a file so that it can be identified. | p. 19 |
| Frequency-Hopping Spread Spectrum (FHSS) | A method used by Bluetooth to send signals by rapidly switching between radio frequencies to reduce interference. | p. 146 |
| General-Purpose Computer | A computer designed to perform many different tasks using different applications, like desktops, laptops, tablets, and smartphones. | p. 32–33 |
| Geographic Information System (GIS) | A type of information system designed to collect, store, process, and share geographic data, often shown as maps. | p. 195 |

| | | |
|---|---|---|
| Google Maps | A widely used GIS application that provides maps, directions, traffic updates, and features like Street View. | p. 194–195 |
| GPU (Graphics Processing Unit) | A processor designed specifically to handle graphics-related instructions, often working alongside the CPU. | p. 107 |
| Graphical User Interface (GUI) | A user interface that uses windows, icons, and menus so people can interact with computers more visually and easily. | p. 62 |
| Gyroscope | A sensor that detects rotation, like rolling, spinning, or tilting movements of a smartphone. | p. 58 |
| Hardware (in IS) | The physical parts of an information system, such as computers, servers, network devices, and IoT equipment. | p. 197 |
| Hop Count | The number of routers a signal must pass through to reach its destination. | p. 147 |
| Human-Computer Interaction (HCI) | A field of study that focuses on how people design, use, and interact with computers and technology. | p. 122 |
| Human-Computer Interface | The boundary where humans and computers exchange signals, usually through input and output devices. | p. 122 |
| Human-Machine Interface (HMI) | The point where humans interact with machines that have embedded computers, like cash registers or smart devices. | p. 126 |
| Information | Any meaningful content that tells us something about ourselves or the world, such as words, images, or sounds. | p. 12 |
| Information System (IS) | A system that collects, stores, processes, and shares data to provide useful information and support decision-making. | p. 192 |
| Input Device (Sensor) | A device that converts signals from the outside world, like light, sound, or pressure, into internal digital signals a computer can use. | p. 57 |
| Input/Output (I/O) Loop | The continuous cycle where a computer's output becomes the human's input, and the human's output becomes the computer's input. | p. 123 |
| Instruction | A command given to the processor that tells it what operation to perform and on which data. | p. 81 |
| Instruction Cache (I-Cache) | A cache memory that stores instructions separately, making it faster for the processor to find the next instruction. | p. 110 |
| Instruction Cycle | The repeated process of fetching, decoding, and executing instructions in a processor. | p. 91 |
| Instruction Set | The complete collection of instructions that a processor is designed to understand and carry out. | p. 83 |
| Interface | The boundary where a system communicates with the outside world by exchanging signals. | p. 56 |
| Internet of Things (IoT) | A system of devices ("things") connected to a network that can send or receive data automatically without human input. | p. 178 |
| Internal Bandwidth | The data transfer capacity between servers inside a data center. | p. 176 |

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|---|---|---|
| LAN (Local Area Network) | A network covering a small area, like a home, office, or school, where computers and devices share resources. | p. 142 |
| Laptop Computer | A portable personal computer with a built-in screen and keyboard, capable of most tasks a desktop can do. | p. 34–35 |
| Latency | The time it takes for a signal to travel from the source to the destination across a network. | p. 148 |
| Logic Circuit | A collection of transistors arranged to perform a specific operation such as add, multiply, or compare. | p. 78 |
| Machine Language | The language of computers, made up of binary signals (0s and 1s) that represent instructions and data. | p. 85 |
| Magnetometer | A sensor in smartphones that detects the Earth's magnetic field to help determine orientation. | p. 58 |
| Main Memory | The larger but slower memory (RAM) where programs and data are kept for active use. | p. 109 |
| Messaging | A quick way of sending and receiving short pieces of information, often between friends or classmates. | p. 18 |
| Modem | A device that converts digital computer signals into analog signals (and back) so they can travel over telephone lines. | p. 146 |
| Multiplier | A logic circuit that multiplies two input signals and outputs the product. | p. 78 |
| Multiplexer | A logic circuit that selects one signal from several possible inputs and forwards it. | p. 78 |
| Network | A system of interconnected computers and devices that can exchange information. | p. 140 |
| NIC (Network Interface Card) | A hardware card that allows a computer to connect to a network, available in Ethernet, fiber optic, or WiFi versions. | p. 146 |
| Opcode (Operation Code) | The part of an instruction that tells the control unit which operation to perform. | p. 84 |
| Operating System | A group of programs that manage how a computer works and allow other apps to run properly. | p. 20 |
| Output Device | A device that takes internal computer signals and converts them into something external, such as sound, light, or movement. | p. 59 |
| PAN (Personal Area Network) | A small network centered around one person's computer and devices, often using Bluetooth. | p. 140–141 |
| Parallel Processing | When multiple tasks are truly processed at the same time, often by multiple cores or servers. | p. 175 |
| Parallelism | A strategy that speeds up performance by performing multiple tasks at the same time, often using multi-core CPUs or GPUs. | p. 107 |
| Parent Data | Information about students' parents or guardians, such as names and contact details. | p. 198 |

| | | |
|---|---|---|
| Performance | A measure of how fast and efficiently a computer completes tasks, influenced by factors like clock speed, cores, and memory design. | p. 106 |
| Peripheral Device | An input or output device connected to a computer to extend its functions, like a printer, headset, or smartboard. | p. 124 |
| Pointer | A small graphical symbol (like an arrow or hand) controlled by a mouse or touchpad to select and move items on the screen. | p. 63 |
| Prediction | A design strategy where the processor guesses what instructions or data will be needed next and fetches them early. | p. 110–111 |
| Prefetching | A prediction method where the processor not only fetches the needed data but also nearby data, assuming it will be used soon. | p. 111 |
| Private Information | Information that is restricted and requires permission or a password to access. | p. 17 |
| Processing (in IS) | The stage where raw data is organized, calculated, or analyzed to turn it into meaningful information. | p. 193 |
| Processor (CPU) | The main component of a computer that changes input into output by executing instructions. | p. 76 |
| Proximity | A design principle where components are placed physically closer together, like cache on the same chip as the processor, to speed up communication. | p. 109 |
| Public Information | Information that is open to everyone and does not require a password. | p. 17 |
| Redundancy | Keeping extra copies of data so it isn't lost if transmission errors occur. | p. 151 |
| Register | A very small and fast memory location inside the processor, used to hold data temporarily. | p. 88 |
| Register File | A group of registers managed together within the processor to store data and addresses close to the ALU. | p. 88, 109 |
| Repeater | A device that boosts weakened signals so they can travel longer distances without losing strength. | p. 151 |
| Response Time | The total time it takes for a user to send a request and receive a response back from a server. | p. 176 |
| Robot | A machine with sensors and actuators that can perform tasks either autonomously or with some level of human control. | p. 128 |
| Router | A device that directs signals between networks, acting as both a gateway to the internet and a traffic controller. | p. 143 |
| Satellite Imagery | Images of Earth taken from satellites, used in GIS to provide wide-area geographic information. | p. 199 |
| School Information System (SIS) | An information system that manages school-related data such as schedules, grades, attendance, and communication with students, parents, and staff. | p. 193 |
| Semi-Autonomous Robot | A robot that can make some decisions on its own but still relies partly on human guidance. | p. 128 |

| | | |
|---|---|---|
| Server | A powerful computer in a data center that stores data or runs applications to provide services for other computers and users. | p. 38, 173 |
| Shared Cache | A cache memory used by multiple cores in a processor, storing data that any of them may need. | p. 110 |
| Shielding | A protective layer around wires or circuits to block outside interference. | p. 151 |
| Shifter | A logic circuit that moves the bits of a signal left or right by a certain number of places. | p. 80 |
| Signal | A message carried in physical form, such as electrical energy, that allows parts of a system to communicate and coordinate. | p. 55 |
| Smart Device / Smart Thing | A device with sensors and connectivity that links to a network (like smart watches, fridges, or cameras) to automate or improve tasks. | p. 179 |
| Smart Farming | The use of IoT devices such as soil sensors, drones, and livestock trackers to optimize farming practices. | p. 180 |
| Smart Home | A home equipped with IoT devices (like smart lights, thermostats, or security cameras) connected to a hub that automates daily tasks. | p. 179 |
| Smart Machine | A machine with an embedded computer that adds new features and makes it more functional and user-friendly. | p. 126 |
| Smartphone | A general-purpose personal computer in the form of a phone, capable of running many applications beyond calls and texts. | p. 126 |
| Smart Waste Management | A city system using IoT sensors in bins and trucks to optimize garbage collection and recycling. | p. 181 |
| Software (in IS) | Programs and applications, including databases, that process and manage data within an information system. | p. 197 |
| Source Address | The location where input data for an instruction is stored before processing. | p. 82 |
| Specific-Purpose Computer | A computer designed to perform one set of tasks very well, such as a supercomputer, server, or embedded computer. | p. 36 |
| Specific-Purpose Machine | A machine designed to perform only one type of task, like a cash register at a store. | p. 127 |
| Storage (in IS) | The function of keeping collected data in databases, usually on servers or in the cloud, so it can be accessed later. | p. 192 |
| Storage | The largest and slowest form of memory, used for long-term keeping of data and programs (e.g., hard drives or SSDs). | p. 109 |
| Stored Program | The concept that a computer keeps instructions inside it, allowing it to perform different tasks by running different programs. | p. 19–20 |
| Street View Data | 360° street-level images collected by special vehicles for use in GIS platforms like Google Maps. | p. 199 |
| Student Data | Information about students, including names, IDs, grades, test scores, and attendance. | p. 198 |

| | | |
|---|---|---|
| Supercomputer | A massive and very powerful computer built to solve one very complex problem at a time, such as weather forecasting or scientific research. | p. 36–37 |
| Synchronization | Keeping all computer components working together in step using timing signals from the clock. | p. 104 |
| System | A set of interconnected parts working together to serve a particular purpose. | p. 54 |
| System Administrator | A professional who oversees and maintains an information system, manages user accounts, installs updates, and ensures smooth operation. | p. 195 |
| Tablet | A portable personal computer that uses a touchscreen instead of a physical keyboard, making it lighter but usually less powerful than a laptop. | p. 34–35 |
| Teacher Data | Information about teachers, such as their names, subjects taught, schedules, and contact details. | p. 198 |
| Technology | Tools, machines, or systems created to help us perform tasks, from pencils to computers. | p. 10 |
| Throughput | A measure of server performance, defined as the number of requests a server can complete in one second. | p. 174–175 |
| Usability | How easy it is for people to use a computer or interface to complete a task. | p. 64 |
| User Interface (UI) | The software part of the human-computer interface that lets users give instructions to a computer and see results. | p. 61 |
| Voice User Interface (VUI) | A user interface where people interact with computers by speaking, using natural language commands. | p. 63 |
| WAN (Wide Area Network) | A larger network covering a city, country, or even the whole globe. The internet is the biggest WAN. | p. 143–144 |
| Web Address | A special type of address used to locate a file or webpage on the World Wide Web. | p. 17 |
| WiFi NIC | A network card that connects a computer to wireless networks using radio signals. | p. 146 |
| WIMP Interface | A type of GUI that uses Windows, Icons, Menus, and a Pointer to interact with the computer. | p. 63 |
| Word (in computing) | The largest group of bits a processor can handle at once; in a 32-bit processor a word is 32 bits, in a 64-bit processor it is 64 bits. | p. 77 |
| World Wide Web (Web) | The collection of files, websites, and content (text, images, videos) that can be accessed through the Internet. | p. 17 |

# Index

complete a program. A measure of performance (p.112, 117).

**External drives** – Storage devices such as HDDs or SSDs that provide large capacity but risk damage or loss (p.171–172).

**Fiber optic cables** – Backbone of the internet, transmitting light signals across continents (p.144).

**Fields (Instruction)** – Sections of a machine language instruction, typically including opcode, data, and addresses (p.87).

**File** – A unit of storable information (p.19).

- Filename – Label assigned to identify a file (p.19).
- File servers – Servers that store and share files (p.40).

**Game servers** – Host online games and manage player accounts (p.40).

**General-purpose computers** – Flexible computers that run a wide range of applications; include personal computers such as desktops, laptops, tablets, and smartphones (p.32–33).

**Geographic Information System (GIS)** – Information system for collecting, storing, processing, and sharing geographic data, often visualized as maps (p.194–196).

**Graphical User Interface (GUI)** – Uses windows, icons, and menus to make computers more intuitive and accessible (p.62).

- WIMP Interface – GUI controlled by mouse or touchpad with windows, icons, menus, and pointer (p.63).
- Touchscreen Interface – GUI where users interact directly with the screen through gestures like tapping or swiping (p.63).

**GPU (Graphics Processing Unit)** – Specialized processor that executes graphics-related instructions. Works in parallel with the CPU (p.107).

**Gyroscope** – Smartphone sensor that detects rotational movement: clockwise/counterclockwise, rolling, pitching (p.58).

**Hardware** – The physical components of a computer, including input and output devices, that handle the physical (signal) layer of the interface (p.60).

**Hop count** – The number of routers a signal passes through on its path to a destination (p.147–149).

**Human-Computer Interaction (HCI)** – A multidisciplinary field studying how humans interact with computers and other technologies (p.122).

**Human-Computer Interface** – The boundary where humans and computers interact via input/output devices (p.122–123).

**Human-Machine Interface** – The boundary where humans interact with machines, often through embedded computers (p.126).

**I-cache (Instruction cache)** – Cache memory dedicated to storing instructions for faster execution (p.110).

**Information** – Meaningful content about ourselves or the world, processed and used by humans and computers (p.12–13).

- Creating (p.15).
- Sharing (p.16–18).
- Storing (p.19).

**Information systems** – Systems that collect, store, process, and share data to support decision-making (p.192).

- Functions – Collection, storage, processing, sharing (p.192).
- Components – Hardware, software, data, people (p.197–199).
- Examples – School Information Systems (SIS) (p.193–195); Geographic Information Systems (GIS) (p.194–196).

**Input devices** – Devices that convert external signals (light, sound, pressure) into internal signals; e.g., keyboard, mouse, microphone, camera, touchscreen, sensors (p.57–58).

**Input/Output (I/O) loop** – Continuous cycle of input and output exchanged between humans and computers until a task is completed or interrupted (p.123).

**Instruction cycle** – The repeated process of fetch, decode, and execute, performed continuously by the processor (p.91).

**Instruction set** – The complete set of instructions a processor is designed to perform (p.83, 92).

through multi-core CPUs and GPUs (p.107).

**Performance measures** – Metrics such as clock frequency, instruction count per second, and program execution time (p.106, 112).

**Peripherals** – External input, output, or I/O devices connected to computers to extend functionality (p.124–125). Examples: printer, projector, smartboard, webcam, scanner, headset.

**Personal computers (PCs)** – General-purpose computers for individual use, including desktops, laptops, tablets, and smartphones (p.34–35).

- Distinguished by portability (size, weight, power source) and capability (computational power) (p.34).
- Designed to balance flexibility, usability, capability, and affordability (p.35).

**Physical layer** – The actual signals transmitted and received, handled by hardware (p.60).

**Ping** – Networking tool to measure round-trip time between source and destination (p.148–149).

**Portability** – The ease with which a computer can be carried, depending on size, weight, and power supply (p.34).

**Prediction** – Optimization strategy that anticipates the next instruction or data likely to be needed, using separate caches or prefetching (p.110–111).

**Prefetching** – A prediction strategy where the processor fetches adjacent data items in advance, improving performance (p.111).

**Printers** – Output devices producing receipts or documents; use actuators like motors and printheads (p.127).

**Processing** – Structured change from one form to another, carried out by a series of repeatable steps (p.11–13).

- Information processing – Comparison with memory to identify and interpret (p.13–15).

**Processing (cloud-based)** – Service where computational power is provided via the cloud rather than on local devices (p.172).

**Processor (CPU)** – The central unit that changes input into output. Composed of ALU, CU, and register file (p.76, 92).

**Program** – A set of instructions to perform a specific task (p.20–21).

- Stored program – Concept that made modern computers possible (p.19–21).

**Projector** – Peripheral output device that displays visuals for large audiences (p.125).

**Public vs. Private Information** –

- Public – Accessible to everyone (p.17, 21).
- Private – Restricted by password access (p.17, 21).

**Purpose (robots)** – The specific tasks robots are designed to perform, from repetitive labor to hazardous work (p.128–129).

**Redundancy** – Strategy of keeping multiple copies of information to protect against loss (p.151).

**Registers / Register file** – Small, fast memory components located within the processor, managed by the CU, used to hold data and addresses during execution (p.88–90, 92).

**Repeaters** – Devices that amplify and retransmit weakened signals to prevent loss (p.151).

**Research** – Task example for general-purpose computers, such as using a web browser (p.33).

**Response time** – The time it takes for a server to respond to a user's request, as seen from the user's perspective (p.176).

**Robots** – Machines capable of performing tasks autonomously or with human intervention, using sensors (input) and actuators (output) (p.128–130).

- Benefits – Perform repetitive or dangerous tasks (p.129).
- Concerns – Job displacement, ethical behavior (p.130).

**Routers** – Devices that connect LANs to WANs, serving as gateways and traffic controllers; determine best routes using hop count and latency (p.143–149).

**School Information Systems (SIS)** – Information systems for managing school operations; provide schedules, attendance, grades, communication, and reports (p.193–195).

**Screen (cash register)** – Output device displaying transaction details (p.127).

building blocks of logic circuits; their arrangement determines operations (p.78).

**Trust in networks** – The expectation that transmissions are accurate, secure, and private (p.149–150).

**Usability** – The ease with which a device or interface can be used to complete a task. Higher in GUIs and touchscreens, lower in CLIs (p.64, 126).

**User Interface (UI)** – The software component of the human-computer interface, translating input into instructions and output into results (p.61).

- CLI, GUI, VUI – Major types of user interfaces (p.61–63).

**Voice User Interface (VUI)** – Interface that allows interaction using voice commands, e.g., Siri, Alexa, Google Assistant (p.63).

**WAN (Wide Area Network)** – Large networks covering cities, countries, or the globe; the internet is the largest example (p.143–144).

**Web (World Wide Web)** – Internet-based information system of files and pages (p.17).

- Web address – Location identifier for content on the Web (p.17, 19).
- Web servers – Host websites and deliver web pages (p.40).

**Webcam / Camera** – Input devices that capture images or video, extending the computer interface (p.125).

**WiFi** – Wireless networking technology for LAN and internet connectivity; typical range 45–150 m (p.146).

**WIMP Interface** – GUI type using windows, icons, menus, and pointer controlled by mouse/touchpad (p.63).

**Word (processor architecture)** – The maximum number of bits transmitted simultaneously (e.g., 32-bit word, 64-bit word) (p.77).